

Omics with Johnson 101

A practical guide to bioinformatics

Johnson

2026-06-05

TIER 1: Foundations: Computing and Mathematical Logic

The First Principles Paradigm

Welcome to the Pod

Welcome!

If you are here, because you want to learn how to write a Python script that runs a pre-packaged alignment tool, or if you expect a collection of copy-paste code blocks to process your datasets, you are in the *wrong* room.

There is a fundamental *crisis* in modern bioinformatics. The discipline has been flooded by “pipeline runners”, where individuals who treat complex biological software as magical black boxes. They pass default arguments to aligners, feed raw count matrices into statistical packages, and accept the output without questioning the underlying physics, mathematics, or hardware bottlenecks. When their pipeline crashes with a silent memory error, or when their results fail to replicate in a clinical trial, they are helpless.

This book is the antidote. It is designed to train you not as a tool operator, but as a computational architect.

The Syllabus: From Transistors to Mass-Based Omics

To master this discipline, you must understand the data path at every level of resolution. I organize this journey into a rigorous four-tier curriculum:

1. **Tier 1: Foundations** (Chapters 1–6): Before you analyze a genome, you must understand the machine. I begin with hardware limits (CPU, RAM, disk I/O), environment isolation (containers and DAGs), and the fundamental mathematical languages of omics: probability distributions (Negative Binomial overdispersion), linear algebra (Singular Value Decomposition), and statistical inference.
2. **Tier 2: Core Omics** (Chapters 7–14): I will dissect the physics of measurement. I will teach you how fluorescence chemistry (Illumina) and electrical current changes (Nanopore) are converted into sequences, followed by BWT alignment heuristics, de Bruijn assembly graphs, transcript abundance estimation, metagenomics, and statistical genetics.
3. **Tier 3: High-Resolution & Systems** (Chapters 15–21): I will transition to single cells, spatial coordinates, and networks. I will teach you manifold learning (t-SNE/UMAP), trajectory inference, spatial transcriptomic coordinates, network graph theory, multi-omics integration, 3D genome folding, and functional genomic perturbations.

4. **Tier 4: Mass-Based Omics** (Chapters 22–24): I will model biology at the functional endpoint. I will teach you mass spectrometry physics, proteomics acquisition workflows (DDA vs. DIA), metabolomic profiling, and lipid class separations.

The Course Rules

This curriculum enforces five non-negotiable rules:

Rule 1: First Principles First

You will not run a tool in this book until you have derived or conceptualized the mathematics and physics that justify its existence. I will teach you:

- The **physics** of the measurement instrument before the data format.
- The **mathematics** of the algorithm before the alignment command.
- The **statistics** of the noise before the analysis package.

Rule 2: Cultivate the Reviewer Mindset

Every published dataset is flawed until proven otherwise. As a computational biologist, your role is that of a forensic auditor. Throughout this course, you will be trained to look for technical batch effects, index hopping, mapping bias, and unconstrained permutations. You are the firewall against bad science.

Rule 3: Isolation is Scientific Integrity

A pipeline that runs only on your local machine is not a scientific instrument; it is a liability. Every analysis you publish must be version-controlled, containerized, and orchestrated via a Directed Acyclic Graph (DAG). You will learn to treat host environment dependency drift as a physical contaminant that must be rigorously excluded.

Rule 4: Reject Language Tribalism

The debate between programming languages is anti-scientific. You will choose your platform based on the mathematical and computational constraints of the problem, not your comfort zone. We use the UNIX shell for memory-efficient stream processing, Python for OOP and systems engineering, and R for vector algebra and statistical inference. An elite bioinformatician is multi-lingual by necessity.

Rule 5: Replicates Before Algorithms

A single high-throughput sequencing run is not a measurement; it is a random variable. Bioinformatics is the study of variance, and variance cannot be modeled from a single point. You will not analyze data without biological replicates, and you will not trust a statistical signal without assessing its dispersion parameter. If your experimental design is broken, no downstream algorithm can salvage it.

An Ongoing Project: The Road Ahead

This book is a living, ongoing project. The first four tiers lay the absolute physical and mathematical foundations of bioinformatics. Because the computational landscape shifts constantly, I treat this curriculum as a dynamic document. I am actively drafting future tiers that will extend this foundation to clinical environments, covering clinical pipeline validation, somatic variant calling in cancer genomes, and ACMG interpretation guidelines. These additions will be integrated in subsequent quarterly updates.

There are no step-by-step labs here. Code blocks exist solely as conceptual illustrations of production-grade logic. If you are ready to study bioinformatics at this level of rigor, turn to Chapter 1.

Chapter 1: The Bioinformatic Computing Environment

Compute is a Physical, Not a Magical Resource

All bioinformatics is performed on computers. Every command you run—whether aligning millions of sequencing reads, filtering genomic variants, or plotting gene expression—must direct physical electrons through a silicon chip. If you do not understand the hardware limits of your computer, your analysis will fail. You will try to open a 10-gigabyte sequencing file in a basic text editor or Excel, your system will freeze, and you will not understand why. A professional understands that hardware capacity governs data capacity.

Core Concepts

I suggest you think of your computing environment not as a software sandbox, but as a collection of physical resources. As a biological researcher, you will rarely write complex, multi-node supercomputing scripts. Most of your daily work will involve processing a single dataset at a time on your local laptop, a high-performance desktop workstation, or a remote institutional server. However, biological data is inherently large. Even a single RNA-seq or metagenomic sample can easily exceed 5 to 50 gigabytes of raw data.

To manage this volume without crashing your computer, I will teach you to understand the three physical bottlenecks of computation: **CPU (processing power)**, **RAM (working memory)**, and **Storage I/O (disk read/write speed)**.

The Three Bottlenecks

I recommend you monitor these three resources constantly during any pipeline execution:

1. **CPU (Central Processing Unit):** The brain of the computer that executes mathematical instructions. CPU bottlenecks occur when you run computationally intensive algorithms (such as sequence alignment or graph-based assembly). If your CPU utilization is running at 100% across all allocated cores, your task is CPU-bound; it is computation-limited.

2. **RAM (Random Access Memory):** The temporary, high-speed workspace of the computer. When a program runs, it loads data into RAM so the CPU can access it instantly. RAM is volatile (erased when the computer turns off) and finite. If your data volume exceeds your physical RAM, the operating system will crash or force-kill the program.
3. **Storage I/O (Input/Output):** The read and write channels connecting your long-term storage (SSD or HDD) to the motherboard. Disk I/O bottlenecks occur when the CPU sits idle because it cannot read data off the disk fast enough. If you read a massive file from a slow mechanical hard drive, your pipeline stalls.

The Memory Wall: Why Opening a 10GB File Crashes Your Laptop

A common mistake for young researchers is trying to view a raw biological data file (like a `.fastq` or `.sam` file) using default office software (like Microsoft Word, Excel, or TextEdit).

I suggest you understand why this fails: standard applications are designed to load the **entire file** into RAM before displaying it. If you try to open a 15GB FASTQ file on an 8GB RAM laptop, the operating system is forced to search for space. When physical RAM is exhausted, the OS tries to write temporary data to your hard drive (called “swapping”), slowing your computer to a crawl. Eventually, the Linux kernel invokes a mechanism called the **OOM (Out-of-Memory) Killer** to force-terminate the text editor to save the machine from freezing.

[!NOTE] **The OOM Killer:** This is a kernel safety feature. It acts as an automated emergency brake. If you run a script and it suddenly terminates with the word `Killed`, it almost always means you exceeded your machine’s physical memory.

To process genomic scale datasets, you cannot use office software. I will teach you to use the UNIX command line and the concept of **stream processing**.

The Leverage of the Terminal: UNIX Streams and Pipes

The UNIX operating system dominates bioinformatics because of a simple, elegant abstraction: **everything is treated as a file**.

Instead of loading a massive file into memory all at once, the terminal allows you to stream data line-by-line. This is called **stream processing**—processing data in small, continuous chunks rather than in a single batch. The memory footprint of the program remains constant (usually just a few megabytes), regardless of whether the file is 10 megabytes or 10 terabytes.

The key leverage tool is the **pipe operator (|)**. It redirects the output of one program directly into the input of another, entirely in RAM, without writing intermediate temporary files to your disk.

Let us look at a practical example:

```
# Novice Approach: writes huge intermediate files to disk, wasting I/O and RAM
grep "chr1" large_dataset.sam > temp_chromosome1.sam
wc -l temp_chromosome1.sam
rm temp_chromosome1.sam

# Professional Approach: streaming data in memory via a pipe
grep "chr1" large_dataset.sam | wc -l
```

In the professional approach, `grep` reads `large_dataset.sam` line-by-line, filters for rows containing "chr1", and pushes only those matching lines directly to `wc` (word count). The data flows through a small memory buffer in RAM. No intermediate files are written to the disk, saving time and storage space.

Essential Command-Line Streams

I suggest you master a small set of general-purpose UNIX utilities to navigate biological data:

- `cat` and `zcat`: Display files (or compressed `.gz` files) as a stream.
- `head` and `tail`: View the first or last few lines of a file (essential for checking file formats without loading the whole file).
- `grep`: Search for matching patterns (e.g., finding a barcode or primer sequence).
- `awk`: A powerful column-oriented text processor (perfect for filtering tab-delimited coordinate files like BED or GFF).
- `sed`: A stream editor used to replace text patterns on the fly.

Remote Server Basics: SSH and tmux

As a sole researcher, your laptop will often lack the RAM or CPU cores to run genomic alignments. Your institution will provide access to a remote Linux server.

You connect to this server using **SSH (Secure Shell)**:

```
ssh username@server.university.edu
```

Once connected, your terminal is executing commands directly on the remote machine. However, if your internet connection drops—even for a split second—your SSH session will disconnect. The operating system sends a termination signal (SIGHUP) to your running programs, killing your analysis midway.

To prevent this, I recommend you run your commands inside a **persistent terminal multiplexer**—a utility that maintains active shell sessions independently of your network connection—such as `tmux` (or `screen`).

`tmux` creates an isolated shell session that runs independently of your SSH connection. If your laptop loses power or internet, the process on the server continues to execute uninterrupted.

- `tmux new -s my_analysis`: Create a new session named "my_analysis".
 - `Ctrl+b` followed by `d`: Detach from the session (leaving your script running in the background).
 - `tmux attach -t my_analysis`: Reconnect to your running session from any computer at any time.
-

Biological Interpretation

When your analysis runs slowly or crashes, I recommend you diagnose the hardware bottleneck before trying to alter your biological parameters or software versions. Let me ask you: if a pipeline runs overnight and fails with a generic error code, how do you distinguish an algorithmic error from an Out-Of-Memory termination? If you cannot answer this, you are wasting compute time.

How to Diagnose Bottlenecks Locally

1. **Check Resource Utilization with `htop`**: I suggest you open a terminal window and run `htop` to display a real-time visual representation of your CPU cores and RAM usage.
 - If you see all CPU bars pegged at 100%, your task is CPU-bound. If your tool supports multi-threading (e.g., using `-t` or `-p` flags in aligners), you can accelerate it by allocating more cores.
 - If you see your memory bar (RAM) full, and the computer is unresponsive, your task is memory-bound. I recommend you rewrite your pipeline to stream the files in smaller chunks rather than loading them in their entirety.
2. **Verify Memory Crashes with `dmesg`**: If your script abruptly stopped and displayed the word `Killed`, I recommend you run the following command to inspect the kernel buffer:

```
dmesg -T | grep -i -E "oom|kill"
```

If you see a line stating `Out of memory: Kill process`, you have hit the memory wall. You must rewrite your code to process data in chunks or migrate to a server with higher memory capacity.

3. **Monitor Disk Capacity with `df -h`**: Before downloading massive FASTQ datasets from the Sequence Read Archive (SRA), I recommend checking your remaining storage:

```
df -h
```

Running out of disk space midway through a sequence alignment will corrupt your output files, forcing you to restart the process.

Current Landscape

While raw command-line tools are essential, modern bioinformatics has evolved to make remote development more user-friendly:

- **VS Code Remote Development:** Instead of editing files on a remote server using complex terminal editors (like `vim` or `nano`), you can use Visual Studio Code with the **Remote - SSH** extension. This allows you to edit files, view plots, and run terminal commands on the remote server using a standard, familiar desktop interface.
 - **Mamba/Conda Environment Managers:** Software installation has moved away from manual compilation. As we will discuss in the next chapter, package managers allow sole researchers to set up complete environments with a single command.
-

Summary and Required Reading

1. **Compute is physical:** Every command consumes CPU cycles, RAM, and Disk I/O.
2. **Do not load large files into text editors:** Use `head` or `zcat` to inspect files.
3. **Pipes prevent memory bottlenecks:** Use the pipe operator (`|`) to stream data line-by-line in memory without writing intermediate files to disk.
4. **tmux is your safety net:** Run all server analyses inside a `tmux` session to prevent connection-drop crashes.
5. **Diagnose first:** Use `htop` to identify CPU vs. RAM bottlenecks, and `dmesg` to check for OOM memory kills.

Required Reading

- UNIX Shell Basics: Learn the usage of `head`, `tail`, `grep`, `awk`, and pipes (`|`).
- SSH Basics: Understand how to securely copy files between your local machine and a server using `scp` or `rsync`.

Johnson's Rule: A 100-gigabyte genomic dataset does not fit into an Excel sheet, but it fits perfectly into a UNIX pipe. Master stream processing, or you will spend your career waiting for frozen laptops to reboot.

Chapter 2: Organizing and Securing Your Analysis (Git & Conda)

An Unrecorded Step is Scientific Fiction

In bioinformatics, your code is your experimental record. If you run a script, generate a figure, and cannot trace the exact code version and software packages that produced it, your results are not science—they are a draft. Bioinformatics reproducibility is not about making your code pretty; it is about ensuring that your future self, your principal investigator (PI), and scientific reviewers can recreate your exact results from your raw data.

Core Concepts

When you start analyzing a biological dataset, the immediate temptation is to write quick, messy scripts and save files all over your desktop. This habit leads to the “reproducibility crisis”: a year later, when a reviewer asks you to re-plot a figure with a new threshold, you will find yourself staring at files named `script_v2_final_edit.R` or `results_table_new_corrected.csv`, unable to recall which one is correct.

Reproducibility for a sole researcher requires two simple, non-negotiable tools: **project organization (directories)**, **version control (Git)**, and **environment isolation (Conda/Mamba)**.

Structuring Your Project Directory

Before writing a single line of code, you must establish a standard, consistent folder structure. Do not store your raw sequencing files in the same directory as your scripts or figures.

A standard biological project directory should look like this:

```
my_project/
├── .gitignore      # Tells Git what files to ignore (e.g., massive fastq files)
├── environment.yml  # Lists all software packages and versions used
└── README.md       # Documents the project overview and execution order
```

```
|— data/
|   |— raw/          # Raw sequencing reads (read-only, never edit)
|   |— processed/    # Intermediate filtered files (e.g., cleaned BAMs)
|— scripts/          # Python/R scripts that execute the analysis
|— results/          # Final figures, tables, and statistics
```

By keeping raw data strictly read-only and separate from your scripts, you ensure that your raw input remains untainted, and your results are always a predictable output of your scripts.

Layer 1: Version Control with Git

Git is a tool that records changes to your files over time. It creates a cryptographic history of your code. If you make a mistake that breaks your analysis, Git allows you to instantly roll back your files to a working state.

For a sole researcher, the core Git workflow is simple:

1. **Initialize:** Create a Git repository in your project folder:

```
git init my_project
```

2. **Inspect:** Check which files have changed:

```
git status
```

3. **Stage:** Mark changes you want to save:

```
git add scripts/normalize_counts.R
```

4. **Commit:** Save the staged snapshot to history with a clear description:

```
git commit -m "Add DESeq2 normalization script"
```

5. **Review:** Look at the changes between your current file and the last saved version:

```
git diff scripts/normalize_counts.R
```

What NOT to Track with Git: The .gitignore file

Git is designed for text files (code scripts). It is not designed to track massive binary datasets (such as raw `.fastq.gz` sequencing reads, 50GB alignment `.bam` files, or binary R workspaces). If you try to commit large data files to Git, your repository will become bloated, slow, and impossible to upload to repositories like GitHub.

You must create a file named `.gitignore` at the root of your project folder to tell Git to ignore large data and temporary files:

```
# Exclude raw and processed data
data/
results/
```

```
# Exclude large sequencing formats
*.fastq
*.fastq.gz
*.bam
*.bai
*.vcf
*.sam

# Exclude temporary system files
.DS_Store
.Rhistory
.RData
```

Layer 2: Environment Isolation with Conda and Mamba

Git tracks the code you write, but it does not track the software packages you run.

If you write an R script that uses the package DESeq2 version 1.38, and a collaborator runs the same script on their laptop with DESeq2 version 1.44, the script may crash or produce slightly different results due to changes in default parameters. This is the package dependency conflict problem.

Conda (and its faster drop-in replacement, Mamba) solves this by creating **isolated software environments**. Mamba installs tools and libraries inside a dedicated directory, keeping them completely separate from your computer's default system libraries.

Pinning Your Environment with `environment.yml`

To make your analysis reproducible, you must document every software package and version you used. You do this by exporting your environment to a configuration file called `environment.yml`:

```
name: rna_seq_analysis
channels:
  - bioconda
  - conda-forge
  - defaults
dependencies:
  - python=3.9
  - r-base=4.2.3
  - bioconductor-deseq2=1.38.0
  - bioconductor-salmon=1.9.0
  - samtools=1.17
```

A collaborator (or your future self) can recreate your exact software environment on a new machine with a single command:

```
mamba env create -f environment.yml
```

This guarantees that the exact same package versions are resolved and installed, preventing library conflicts.

Biological Interpretation

For a researcher, reproducibility is not a theoretical exercise; it has direct consequences during peer review.

The Reviewer Mindset: Guarding Against “Silent Shifts”

When you submit a manuscript, a reviewer may request that you rerun your analysis using an updated reference genome or different threshold parameters. If you have not pinned your environment and tracked your code:

- You will spend weeks trying to reconstruct which versions of software you used.
- Updating a package globally on your computer to solve one problem will break a different project’s scripts.
- Your output plots may shift slightly, raising doubts about the robustness of your findings.

A robust directory structure with a version-controlled Git history and a pinned environment `.yaml` means you can address reviewer comments in hours rather than weeks.

Current Landscape

While Conda/Mamba environments are the baseline standard for sole researchers, larger core facilities and clinical labs use **containers** (like Docker or Singularity) and **workflow managers** (like Nextflow) to scale analysis across thousands of samples in the cloud. However, as a researcher processing one or two datasets at a time, local Conda/Mamba environments provide the ideal balance of simplicity, speed, and absolute reproducibility.

Summary and Required Reading

1. **Code is your experimental record:** Always write scripts; never perform manual Excel formatting on raw data.
2. **Separate data from code:** Keep raw data read-only in `data/raw/` and script code in `scripts/`.
3. **Git tracks code, not data:** Never commit massive biological files (`.fastq`, `.bam`) to Git. Use a `.gitignore` file.

4. **Mamba isolates environments:** Create isolated software environments and pin exact package versions in `environment.yml`.
5. **Recreate in one step:** Use `mamba env create -f environment.yml` to clone environments across machines.

Required Reading

- Conda / Mamba Guide: Learn how to create, activate, and export environments.
- Git Basics: Practice initializing a repository, staging files (`git add`), and committing changes.

Johnson's Rule: A peer-reviewed paper whose code cannot be run by a reviewer on a cold machine is a scientific claim without evidence. Pin your software environment, commit your scripts, and treat your analysis as a published protocol.

Chapter 3: Programming Paradigms and Project Organization (Python & R)

Choose the Language for the Task, Not the Tribe

Language tribalism (“I only program in R” or “I only use Python”) is a hallmark of the amateur. R and Python are not religions; they are tools engineered for distinct tasks. Python was built for general-purpose programming, file parsing, and automation. R was built by statisticians for data analysis, modeling, and plotting. A successful researcher uses both, selecting R to test hypotheses and Python to wrangle raw text.

Core Concepts

As a researcher, you do not need to build complex software packages. However, you must write code that is clean, readable, and does not corrupt your data.

To achieve this, you must understand when to deploy Python versus R, how biological files are structured as tables, and how to save your data objects safely.

Python: The Wrangle and Text Processing Tool

Python excels at handling raw files. In genomics, raw data typically arrives as massive text files (FASTQ files of raw reads, FASTA files of genomes, or tabular annotation files like GFF and BED).

Python’s syntax and built-in libraries make it the ideal language for:

1. **Reading files line-by-line:** Parsing text files without loading them fully into RAM (as discussed in Chapter 1).
2. **Regular Expressions (Regex):** Finding specific sequence motifs, adapters, or primers inside DNA strings.
3. **Command-Line Interfaces (CLIs):** Writing scripts that take input file arguments so you can run the same script on multiple samples.

R: The Statistical and Plotting Tool

R is built around the **DataFrame**—a tabular structure where rows are observations (e.g., genes) and columns are variables (e.g., samples). R is the industry standard for:

1. **Statistical Modeling:** Running differential expression (e.g., DESeq2), regression, and multiple-testing corrections.
2. **Visualization:** Creating publication-grade plots using `ggplot2` (a visual language built on the “Grammar of Graphics” where plots are built in composable layers).
3. **Bioconductor:** Accessing a vast repository of packages designed specifically for biological coordinate and genomic data.

Tabular Genomic Formats: BED and GFF as Data Frames

Most genomic annotation files are simply tab-delimited text tables. Understanding how to load these files as data frames is the first step in coordinate math:

- **BED (Browser Extensible Data):** A simple table mapping genomic regions. It requires three core columns: chromosome, start position, and end position. Note: BED files use **0-based, half-open coordinate indexing** (the start position is 0-indexed, and the end position is not included in the range).
- **GFF (General Feature Format):** A more detailed annotation table detailing exons, genes, and transcripts. It uses **1-based, closed coordinate indexing** (both start and end coordinates are inclusive and start at 1).

In R, Bioconductor packages translate these text files into validated objects (like `GRanges`) that handle the coordinate differences automatically, preventing off-by-one errors when computing gene overlaps.

Serialization: R Objects and `saveRDS` Safety

After running an expensive analysis (like normalizing a single-cell count matrix), you will want to save your R variables to disk so you do not have to rerun the pipeline tomorrow.

In R, there are two primary ways to save files, and one of them is highly dangerous:

The Dangerous Way: `save()` and `.RData`

```
# DO NOT USE THIS FOR SCIENTIFIC DATA
save(my_normalized_counts, file = "my_analysis.RData")
```

`save()` exports your variable along with its exact variable name into a binary workspace file. When you load it tomorrow using `load("my_analysis.RData")`, R restores the variable `my_normalized_counts` directly into your active workspace. If you already had a different variable named `my_normalized_counts` in your active session, it will be silently overwritten and erased.

The Safe Way: `saveRDS()` and `readRDS()`

```
# THE CORRECT WAY
saveRDS(my_normalized_counts, file = "my_normalized_counts.rds")
```

`saveRDS()` serializes the data structure itself, *independent of the variable name*. When you load it, you must explicitly assign it to a new variable:

```
counts_subset <- readRDS("my_normalized_counts.rds")
```

This preserves your workspace namespaces, prevents accidental variable overwrites, and makes your scripts transparent and readable.

Biological Interpretation

The “Notebook” Pitfall

Many freshman researchers perform their entire analysis in interactive environments like Jupyter Notebooks or RStudio, running blocks of code in random orders. While interactive notebooks are excellent for exploration, they are dangerous for publishing:

- **Out-of-order state:** If you run cell 5, then edit cell 2, then run cell 10, your active workspace depends on a sequence of actions that cannot be replicated.
- **Hidden variables:** You may delete a line of code, but the variable remains in RAM, letting your script run until you restart the session and it crashes.

The Rule of the Script

Always verify your analysis by restarting your R or Python session and running your scripts sequentially from top to bottom. If the script does not run successfully on a clean workspace, the analysis is not complete.

Summary and Required Reading

1. **Select the tool for the job:** Use Python for file preparation and text parsing; use R for statistics and data plotting.
2. **Understand coordinates:** BED files are 0-indexed; GFF files are 1-indexed. Use coordinate-aware libraries to handle overlaps.
3. **Use `saveRDS`:** Never use `save()` to dump workspace images. Use `saveRDS()` and `readRDS()` to maintain variable namespace safety.
4. **Enforce sequential execution:** Interactive notebooks are for drafts; sequential scripts are for reproducible science.

Required Reading

- Bioconductor Overview: Familiarize yourself with standard container objects like `GRanges` and `SummarizedExperiment`.
- R ggplot2 tutorial: Learn the layers of the Grammar of Graphics (Data `→` Aesthetics `→` Geometries).

Johnson's Rule: Excel is not a database, and an RWorkspace save file is not an archive. Write clean scripts that read raw tables and output plots. If you cannot replicate your final figures by running a single script on a clean session, your analysis does not exist.

Chapter 4: Statistical Foundations of Sequencing Counts (Overdispersion)

Read Counts are Random Variables

If you take a tube of RNA, prepare two biological libraries, and sequence them under identical conditions, you will not get the same number of reads for any gene. A gene that yields 450 reads in one sample might yield 420 in another. This is not measurement error; the sequencer did not malfunction. The difference arises from biological variation and random sampling. You cannot declare a gene “upregulated” unless you prove that the observed difference exceeds the expected noise of its statistical distribution.

Core Concepts

In genomics, data comes in the form of **counts** (the number of sequencing reads mapping to a gene). A gene is observed 0, 1, 2, or 500 times—it cannot be observed 450.7 times.

This discrete, integer-valued nature of genomic data means you **cannot use standard normal-distribution statistics** (like t-tests or ANOVA) for two primary reasons:

1. **Negative and fractional values:** The normal distribution assigns probability to impossible values (like -5 reads or 12.3 reads).
2. **The mean-variance relationship:** In sequencing data, variance scales with the mean. Highly expressed genes have much larger absolute variance than lowly expressed genes, violating the normal distribution’s assumption of equal variance.

To analyze counts, you must use statistics tailored to discrete data.

The Poisson Model: Technical Noise Only

Sequencing is a sampling process. The sequencer extracts a tiny subset of reads from a pool of billions of molecules. This process is modeled by the **Poisson distribution**, which describes independent events occurring at a constant rate.

The defining property of the Poisson distribution is that its **variance equals its mean**:

$$\text{Variance} = \text{Mean}$$

This model fits technical replicates perfectly (re-sequencing the *same* tube of RNA). Here, the only variation is the random chance of which molecule gets read.

However, the Poisson model **fails completely on biological replicates** (sequencing samples from different mice or different patient biopsies).

Overdispersion: When Variance Exceeds the Mean

Consider a real-world scenario. You sequence three biological replicates of healthy mice and measure the read counts for a single gene:

- **Mouse 1:** 150 reads
- **Mouse 2:** 300 reads
- **Mouse 3:** 450 reads
- *Mean (μ):* 300
- *Variance (σ^2):* 22,500

The Poisson model predicts that if the mean is 300, the variance should be 300. In reality, the observed variance (22,500) is **75 times larger** than the mean.

This phenomenon is called **overdispersion** (where variance is significantly greater than the mean). It occurs because biological replicates are not identical; they differ due to genetic backgrounds, micro-environments, and cellular stochasticity. This true biological variation is layered on top of the technical Poisson sampling noise.

The Negative Binomial Model: The Conclusion of Overdispersion

To model biological replicates, differential expression software (such as DESeq2 and edgeR) uses the **Negative Binomial distribution**.

The Negative Binomial distribution introduces a second parameter, the **dispersion parameter (α)**, to capture this biological variation. Its variance is defined by the following conclusion:

$$\text{Variance} = \text{Mean} + \alpha \times \text{Mean}^2$$

- When $\alpha = 0$, the quadratic term drops out, and the distribution collapses back to the Poisson (variance = mean).
- When $\alpha > 0$, the term $\alpha \times \text{Mean}^2$ accounts for the biological variation across replicates.

Why the Dispersion Parameter (α) Prevents False Discoveries

If you use a Poisson-based model to analyze biological replicates, the model assumes the variance is much smaller than it actually is (it assumes variance = 300 instead of 22,500).

Because the model underestimates the noise, it will misinterpret random biological fluctuations as highly significant biological discoveries. You will report thousands of “differentially expressed genes” that are actually just noise. This is why the Negative Binomial model is the standard in transcriptomics: it correctly adjusts the statistical threshold to account for true biological variation.

Biological Interpretation

When analyzing a differential expression table, you must look at both the fold change and the dispersion of your target genes.

High-Dispersion vs. Low-Dispersion Genes

Different classes of genes exhibit different biological dispersion (α):

- **Housekeeping Genes:** Genes required for basic cellular maintenance (e.g., ribosomal proteins, actin) have very low dispersion ($\alpha \approx 0.01$). Their expression is stable and uniform across biological replicates.
- **Environmental Response/Immune Genes:** Genes that respond to stress, cytokines, or activation states often have very high dispersion ($\alpha \geq 0.5$). Their expression varies wildly across individuals, even in the same treatment group.

If a gene of interest has high dispersion, you will require a larger sample size (more replicates) to achieve the statistical power necessary to prove its change is significant.

Summary and Required Reading

1. **Read counts are discrete:** Do not use normal-distribution statistics (t-tests) on sequencing read counts.
2. **Poisson models technical noise:** It assumes variance equals the mean, which holds true only for technical replicates of the same sample.
3. **Overdispersion is biological:** Variance across different organisms always exceeds the mean.
4. **Negative Binomial models overdispersion:** It uses a dispersion parameter (α) to inflate the expected variance: $\text{Var} = \mu + \alpha\mu^2$.

5. **Ignore dispersion at your peril:** Neglecting the α parameter leads to massive rates of false-positive biomarker discovery.

Required Reading

- DESeq2 Methodology paper: Read the introduction explaining why Poisson models fail on biological replicates and how dispersion is estimated.

Johnson's Rule: Organisms are not cells in a tissue culture plate, and biological replicates are not technical runs. If your software uses a Poisson model for biological differential analysis, it is lying to you. Use the Negative Binomial model to ensure your discoveries survive replication.

Chapter 5: Visualizing High-Dimensional Biology (PCA and Clustering)

Dimensionality Reduction Compresses Space, Not Signal

An RNA-seq dataset measuring 20,000 genes across 50 samples is not a simple spreadsheet; it is a mathematical matrix in 20,000-dimensional space. Every sample is a vector positioned along 20,000 gene coordinate axes. The human brain cannot visualize anything beyond three dimensions. We must rely on linear algebra to rotate and compress this high-dimensional space into a two-dimensional plot, allowing us to see how biological samples relate.

Core Concepts

In genomics, the data you analyze has far more dimensions (variables) than samples. This is the **high-dimensional data problem**.

However, genes do not act independently. Biological processes (like cell division, immune activation, or tissue differentiation) involve the coordinated expression of hundreds of genes simultaneously. This means the true biological signal does not occupy 20,000 independent dimensions—it is low-rank, meaning it is governed by a small number of coordinated biological pathways.

We use **Principal Component Analysis (PCA)** to find these coordinated signals and project them onto a 2D or 3D plot.

PCA: Intuitive Coordinate Rotation

Imagine a cloud of points representing your samples. PCA rotates this cloud of data points in high-dimensional space until it finds the direction of **maximum variation**.

- **Principal Component 1 (PC1):** The single coordinate axis along which the data points are most spread out. It captures the strongest signal (variance) in the dataset.

- **Principal Component 2 (PC2):** The second axis, chosen to be completely perpendicular (orthogonal) to PC1, capturing the next largest amount of variation.

By plotting your samples using PC1 as the X-axis and PC2 as the Y-axis, you compress 20,000 genes into a simple 2D scatter plot.

How to Read a PCA Plot

1. **PCA Scores (The Sample Points):** Each point on a PCA plot represents a single biological sample. If two sample points are close to each other on the plot, it means they have highly similar transcriptomes. If they are far apart, their gene expression profiles differ.
2. **PCA Loadings (The Genes):** Loadings are weights that define how much each gene contributes to a given PC. A gene with a high positive loading on PC1 is highly expressed in samples positioned on the right side of the plot and lowly expressed in samples on the left. Loadings connect the mathematical axes back to biological pathways.
3. **Variance Explained:** Every PC has an associated percentage (e.g., "PC1: 45% variance explained"). This tells you what proportion of the total transcriptomic variation is captured by that PC. If PC1 and PC2 together explain 60% of the variance, a 2D scatter plot represents the majority of the signal in your data.

The Scree Plot: Deciding How Many PCs Matter

A scree plot shows the percentage of variance explained by each Principal Component in descending order.

You look for the **"elbow"** in the plot—the point where the variance explained drops sharply and flattens out. The PCs before the elbow contain the biological signal; the flat tail after the elbow contains random technical noise. For downstream analyses (like clustering), researchers typically keep only the top PCs (usually 10 to 30) before the elbow, discarding the noise-only dimensions.

Biological Interpretation

PCA is the primary quality control tool in genomics. Before you analyze your data for biological significance, you must audit the PCA plot.

Rule 1: Color by Batch Before You Color by Biology

When you receive your sequencing data, the first thing you should do is plot a PCA of the samples and color the points by **technical variables** (e.g., sequencing run day, library preparation batch, or technician).

- If the samples cluster cleanly by biological group (e.g., tumor vs. normal), your biological signal is strong.

- If the samples cluster by the date they were prepped, the primary driver of variation in your dataset is a **technical batch effect**. You must correct for this batch effect before you can trust any biological discoveries (as discussed in Chapter 6).

Rule 2: Inspect the Loadings for Biological Sanity

Do not treat the principal components as abstract numbers. Look at the top 10 genes with the highest loadings on PC1 and PC2.

- In cancer samples, PC1 is often driven by cell proliferation genes (e.g., *MKI67*, *PCNA*).
 - If your top loadings are dominated by mitochondrial genes (e.g., *MT-CO1*), it is a technical warning sign: high mitochondrial expression indicates dying cells, meaning the primary variation in your experiment is cell death, not biology.
-

Summary and Required Reading

1. **Genomic data is high-dimensional:** PCA compresses thousands of genes into a low-dimensional space we can visualize.
2. **PCs are directions of maximum variation:** PC1 captures the strongest signal; PC2 captures the second strongest.
3. **Distance equals similarity:** Samples that cluster close together on a PCA plot have similar expression profiles.
4. **Use PCA for quality control:** Color by technical variables first to detect batch effects, and audit gene loadings for biological sanity.

Required Reading

- Bioconductor PCA Workflow: Practice loading a count table in R, centering the data, running `prcomp()`, and plotting the results.

Johnson's Rule: A PCA plot is the mirror of your experiment. If your samples cluster by the day they were prepped rather than the drug they were treated with, no statistical trick can save your analysis. Find the batch confounder before you write the manuscript.

Chapter 6: Hypothesis Testing and Confounder Management (FDR & Batch Effects)

Every P-value is Conditional on What You Model

If you run a differential expression test comparing 10 cancer patients against 10 healthy controls and produce a list of 5,000 “significant” genes, you have not discovered the genomic signature of cancer. If the cancer patients were sequenced in January and the controls in March, you have discovered the signature of a sequencing batch. A p-value is always conditional on your experimental design, and every technical confounder you omit will inflate your variance, creating false discoveries that mimic biological signal.

Core Concepts

In genomics, we rarely measure one variable at a time. We measure 20,000 genes or millions of genetic variants simultaneously. This scale introduces two major statistical challenges: **the multiple testing problem** and **technical batch effects**.

P-values and the Multiple Testing Problem

When you run a statistical test (e.g., comparing gene expression between a control and a treated group), the output is a **p-value**.

- **What a P-value is:** The probability of observing a difference as large or larger than what you measured, assuming the null hypothesis (H_0 : no true difference) is true.
- **What a P-value is NOT:** It is not the probability that your discovery is real or that the drug worked.

If you set a standard statistical significance threshold of $p < 0.05$:

- For a single gene, there is a 5% chance of calling it significant by pure random noise.
- For 20,000 genes, you are guaranteed to find $20,000 \times 0.05 = 1,000$ significant genes by pure random chance, even if the drug had absolutely no effect.

This is the **multiple testing problem**. If you do not correct for it, your biomarker list will be dominated by random false discoveries.

Controlling the False Discovery Rate (FDR)

To solve the multiple testing problem, we adjust our p-values. The industry standard in genomics is the **Benjamini-Hochberg False Discovery Rate (FDR)** correction, which outputs an **adjusted p-value (or q-value)**.

FDR (q-value) = The expected proportion of false discoveries among your selected hits

- If you select genes with a raw $p < 0.05$, you expect 1,000 of them to be false discoveries.
- If you select genes with a adjusted $q < 0.05$, you accept that **5% of your final list** (e.g., 5 out of 100 genes) will be false discoveries. This preserves your statistical power to find real biological signals while keeping false discoveries under strict control.

[!IMPORTANT] **Ignore Raw P-values:** In genomics, raw p-values are statistical illusions. You must always use FDR-adjusted q-values (in R, often labeled `padj` or `adj.P.Val`) when reporting differential expression or biomarker hits.

Confounders and Batch Effects: The Silent Killers of Science

A **confounder** is a technical variable that correlates with your biological variable of interest, distorting your statistical tests. The most common confounder in genomics is the **batch effect**.

A batch effect occurs when samples are processed, extracted, or sequenced at different times, by different technicians, or using different reagent lots. Because sequencing instruments are highly sensitive, these minor technical differences introduce massive, systematic variation in read counts.

If you confound your study design (e.g., preparing all control libraries on Monday and all treated libraries on Friday), the batch effect is completely mixed with your biology. No statistical method can separate the two. Your differential analysis will identify genes driven by “Friday’s prep conditions” rather than your drug.

Batch Management and Covariate Adjustment

If you have balanced your design (processing both controls and treatments across batches), you can mathematically remove the technical variation:

1. **Add Batch to Your Linear Model:** When running differential expression software (like DESeq2), you can include the batch variable as a covariate in your design matrix:

```
# Correct formula: controls for batch while testing for condition  
design <- ~ batch + condition
```

This instructs the statistical model to estimate and absorb the variance driven by the batch, letting you test the true effect of the condition with maximum power.

2. **Run Batch Correction (ComBat):** For visualization (like plotting PCAs or heatmaps), you can use tools like ComBat (from the R package *sva*) to mathematically align the sample counts across batches, removing the visual clustering driven by technical dates.
-

Biological Interpretation

The P-value Histogram Audit

Before performing FDR corrections or biological pathway enrichment, you must inspect the histogram of your raw, uncorrected p-values. The shape of this histogram acts as a diagnostic audit of your experiment:

- **Anti-Conservative (Good):** A sharp spike near zero (representing true biological discoveries) tapering off into a flat, uniform distribution from 0.1 to 1.0 (representing null genes).
 - **Broad Left Skew (Warning):** A hill-like curve peaking between 0.01 and 0.15 without a sharp spike at zero. This indicates unmodeled technical variation (batch effects) inflating the significance of hundreds of null genes. Do not trust your FDR corrections on this data until you inspect the PCA and adjust for confounders.
-

Summary and Required Reading

1. **P-values are probabilities under the null:** They do not measure the probability that your biological hypothesis is true.
2. **Multiple testing multiplies errors:** Testing 20,000 genes at $p < 0.05$ produces 1,000 false positives by pure chance.
3. **FDR controls the false discovery proportion:** Always use adjusted q-values ($q < 0.05$) to control the ratio of false discoveries in your final biomarker list.
4. **Balance your batches:** Never mix technical batch boundaries with biological boundaries. Process controls and treatments together in every batch.
5. **Model technical variance:** Include batch covariates in your regression formulas to absorb technical noise and reveal true biological signals.

Required Reading

- Benjamini-Hochberg FDR Paper: Read the conceptual section describing why controlling FDR is preferred over conservative FWER (Bonferroni) corrections in high-dimensional discovery.

Johnson's Rule: A p-value is a mathematical calculation, not a biological truth. If you do not balance your experimental batches and control for multiple testing, your statistical software will happily deliver highly significant p-values for random noise. Correct your model before you declare your discovery.

TIER 2: The Core Omics: Sequencing Physics and Algorithms

Chapter 7: Sequencing Physics, Chemistry, and Raw Formats

You Cannot Process a File Correctly If You Do Not Understand the Physics of the Machine That Generated It

A sequencing file (such as a FASTQ file) is not just abstract text on your screen; it is a digital recording of a physical chemistry experiment. The numbers, characters, and coordinates inside these files represent light signals, electrical currents, and chemical reactions that occurred inside a sequencing instrument. If you treat these files as generic databases without understanding the hardware chemistry under the hood, your downstream analyses will be fundamentally blind to technical errors.

Core Concepts

A Brief History: From Capillaries to Massive Parallelization

For decades, the gold standard of sequencing was **Sanger Sequencing**. Developed by Frederick Sanger in 1977, this method relies on chain-termination chemistry.

1. **The Chemistry:** DNA polymerase copies a target strand using standard nucleotides (dNTPs) mixed with a small concentration of fluorescently labeled chain-terminating nucleotides (ddNTPs).
2. **The Measurement:** Whenever a ddNTP is incorporated, synthesis stops. This creates a mixture of DNA fragments of every possible length, each ending with a colored label. These fragments are separated by size using capillary electrophoresis, and a laser reads the color of each fragment as it passes by.

While Sanger sequencing is highly accurate and yields long reads (up to 900 base pairs), it has a catastrophic bottleneck: it can only sequence one DNA molecule (or one PCR product) per capillary run. It is targeted, slow, and expensive.

The Next-Generation Sequencing (NGS) revolution solved this bottleneck by replacing single-capillary runs with **Massive Parallelization**. Instead of reading one long fragment at a time, NGS instruments sequence billions of short DNA fragments simultaneously on a single glass surface (a flow cell). This dropped the cost of sequencing a million bases

by more than 100,000-fold, transitioning genomics from a boutique academic luxury to a central pillar of clinical medicine.

Why NGS Dominates the Field: Clinical and Basic Applications

The high throughput of NGS makes it the default tool for both discovery research and medical diagnostics:

- **Pathogen Tracking (Genomic Epidemiology):** During the COVID-19 pandemic, researchers worldwide used NGS to sequence SARS-CoV-2 genomes directly from patient samples. By comparing mutational changes across thousands of viral genomes, systems like Nextstrain mapped the evolution and spread of variants (like Omicron or Delta) in real time.
- **Non-Invasive Prenatal Testing (NIPT):** Clinicians sequence cell-free DNA (cfDNA) circulating in the mother's blood. By counting the relative proportion of reads mapping to Chromosome 21 versus other chromosomes, software can identify fetal trisomies (Down syndrome) without invasive amniocentesis.
- **Oncology Liquid Biopsies:** Sequencing circulating tumor DNA (ctDNA) in a patient's blood to detect rare somatic mutations, enabling early cancer detection and real-time treatment monitoring.

Illumina Chemistry: Sequencing-by-Synthesis Under the Hood

The dominant technology in the NGS market is Illumina's **Sequencing-by-Synthesis (SBS)**.

To understand what you are measuring, you must understand the four steps of the Illumina protocol:

1. **Library Preparation:** DNA is sheared into short fragments (~200–500 bp). Synthetic adapter sequences are ligated (attached) to both ends of each fragment. These adapters contain **barcodes (indices)** that allow you to mix multiple samples together in the same run, along with sequencing primer binding sites.
2. **Cluster Generation (Bridge Amplification):** The DNA library is loaded onto a glass **flow cell** covered in short oligonucleotide lawns. The adapters bind to these oligos. Through a series of isothermal PCR cycles, each single DNA molecule is amplified into a dense physical cluster of ~1,000 identical clonal copies. This clustering is necessary because a single molecule does not emit enough light to be detected by the camera.
3. **Reversible Terminator Incorporation:** The instrument floods the flow cell with DNA polymerase and four modified nucleotides. Each nucleotide has a unique fluorescent dye (red, green, blue, or yellow) and a **reversible blocking group** that prevents the polymerase from adding more than one base at a time.
4. **Imaging and Cleavage:** The unbound nucleotides are washed away. A laser excites the fluorescent dyes, and a high-resolution camera takes a picture of the flow cell. The color of the light emitted by each cluster tells the instrument which base

was added. Finally, the blocking group and dye are chemically cleaved (removed), allowing the next nucleotide to bind in the next cycle.

Alternatives and Technology Evolution: Long-Read sequencing

While Illumina's short-read chemistry is highly accurate and cheap, short reads (~150 bp) cannot easily span large structural variants, repeat regions, or highly homologous genes. This bottleneck drove the evolution of long-read sequencing technologies:

- **Pacific Biosciences (PacBio) SMRT Sequencing:** This chemistry uses **Zero-Mode Waveguides (ZMWs)**—billions of tiny metallic wells where a single DNA polymerase is anchored at the bottom. As fluorescently labeled bases are incorporated in real time, the camera records the light pulse. By attaching circular adapters to the DNA (Circular Consensus Sequencing or CCS), the polymerase circles the template multiple times, allowing the random errors to be averaged out, yielding highly accurate (Q30) long reads (>10 kb).
- **Oxford Nanopore Technologies (ONT) Sequencing:** Instead of using light, ONT measures **ionic current**. A motor protein feeds a single strand of DNA through a protein pore embedded in an electrically charged membrane. Each combination of bases (k-mers) blocking the pore produces a distinct drop in the electrical current. A neural network basecaller decodes these electrical fluctuations directly into sequence strings. ONT can sequence molecules of arbitrary length (up to megabases), but it is prone to systematic errors in homopolymer regions (e.g., distinguishing a run of 6 Adenines from 7 Adenines).

From Raw Camera Signals to FASTQ

The direct output of an Illumina sequencer is a directory of binary base call files called **BCL files**. BCL files record the fluorescent color detected at each cluster coordinate for every cycle of the run.

To convert these raw coordinate signals into a usable format, you must run a tool like `bcl2fastq` or `bcl-convert`. This tool performs two critical steps:

1. **Base Calling:** Converts the color signals into DNA letters (A, C, G, T) and assigns a quality score to each call.
2. **Demultiplexing:** Reads the short index barcode sequences on the adapters and assigns each read to its corresponding sample file.

The output of this process is a set of raw **FASTQ files**.

Decoding the FASTQ Specification

A FASTQ file contains four lines per read:

```
@SRR1234567.1 1/1
GATCGATCGATCGATCGATCGATCGATCGATCGATC
+
F:FFA9FEE,EE.E/FFEF<F:FF/FF:F8FFF9FFDFD/
```

- **Line 1 (The Header):** Always starts with @ and contains the read name, sequencing instrument ID, flow cell coordinates, and sample index/lane details.
- **Line 2 (The Sequence):** The raw base calls (A, C, G, T, or N for undetermined bases).
- **Line 3 (The Spacer):** Always starts with + and is sometimes followed by a repetition of the header.
- **Line 4 (The Quality Scores):** A string of ASCII characters matching the length of the sequence on Line 2. Each character represents a Phred quality score.

Understanding Quality Scores (The Phred Scale)

The quality string encodes the probability (P_e) that a base call is incorrect. The **Phred score (Q)** is calculated using a logarithmic scale:

$$Q = -10 \log_{10}(P_e)$$

A higher Phred score means a lower error probability. To represent these values as a single character in text files, the Phred score is offset by 33 (or 64 in older pipelines) and mapped to the standard ASCII character table (Phred+33).

For example, a Phred score of 30 ($Q = 30$) has an ASCII value of $30 + 33 = 63$, which corresponds to the character `?`.

Character	Phred Score (Q)	Error Probability (P_e)	Accuracy	Meaning in Practice
*	9	12.5%	87.5%	Highly unreliable
5	20	1.0%	99.0%	Standard minimum
?	30	0.1%	99.9%	High-quality call
I	40	0.01%	99.99%	Excellent accuracy

Clues for Quantitative Analysis: The Relative Nature of Counts

Before we align these reads, you must internalize two biological and mathematical realities of sequencing data. We will rely heavily on these clues in later chapters:

1. **Reads are Discrete Count Data:** A FASTQ file contains individual, integer counts of reads. You cannot observe a fraction of a read. Therefore, continuous normal statistics are fundamentally inappropriate for raw sequence data (Chapter 4).
2. **Sequencing yields Relative (Compositional) Quantifications:** The sequencer has a fixed capacity (e.g., 300 million reads per lane). If Sample A has twice as many reads as Sample B, it does not mean Sample A had a higher starting concentration of molecules in the tube. It only means Sample A took up a larger *fraction* of the sequencer's total output. Read counts are relative measurements, and they must

be normalized for sequencing depth and library composition before any biological comparison can be made.

WGS vs. WES: The Physical Mechanics of Target Enrichment

Whole Genome Sequencing (WGS) and Whole Exome Sequencing (WES) are not just different computational analysis pathways; they represent entirely distinct physical library preparations with different biochemical constraints.

- **The Physical Bias of Capture:** WGS sequences the entire genome, typically using a PCR-free library preparation that minimizes GC-bias and PCR amplification artifacts. WES, by contrast, relies on a hybridization capture step. After DNA fragmentation, biotinylated oligonucleotide probes (baits) designed to target the exonic regions (representing ~1.5% of the human genome) are hybridized to the library. These probe-bound fragments are then physically pulled down using streptavidin-coated magnetic beads.
 - **Targeted Efficiency vs. Genomic Fairness:** WES is an “efficiency-first” approach based on the empirical observation that ~85% of known disease-causing mutations reside in coding exons. However, this hybrid-capture process introduces severe physical limitations:
 1. **GC Bias and Capture Efficiency:** Hybridization is highly sensitive to melting temperature (T_m), resulting in poor capture efficiency in extremely GC-rich or AT-rich exons, causing “cold spots” where sequencing coverage is zero despite the region being coding.
 2. **Regulatory and Intronic Blind Spots:** WES is completely blind to deep intronic mutations that cause aberrant splicing, and to regulatory mutations in enhancers, promoters, and non-coding RNAs.
 3. **Evenness of Coverage:** WGS provides “fair” and relatively even coverage across the genome. WES coverage is highly uneven, with peak-and-valley distributions driven by probe binding affinities. Thus, while WES is cost-effective for identifying coding mutations, WGS is far superior for structural variant detection, copy number analysis, and identifying novel non-coding regulatory variants.
-

Biological Interpretation

When auditing a raw dataset, you must look beyond the raw sequence and examine the quality score patterns.

Positional Quality Decay (The 3' Drop)

If you plot the average Phred scores across the length of Illumina reads, you will notice a systematic decay: quality scores are highest at the 5' beginning of the read and drop at the 3' end.

This drop is driven by the physical chemistry of the flow cell. Over 150 cycles of synthesis, two technical issues accumulate:

- **Phasing:** A small percentage of clonal copy molecules fail to incorporate a base during a cycle, falling behind.
- **Pre-phasing:** Some molecules incorporate two bases, skipping ahead.

As the molecules in a cluster lose synchrony, the fluorescent signal becomes mixed (blurry). The instrument can no longer confidently distinguish which base is being added, and the Phred score drops. If your quality scores drop below Q20 at the 3' end, you may need to trim the reads before alignment to prevent false-positive mapping.

Summary and Required Reading

1. **Sanger is targeted; NGS is parallel:** Massive parallelization dropped sequencing costs and made genome-wide assays clinical realities.
2. **Illumina sequences by synthesis:** Reversible terminators incorporate one fluorescent base per cycle; coordinate images are translated from BCL files to FASTQ.
3. **Phred scores are logarithmic:** $Q = -10 \log_{10}(P_e)$. A Q30 score represents a 99.9% base call accuracy.
4. **Counts are relative:** Read counts reflect the proportion of a molecule in the sequencing library, not its absolute biological concentration.

Required Reading

- Cock et al.: "The Sanger FASTQ file format for sequences with quality scores" (Nucleic Acids Research, 2010). A historical and technical breakdown of Phred encoding.

Johnson's Rule: A FASTQ file is the raw signal of a physical chemistry reaction. If your quality score plots show sudden drops in the middle of a run, the chemistry failed on the flow cell. Never trust an aligner to correct for a failing instrument.

Chapter 8: Alignment Mechanics and Mapping Formats (SAM/BAM)

Mapping Billions of Reads Perfectly is Computationally Impossible

If you attempt to align a single 150 bp sequencing read to the 3.2 billion bp human genome using exact mathematical models, you will compute billions of alignment combinations. If you multiply this by 300 million reads from a single sequencing lane, your pipeline will run for decades. Bioinformatics is the science of engineered heuristics—sacrificing mathematical perfection for speed. Every aligner is wrong; some are useful.

Core Concepts

Once your sequencer outputs raw FASTQ files, your next task is to determine where each read originated in the reference genome. This step is called **sequence alignment (or mapping)**.

To choose the right aligner and understand its output, you must understand the mathematical limits of alignment, the heuristics used to bypass them, and the resulting file formats.

The Computational Wall: Dynamic Programming

The mathematically “perfect” way to align two sequences is through **Dynamic Programming (DP)**, specifically the **Needleman-Wunsch** (for global alignment) and **Smith-Waterman** (for local alignment) algorithms. These algorithms build a massive scoring matrix where every cell represents a comparison between a base in the read and a base in the genome.

For a read of length L and a genome of length G , the complexity of dynamic programming is:

$$\text{Time Complexity} = O(L \times G)$$

For a single 150 bp read against the 3.2 billion bp human genome, the matrix contains $150 \times 3.2 \times 10^9 = 480$ billion cells. To calculate this for millions of reads is computationally impossible. Therefore, aligners do not search the genome using dynamic programming from scratch.

The Heuristic Escape: Seed-and-Extend

To bypass this computational wall, modern aligners use the **Seed-and-Extend** heuristic:

1. **Seeding:** The aligner identifies short, exact matches (e.g., 20 bp) between the read and the genome. These exact matches are called “seeds”. To do this quickly, the aligner uses a pre-built index of the reference genome.
2. **Extension:** Once the aligner finds a seed location, it halts the genome-wide search and performs local dynamic programming (Smith-Waterman) in the immediate flanking region around the seed to align the rest of the read, allowing for mismatches and gaps (indels).

Classifying Aligner Methods: Choosing the Right Tool

Different biological assays require different alignment behaviors. Aligners are classified into three primary categories based on the prior assumptions they make:

1. DNA Aligners (BWA-MEM & Bowtie2)

- **The Assumption:** Reads align contiguously to the genome with few gaps or mismatches.
- **The Technology:** These tools index the genome using the **Burrows-Wheeler Transform (BWT)** and the **FM-index**, which compresses the 3 Gb human genome into a ~2 GB index that fits in RAM, allowing exact seed searches in linear time.
- **Usage:** BWA-MEM is the gold standard for variant calling (WGS/WES); Bowtie2 is preferred for short reads with minimal gaps (e.g., ChIP-seq or metagenomic profiling).
- **The BWA-MEM Parameter Choices:** In production genomic pipelines, running BWA-MEM with default parameters is a rookie mistake. I recommend using two crucial flags:
 1. **-M:** This flag marks shorter split hits as secondary (rather than supplementary). This is a critical prerequisite for downstream duplicate marking tools (like Picard MarkDuplicates or GATK). If you omit the **-M** flag, these tools will fail to recognize and eliminate PCR duplicates from split-aligned reads, artificially inflating your depth and causing false somatic variant calls.
 2. **-K <integer>:** In multi-threaded execution, BWA-MEM processes reads in chunks. By default, the chunk size changes dynamically based on thread scheduling, leading to slightly different alignment outputs depending on the thread count (non-deterministic runs). Specifying a fixed chunk size (e.g., **-K 1000000000**) forces BWA-MEM to process identical chunks regardless of threading, guaranteeing 100% computational reproducibility across runs.

2. RNA-Seq Splice-Aware Aligners (STAR)

- **The Assumption:** Reads originate from spliced messenger RNA transcripts. Because splicing removes introns, a single read can cross an exon-exon junction,

meaning one half of the read maps to Exon 1, and the other half maps to Exon 2 (separated by thousands of base pairs in DNA space).

- **The Technology:** STAR identifies **Maximal Mappable Seeds** on either side of the junction and aligns them separately, bridging the intron gap and looking for canonical splice site signatures (e.g., GT-AG).
- **Usage:** Standard transcriptome profiling (RNA-seq).

3. Long-Read Aligners (Minimap2)

- **The Assumption:** Reads are long (10 kb to 1 Mb) but have high error rates (1% to 5%).
- **The Technology:** Minimap2 uses **minimizers**—a subset of representative k-mers—to index the reference and seed alignments. This indexing strategy is highly tolerant of mismatches and gaps, allowing noisy long reads to find their correct genomic coordinates.
- **Usage:** Long-read DNA/RNA alignment (PacBio/Oxford Nanopore).

The Mapping Output: SAM/BAM Format

The output of an aligner is a **SAM (Sequence Alignment/Map) file** (or its binary, compressed version, the **BAM file**). A BAM file is the foundational input for almost all downstream assays (ChIP-seq peak calling, RNA-seq quantification, and Variant Calling).

The Anatomy of a BAM File

A BAM file consists of a header section followed by tab-delimited alignment records.

The Header

Always starts with @ and contains crucial metadata:

- @HD: Format version and sorting order (e.g., coordinate sorted).
- @SQ: Reference sequence dictionary (chromosome names and lengths).
- @RG: Read Groups, mapping which reads belong to which biological sample (essential for batch identification).

The Alignment Records

Every read has a record with 11 mandatory fields:

SRR1234567.1	83	chr1	10045	30	150M	*	0	0	GATC ...
--------------	----	------	-------	----	------	---	---	---	----------

1. **QNAME (Query Name):** The unique name of the read from the FASTQ header.

2. **FLAG (Bitwise Flag):** A single integer that encodes multiple mapping properties (see below).
3. **RNAME (Reference Name):** The chromosome name where the read aligned.
4. **POS (Leftmost Position):** The 1-based start coordinate of the alignment.
5. **MAPQ (Mapping Quality):** The probability that the read is placed at the wrong locus (see below).
6. **CIGAR (CIGAR String):** A string summarizing the match, mismatch, insertion, and deletion operations (e.g., 150M = 150 matches; 30M2D120M = 30 matches, 2 deletions in reference, 120 matches).
7. **RNEXT:** Name of the reference for the mate read (if paired-end).
8. **PNEXT:** Leftmost position of the mate read.
9. **TLEN (Template Length):** Estimated size of the sequenced DNA fragment.
10. **SEQ:** The read sequence.
11. **QUAL:** The quality string.

Understanding MAPQ (Mapping Quality)

The MAPQ score represents the probability (P_m) that the aligner has placed the read at the wrong location:

$$\text{MAPQ} = -10 \log_{10}(P_m)$$

- **MAPQ 30:** A 1-in-1,000 chance that the read is mismapped (99.9% accuracy).
- **MAPQ 0:** The read maps equally well to multiple locations in the genome (e.g., repetitive regions). The aligner picked one position at random.

[!WARNING] **Filter MAPQ 0 in Variant Calling:** If you do not filter out MAPQ 0 reads before calling variants, your pipeline will identify thousands of false mutations that are actually just misplaced reads from repeat regions. I recommend filtering for $\text{MAPQ} \geq 20$ or $\text{MAPQ} \geq 30$ in germline pipelines.

Decoding the Bitwise FLAG

The FLAG field is a bitwise combination of 12 distinct binary flags. Each flag represents a boolean question. Rather than writing 12 columns, the aligner sums the binary values of the active flags:

Bit	Decimal Value	Meaning
1	1	Read is paired-end
2	2	Read mapped in a proper pair (correct distance and orientation)
3	4	Read is unmapped

Bit	Decimal Value	Meaning
5	16	Read mapped to the reverse strand
7	64	This is Read 1 of the pair
8	128	This is Read 2 of the pair

For example, a FLAG value of **83** represents:

$$83 = 1 + 2 + 16 + 64$$

This means: the read is paired (1), mapped in a proper pair (2), mapped to the reverse strand (16), and is Read 1 of the pair (64). You do not need to calculate this manually; tools like `samtools` use these flags to filter reads (e.g., `samtools view -f 2` extracts only properly paired reads).

Biological Interpretation

When analyzing BAM files, you must be alert to **alignment bias**—the tendency of aligners to preferentially map reads that match the reference genome perfectly over reads that carry non-reference variants.

Soft-Clipping vs. Hard-Clipping

When the end of a read contains sequence that does not match the reference (e.g., unremoved adapters or highly mutated regions), the aligner will clip the ends.

- **Soft-Clipping (S in CIGAR):** The aligner keeps the clipped bases in the BAM record but ignores them during alignment scoring (e.g., `10S140M`).
- **Hard-Clipping (H in CIGAR):** The aligner removes the clipped bases from the BAM record entirely to save space.

If you see high rates of soft-clipping in your alignments, it is a diagnostic clue: either your adapter trimming step failed, or your sample has a structural insertion that differs significantly from the reference genome.

Summary and Required Reading

1. **Genome-wide exact alignment is too slow:** Aligners use indexing and seed-and-extend heuristics to achieve speed.
2. **Select the correct aligner:** Use DNA aligners for contiguous mapping, STAR for RNA splicing, and Minimap2 for long reads.

3. **MAPQ measures mapping confidence:** MAPQ 0 means the read aligns to multiple repeat copies in the genome.
4. **BAM is the universal format:** A binary alignment format representing the raw input for all variant calling, peak calling, and transcriptomics pipelines.

Required Reading

- Li et al.: "The Sequence Alignment/Map format and SAMtools" (Bioinformatics, 2009). The original specification describing the design of SAM/BAM files.

Johnson's Rule: A BAM file is the foundation of modern sequencing analysis. If your MAPQ scores are low, your reads are mapping ambiguously. Do not proceed to variant or peak calling until you audit your mapping quality.

Chapter 9: De Novo Assembly and Graph Algorithms

Assembly is Not a Solved Problem; It is an Approximate Reconstruction

Reconstructing a genome from fragments is the Shortest Common Superstring Problem, finding the shortest string that contains every read as a substring. This problem is NP-hard. When reads are shorter than repetitive elements, there is no unique mathematical solution regardless of algorithmic sophistication. Assembly is always an approximate reconstruction, and its quality is bounded by the relationship between read length and repeat length.

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Genomic DNA extracted from a target organism or community of organisms.
- **Molecular Level:** The chromosome and physical nucleotide sequence level. De novo assembly seeks to reconstruct the linear, contiguous sequence of the target genome from highly fragmented raw reads without using a reference template.

The Repeat Trap

The human genome is over 50% repetitive: Alu elements (~300 bp, ~1M copies), LINE1 (~6 kb, ~500K copies), and segmental duplications (10-500 kb). If a read is shorter than a repeat, the assembler cannot determine whether it came from copy A or copy B. Consider the genome A—REPEAT—B and C—REPEAT—D. Reads from the repeat are identical regardless of origin; the assembly graph branches into two diverging paths with no information to resolve which is correct:

Repeat length > read length $\implies$ assembly graph breaks at repeat boundaries

This is an information-theoretic limit, not an algorithmic weakness. No algorithmic cleverness can recover the correct path across a repeat when the reads are shorter than the

repeat unit. The only solution is longer reads or mate-pair information that bridges the repeat.

From Hamiltonian to Eulerian: Why k-mers Change the Problem

Two graph formulations dominate assembly, and the difference between them explains the revolution in algorithmic efficiency.

Overlap-Layout-Consensus (OLC) models each read as a node, with edges connecting reads that overlap by a minimum number of bases. The genome is the **Hamiltonian path**: visiting each node exactly once, which is NP-complete ($O(N!)$ in the number of reads). For 10 billion short reads, this is computationally impossible. OLC was viable only for Sanger-era projects with thousands of reads.

De Bruijn graphs transform the problem by fragmenting all reads into k-mers (substrings of length k) and using k-mers as nodes. This changes the complexity fundamentally: the node vocabulary has a fixed upper bound (4^k possible k-mers), regardless of how many reads are sequenced. The genome is now an **Eulerian path**: traversing each edge exactly once; solvable in $O(E)$ time.

Concrete example. Genome = GATTACA with $k=3$. The k-mers are GAT, ATT, TTA, TAC, ACA. The De Bruijn graph represents each $(k-1)$ -mer as a node and each k-mer as a directed edge:

```
Nodes (2-mers): GA  AT  TT  TA  AC  CA
Edges (3-mers): GA→AT  AT→TT  TT→TA  TA→AC  AC→CA
```

The Eulerian path is $GA \rightarrow AT \rightarrow TT \rightarrow TA \rightarrow AC \rightarrow CA$. Reconstructing the sequence: start node GA (the first $k-1$ bases), then append the last character of each edge in path order: G + A + T + T + A + C + A = GATTACA. The genome is recovered.

Eulerian path conditions. For a graph to contain an Eulerian path, node degrees must satisfy: one node has out-degree = in-degree + 1 (start), one node has in-degree = out-degree + 1 (end), and all other nodes are balanced. In the GATTACA graph, GA has out=1, in=0 (start) and CA has in=1, out=0 (end). These conditions are checked in $O(E)$ time.

The NP-hard Hamiltonian problem becomes tractable here because k-mers decouple the problem from read count. OLC has as many nodes as reads (10 billion), and checking all orderings is $O(N!)$. De Bruijn has at most 4^k possible nodes (theoretically $4^{31} \approx 4.6 \times 10^{18}$, but typically millions in practice), and Eulerian path is $O(E)$. The tradeoff is memory: a human genome de Bruijn graph with $k=31$ stores billions of k-mer occurrences and requires 100-300 GB of RAM.

```
# De Bruijn graph from k-mers
def build_debruijn(reads, k=31):
    graph = {}
    for read in reads:
        for i in range(len(read) - k + 1):
            kmer = read[i:i+k]
            left = kmer[:-1]
            right = kmer[1:]
```

```
graph.setdefault(left, set()).add(right)
return graph
```

The k-mer tradeoff. The choice of k determines how the graph balances connectivity against repeat resolution. Small k (21) means each k-mer is shorter and more likely to be shared between reads, creating a more connected graph. But shorter k-mers are also more likely to fall within repeats, collapsing repeat copies into the same node. Large k (63) distinguishes repeat copies more effectively because the k-mer extends beyond the repeat unit, but the stricter match requirement produces a more fragmented graph in low-coverage regions. Many assemblers address this by using multiple k values (SPAdes uses a k-mer sweep from 21 to 127) or by building a multi-sized de Bruijn graph.

Graph Cleaning and Resolution

A raw de Bruijn graph contains artifacts from sequencing errors and biological variation:

- **Tips:** Dead-end paths created by sequencing errors at read ends. A single erroneous base at the end of a read creates a k-mer that does not connect to the rest of the graph. Tips are removed if they are shorter than $2k$ and have no supporting coverage.
- **Bubbles:** Alternative paths caused by heterozygous variants, alternative splicing, or sequencing errors. Two paths diverge and reconverge within a short distance. Bubbles are resolved by keeping the higher-coverage path and removing the lower.
- **Bulges:** Compressed bubbles; short alternative paths caused by a single nucleotide variant. Like bubbles but shorter; removed by the same heuristic.

After cleaning, the graph is resolved into **contigs** by finding unambiguous paths; nodes with exactly one incoming and one outgoing edge. Branch points (nodes with multiple in/out edges) indicate repeats or unresolved variation and terminate contigs.

The density of artifacts in the raw graph is a direct diagnostic of sequencing error rate. A high-quality dataset (HiFi, PCR-free Illumina) produces a clean graph with few tips; a noisy dataset produces a dense artifact graph that requires aggressive cleaning, which in turn reduces contiguity by removing true low-coverage regions along with errors. The ratio of tips removed to contigs produced is a useful quality metric for the input data.

Long-Read Assembly

Long reads (PacBio HiFi 10-25 kb, ONT 10-100 kb) solve the repeat problem that short reads cannot. A read spanning an entire Alu element (~300 bp) or LINE1 element (~6 kb) determines unambiguously which flanking sequence it connects to. Modern long-read assemblers have largely replaced short-read assembly for eukaryotic genomes.

- **Hifiasm** (HiFi-only): Produces haplotype-resolved assemblies by using Hi-C or trio information to phase variants. The standard for human genome assembly; produces fully phased assemblies in hours.

- **Flye** (ONT + HiFi): Uses a repeat graph construction that handles the high error rates of raw ONT reads. Preferred for metagenomic assembly and bacterial genomes where coverage depth varies.
- **Canu** (legacy PacBio CLR): OLC-based with error correction before assembly. Largely superseded by Hifiasm and Flye for most applications.

Validation Metrics

Metric	What It Measures	Caveat
N50	Contiguity: 50% of assembly in contigs of this size or larger	Does not measure accuracy
L50	Number of contigs to reach N50	High L50 = fragmented
BUSCO	Presence of conserved single-copy orthologs	Measures gene content, not genome-wide correctness
QV / Assembly quality	Phred-scaled consensus error rate (via merqury)	Q40 = 99.99% accurate at consensus level

N50 is the most widely reported, and most widely misused; assembly metric. High N50 does not mean high quality. A contiguous but systematically wrong assembly (mis-joined repeats, collapsed segmental duplications) can achieve high N50 while having low biological accuracy. BUSCO scores measure conserved gene content, not structural correctness. QV estimation via k-mer comparison (merqury) provides the most honest accuracy assessment by comparing k-mer frequencies between the assembly and raw reads.

Genome size estimation. Before assembly, the k-mer frequency spectrum (histogram of k-mer occurrence counts) provides a rapid genome size estimate: $G = \frac{\text{total distinct k-mers}}{\text{peak k-mer coverage}}$, where the peak in the distribution corresponds to the haploid sequencing depth. If the estimated size deviates substantially from the expected genome size, the data may be contaminated, have unexpected ploidy, or suffer from systematic sequencing bias, catching this before assembly saves hours of failed computation.

Metagenomic Binning and Assembly Trade-offs

For metagenomic data, the assembly produces a mixture of contigs from multiple organisms. This introduces a major upstream decision: **single-sample assembly vs. co-assembly**.

- **Single-Sample Assembly:** Each sample is assembled independently. This reduces graph complexity and minimizes the risk of generating chimeric contigs (mis-joining homologous regions from distinct bacterial strains). However, it lacks the power to reconstruct genomes of low-abundance taxa.

- **Co-assembly:** Pooling reads from multiple samples to boost sequencing depth. While co-assembly enables the reconstruction of rare genomes, it introduces extreme computational graph complexity and frequently mis-assembles homologous regions across closely related strains into false, non-existent hybrid contigs. I suggest starting with single-sample assembly and moving to co-assembly only for highly correlated sample cohorts (e.g., longitudinal samples from the same individual).

Once contigs are assembled, **binning** groups contigs into Metagenome-Assembled Genomes (MAGs) using two orthogonal signals:

- **Composition:** Tetranucleotide frequency (TNF) profiles; each genome has a characteristic oligonucleotide signature that distinguishes it from other genomes.
- **Coverage:** Differential abundance across samples; contigs from the same genome have correlated coverage profiles across multiple samples or conditions.

Tools: CONCOCT, MetaBAT2, MaxBin2.

Validation and Quality Control: We assess MAG quality using CheckM or CheckM2, which scans for lineage-specific single-copy marker genes. A publication-grade, high-quality draft MAG requires: 1. **Completeness > 90%** (having at least 90% of the lineage-specific single-copy markers present). 2. **Contamination < 5%** (having less than 5% of the marker genes duplicated, which would indicate contamination from other bins).

A MAG is always a population consensus, not a single clonal genome; it represents the average genome of all cells assigned to that taxonomic bin, and within-bin strain diversity is masked.

Biological Interpretation

A genome assembly is a statistical inference, not a ground truth. The N50 metric is widely misused; a high N50 can mask systematic misassembly (collapsed duplications, mis-joined repeats). BUSCO scores measure conserved gene content, not genome-wide structural correctness. A 95% BUSCO score means 5% of conserved genes are missing or fragmented, but it says nothing about whether the remaining 95% are correctly assembled.

Short-read-only assemblies of eukaryotic genomes are fundamentally incomplete. The telomere-to-telomere (T2T) human genome required PacBio HiFi (10-25 kb CCS reads) + ultra-long ONT (>100 kb reads) + manual curation with optical maps. Any eukaryotic assembly derived from short reads alone should be treated as a draft.

For metagenomic binning, completeness and contamination are competing objectives. Co-assembling multiple samples increases assembly depth for rare organisms but risks generating chimeric genomes by mis-assembling homologous regions from distinct bacterial strains; single-sample assembly is safer but misses low-abundance taxa. Validation via CheckM is required to prove a MAG has sufficient marker gene coverage (>90% completeness, <5% contamination) before making downstream ecological inferences.

Summary and Required Reading

1. **Assembly is NP-hard with short reads:** repeats longer than the read length create information-theoretic ambiguities that no algorithm can resolve.
2. **De Bruijn graphs (k-mers)** transform the Hamiltonian (NP-complete) path problem into an Eulerian ($O(E)$) path problem, enabling genome-scale assembly.
3. **k-mer choice is the critical parameter:** small k maximizes connectivity, large k resolves repeats. Multiple k-mer assemblies address both needs.
4. **Long reads bridge repeats:** Hifiasm (HiFi) and Flye (ONT) produce near-complete eukaryotic assemblies. Short-read-only assemblies are drafts.
5. **N50 is a contiguity metric, not an accuracy metric:** BUSCO for gene content, merquy QV for consensus accuracy.
6. **Metagenomic binning** groups contigs by composition and coverage into MAGs, which are population consensus, not individual genomes.

Required Reading

- Compeau, Pevzner & Tesler: "How to apply de Bruijn graphs to genome assembly" (*Nature Biotechnology*, 2011).
- Bankevich et al.: "SPAdes: a new genome assembly algorithm" (*Journal of Computational Biology*, 2012).

Johnson's Rule: Assembly is not a solved problem. It is always an approximate reconstruction.

Chapter 10: Variant Calling and Hidden Markov Models

A Variant is a Probability, Not a Fact

At a position with 25 reference reads and 5 alternate reads, is that “G” a true mutation or 5 sequencing errors? You cannot trust your eyes, and you cannot simply count reads. You must mathematically prove the variant by combining the known error rate of the machine with the evolutionary probability of the genome. A variant call is a posterior probability; the product of the data likelihood (derived from quality scores) and a prior (the population frequency of the variant type).

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Genomic DNA (germline DNA from normal tissue, or somatic DNA from tumor biopsies).
- **Molecular Level:** The single nucleotide and insertion/deletion allele levels at individual base coordinates. Variant calling identifies structural and base changes relative to a reference genome.

Sequencing Depth vs. Coverage Breadth: The Poisson Reality

In sequencing projects, researchers often demand “30x WGS” or “100x WES” as if they are uniform guarantees. I suggest you abandon this naive assumption. Sequencing reads do not fall like tiles on a floor; they distribute across the genome randomly.

We model this using a Poisson process. The probability of a genomic coordinate being covered by exactly k reads, given an average sequencing depth of λ , is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Thus, at an average depth of 30x ($\lambda = 30$), the probability of a base having zero coverage (0x) is $e^{-30} \approx 9.3 \times 10^{-14}$, which is negligible. However, if the average depth is

4x, the probability of zero coverage is $e^{-4} \approx 1.83\%$, meaning over 50 million bases of the human genome will be completely unobserved.

Moreover, the Poisson model assumes “fairness” in coverage, which is physically false. GC-rich regions, repetitive sequences, and chromatin status alter DNA shearing and PCR efficiency, creating severe deviation from Poisson expectations. In practice, at 30x average depth, substantial portions of the genome remain covered at $< 10x$ (under-sequenced, leading to false-negative heterozygous calls), while others are covered at $> 100x$.

Bayesian Genotyping

All modern variant callers use Bayes’ theorem to compute the probability of each possible genotype G given the observed sequencing data D :

$$P(G | D) = \frac{P(D | G) \cdot P(G)}{P(D)}$$

The **prior** $P(G)$ encodes population genetics expectations. In humans, the probability that a given position is heterozygous is approximately 10^{-3} ; most positions match the reference. This prior prevents callers from being overly influenced by rare sequencing errors.

The **likelihood** $P(D | G)$ is derived from the Phred quality scores of the reads covering the position. For a single read with base call b and quality score Q :

$$P(\text{correct base}) = 1 - 10^{-Q/10}$$

For 5 alternate reads out of 30 total, the likelihood ratio comparing the heterozygous genotype (AG) to the homozygous reference (AA) is:

$$LR = \frac{P(5 \text{ alt}, 25 \text{ ref} | \text{het})}{P(5 \text{ alt}, 25 \text{ ref} | \text{hom ref})}$$

If the reads are Q30 (0.1% error rate each), the probability of observing 5 alternate reads under the homozygous reference is a binomial: $\binom{30}{5}(0.001)^5(0.999)^{25} \approx 8 \times 10^{-12}$. The heterozygous model assigns probability $\binom{30}{5}(0.5)^5(0.5)^{25} \approx 4 \times 10^{-4}$. The likelihood ratio is $4 \times 10^{-4} / 8 \times 10^{-12} \approx 5 \times 10^7$ (overwhelming evidence for the heterozygous call. If the same reads are Q10 (10% error rate), the homozygous reference probability becomes $\binom{30}{5}(0.1)^5(0.9)^{25} \approx 0.02$, and the LR drops to $4 \times 10^{-4} / 0.02 \approx 0.02$ evidence for the **reference** call. **The quality scores are what convert a raw read count into a statistically meaningful genotype call.**

Base quality score recalibration (BQSR). Quality scores from the sequencer are systematically biased; they are calibrated globally, not per-base-context. A C followed by

G in a CG dinucleotide context has a different empirical error rate than an A in an AT-rich context, even if both were assigned Q30 by the sequencer. BQSR corrects this by tabulating empirical error rates stratified by: reported quality score, base context (the surrounding 2-3 bases), and read group (flow cell lane). Using known variant sites (dbSNP) to distinguish true variation from sequencing artifacts, BQSR builds a recalibration table and adjusts each base's quality score so that Q30 empirically means 1/1000 error rate in every context. **The critical constraint:** BQSR requires a set of known variant sites. For non-human species where dbSNP-equivalent resources do not exist, BQSR should be skipped; applying it without known variants will recalibrate true variation as error and degrade sensitivity.

PCR Duplicates and Mark Duplicates

During library preparation, DNA fragments are PCR-amplified. If a single template fragment is amplified multiple times and those identical copies bind to the flow cell, they are sequenced as independent reads. These are **PCR duplicates**. * **Identification Logic:** If two reads map to the exact same start coordinate, the exact same end coordinate, and in the same orientation, they are flagged as duplicates. GATK or Picard MarkDuplicates retains only the read pair with the highest sum of base quality scores and marks the others. * **WGS vs. WES Duplication Rates:** WGS libraries typically use PCR-free protocols, resulting in duplication rates of $< 5\%$. WES, requiring hybridization capture and heavy PCR cycles to enrich exons, routinely has duplication rates of $15 - 30\%$. High duplication rates ($> 40\%$) indicate low initial DNA input or over-amplification. If you do not mark duplicates, your variant caller will mistake a single PCR-amplified sequencing error for a high-confidence homozygous mutation.

Joint Genotyping: The Multi-Sample Consensus Paradigm

In multi-sample studies, running HaplotypeCaller to produce individual VCFs and merging them is a major pipeline design error. It creates the "N+1 problem" (adding one sample requires re-running all samples from scratch) and destroys statistical power.

GATK Best Practices solves this by calling variants per sample in **GVCF mode** (retaining block-level non-variant coverage confidence) and then performing joint genotyping via GenotypeGVCFs.

The core mathematical logic: a marginal variant with only 2 supporting reads in Sample A would be filtered out as noise in single-sample calling. However, if the same variant has 2 supporting reads in Sample B and 1 read in Sample C across a cohort of 100 samples, the joint model pools this weak evidence to confidently call a rare heterozygous variant across the cohort. Joint genotyping converts individual mapping uncertainty into cohort-level statistical power.

Variant Filtering: VQSR vs. Hard Filtering

Once raw variants are called, we must filter out false positives: * **Hard Filtering:** Employs arbitrary thresholds: e.g., QualByDepth (QD) < 2.0 , FisherStrand (FS) > 60.0 , StrandOddsRatio (SOR) > 3.0 , MappingQuality (MQ) < 40.0 . I suggest using hard filtering only

when dealing with small datasets (< 30 exomes or non-model organisms). * **Variant Quality Score Recalibration (VQSR):** VQSR uses a machine learning approach (a Gaussian Mixture Model). It takes known, high-confidence variants (e.g., from HapMap, 1000 Genomes) as true positives and training sets, and fits a multi-dimensional GMM across quality annotations (QD, FS, SOR, MQ, MQRankSum, ReadPosRankSum). It then assigns a probability score (VQSLOD) to each novel variant, allowing you to select a threshold that captures 99% of true variants while filtering out noise. **The constraint:** VQSR requires large sample sizes (> 30 WES samples or WGS) to train the GMM; running VQSR on a small sample set will fail to converge, forcing a fallback to hard filtering.

Why an HMM? Alignment Uncertainty at the Haplotype Level

Bayesian genotyping as described above assumes reads are already correctly aligned. But alignment itself is uncertain, especially around indels, where a read may align in multiple equally plausible ways. A read spanning a 2 bp deletion relative to the reference could be aligned with the deletion in any of several positions depending on the local sequence context.

GATK HaplotypeCaller addresses this through a multi-step workflow that uses a **Hidden Markov Model** to account for alignment uncertainty:

1. **Active region detection:** Find positions where reads disagree with the reference beyond expected error rates. These are regions where variation is plausible.
2. **Local de novo assembly:** Extract all reads in the region and assemble candidate haplotypes using a De Bruijn graph. This generates the most likely alternative sequences given the reads, without assuming the reference is correct.
3. **Pair-HMM alignment:** For each read-haplotype pair, compute the probability that the read originated from that haplotype. This is an HMM because it models the alignment path as a sequence of hidden states (match, insert, delete), marginalizing over all possible paths rather than choosing a single best alignment.
4. **Bayesian genotyping:** For each position and each candidate allele, compute $P(G | D)$ using the pair-HMM likelihoods and the population prior.

The pair-HMM is the computational core. Unlike Smith-Waterman, which finds a single optimal alignment, the pair-HMM sums over all possible alignments:

$$P(\text{read} \mid \text{haplotype}) = \sum_{\text{all alignments}} P(\text{alignment}) \cdot P(\text{read bases} \mid \text{alignment})$$

The HMM has three hidden states at each position: **M** (match: read base aligns to haplotype base), **I** (insert: read has an extra base relative to the haplotype), and **D** (delete: haplotype has a base the read skips). Transitions between these states model the biology of sequencing: M→M is the most probable (continuous alignment), M→I and M→D are less probable (an indel event), and the model penalizes rapid state flipping (I→D or D→I transitions are nearly disallowed). Each state emits a base with a probability derived from the Phred quality score: in state M, the read base should match the haplotype base with high probability; in state I, the read base is “extra” and gets a uni-

form or low probability; in state D, no read base is emitted. The full likelihood is computed by the forward algorithm, which sums over all paths through this state machine in $O(\text{read length} \times \text{haplotype length})$ time.

This marginalization accounts for alignment uncertainty. If a read can align to a haplotype in two equally good ways (e.g., a homopolymer where the deletion could be at any position), Smith-Waterman picks one arbitrarily, while the pair-HMM properly averages over both possibilities. The result is a likelihood that reflects true uncertainty rather than a false sense of precision.

VCF Format and Quality Triaging

The VCF (Variant Call Format) encodes each variant call with fields that allow downstream filtering:

- **QUAL:** Phred-scaled probability that the variant is *not* present ($-10 \log_{10} P(\text{no variant})$)
- **FILTER:** Pass or failed quality thresholds (low QUAL, low depth, strand bias)
- **AD (allelic depth):** Count of reads supporting reference and each alternate allele
- **GQ (genotype quality):** Phred-scaled confidence in the assigned genotype
- **PL (phred-scaled likelihood):** Normalized likelihoods of each possible genotype

A VCF with FILTER=PASS for all entries but inadequate depth or poor quality scores is a VCF that should not be trusted. The FILTER field only checks predefined thresholds; it does not guarantee biological validity. Each of the numeric fields (AD, GQ, PL) should be inspected per variant before accepting a call.

Germline vs. Somatic Calling

Germline calling (GATK HaplotypeCaller, DeepVariant) expects variant allele frequencies of approximately 50% (heterozygous) or 100% (homozygous). The genotype likelihood model and the prior are optimized for these expectations.

Somatic calling (Mutect2, Strelka2) must detect subclonal mutations at any VAF from 1-50%, requiring:

- A **matched normal** sample to filter out germline polymorphisms
- A different prior that accommodates low VAFs
- Cross-sample contamination modeling (subclonal mutations in the normal appear as germline variants)

The distinction is critical: a germline caller applied to tumor data will systematically reject subclonal mutations as noise because they violate the 50% VAF expectation. Conversely, a somatic caller applied to germline data may call heterozygous variants as low-VAF mutations.

Biological Interpretation

A variant call at 10% VAF in a tumor sample may be: a true clonal mutation in a heterogeneous tumor, an artifact from FFPE deamination (C>T transitions), a read misalignment around a homopolymer, or a germline polymorphism in a sample with contamination. The distinction requires inspecting read-level evidence; the BAM, not just the VCF.

The GATK “best practices” workflow is optimized for human germline short-read data. Applying it without modification to non-human species, long-read data, or somatic samples introduces systematic biases. Base quality score recalibration (BQSR) requires known variant sites (dbSNP): these do not exist for most non-human species, so BQSR should be skipped or adapted.

A matched normal is not optional for somatic calling. Population databases (gnomAD) cannot substitute for the patient’s own germline; they exclude rare private germline variants that are the most likely to be mistaken for somatic mutations.

Summary and Required Reading

1. **Bayesian genotyping** combines prior mutation rates with Phred-derived likelihoods. A variant without sufficient evidence is a statistical uncertainty, not a fact.
2. **Pair-HMM alignment** marginalizes over alignment uncertainty; unlike Smith-Waterman, it does not commit to a single optimal alignment path.
3. **Germline callers expect 50/100% VAF**: somatic calling requires a matched normal and different likelihood models.
4. **VCF encodes probability, not truth**: FILTER, AD, GQ, and PL fields must be inspected per variant before trusting any call.

Required Reading

- DePristo et al.: “A framework for variation discovery and genotyping using next-generation DNA sequencing data” (*Nature Genetics*, 2011).
- Poplin et al.: “Scaling accurate genetic variant discovery to tens of thousands of samples” (bioRxiv, 2017).

Johnson’s Rule: The confidence in a variant call is only as strong as the weakest link in the quality chain.

Chapter 11: Transcriptomics and Expectation-Maximization

Quantification is a Mixture Model Estimation Problem

A read that maps to three different transcripts is a single physical molecule whose biological origin is ambiguous. A naive algorithm that counts it three times creates data out of thin air. The Expectation-Maximization (EM) algorithm resolves this ambiguity by distributing each read as a fractional probability across its possible origins.

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Total RNA or messenger RNA (mRNA) transcribed from the genome.
- **Molecular Level:** The transcript and isoform abundance level. Transcriptomics measures the expression levels of individual genes and their spliced isoforms in biological samples.

The “Lab Bench” Metaphor: Translating Wet-Lab to Code

To understand computational transcriptomics, you must translate these digital operations into the physical biochemistry of the wet-lab. Treating them as abstract text manipulation leads to fundamental interpretation errors.

- **Biological Replicates ($n \geq 3$) are your Western Blot Loading Controls:** Just as you cannot draw conclusions from a single Western blot lane without a GAPDH loading control, you cannot infer differential expression without biological replicates to measure population variance. A single sample is a pilot, not a result.
- **Sequence Alignment is a Genomic Hybridization Assay:** The reference genome acts as a multiplexed membrane, and your FASTQ reads are fluorescently labeled probes. The aligner determines where each probe “hybridizes” by sequence complementarity.

- **Expression Quantification is a Gel Band Gray-Scale Scan:** Software like `featureCounts` acts as the scanner, measuring the density (read count) at each genomic coordinate “band”.
- **TPM/FPKM Normalization is Ratio-to-Loading-Control:** It represents the relative fraction of a transcript in the library, which is useful for visualization and comparison of relative expressions.
- **Raw Counts are qPCR Ct Values:** Just as you must run delta-delta-Ct calculations on raw cycles rather than normalized percentages, differential expression models (like DESeq2) must run on raw, un-normalized integer counts. Normalizing before statistical testing breaks the negative binomial variance model and inflates false-positive rates.

rRNA Contamination: The BSA Bottleneck

Ribosomal RNA (rRNA) constitutes over 80 — 90% of total cellular RNA. If you sequence a library without removing rRNA, it acts like a massive BSA protein contamination in a Western blot sample—it saturates the detector and compresses the dynamic range of your target signal (mRNA) to nearly zero, leaving few reads for analysis.

You must run tools like `SortMeRNA` to audit rRNA contamination. High rRNA rates (> 10%) indicate the wet-lab depletion (poly-A selection or ribo-depletion) failed.

Strand-Specificity: Aligning the Primer Direction

Modern RNA-seq libraries preserve the strand orientation of the original RNA molecule (using dUTP marking). If you run `featureCounts` or `Salmon` with the wrong strand parameter (e.g., `-s 1` instead of `-s 2` in `featureCounts`, or `ISR` instead of `ISF` in `Salmon`), the software will align reads to the opposite strand. Because many genomic loci have overlapping genes on opposite strands, you will either get zero counts or mis-attribute reads to antisense transcripts. Always audit your library protocol (e.g., dUTP/stranded-reverse vs. directional) before running quantification.

Splice-Aware Alignment: Why DNA Aligners Fail on RNA-seq

DNA aligners (BWA, Bowtie2) assume contiguous alignment. A read spanning an exon-exon junction requires a deletion of the entire intron; thousands of bases. The aligner assigns MAPQ 0 and discards the read, losing up to 50% of RNA-seq data.

STAR (Dobin et al., 2013) solves this with a suffix array (see Chapter 8) that finds **maximal mappable seeds (MMPs)** from each read. When seed extension fails (the read crosses an intron), STAR starts a new seed. Seeds separated by a genomic gap represent splice junctions, validated against canonical splice site motifs (GT-AG = >99% true junctions). Two-pass mode: discover junctions from all samples, then re-align with the combined junction index.

Expectation-Maximization for Transcript Quantification

Consider a gene with three isoforms sharing exons. A read from a shared exon could originate from any isoform, but we cannot determine which one from the read sequence alone. The EM algorithm resolves this ambiguity iteratively.

The model. For each read r and transcript t , $P(r \mid t) = 1$ if the read maps to the transcript (accounting for fragment length and position), 0 otherwise. This defines the compatibility between reads and transcripts; the only data available to resolve ambiguity.

E-step (Expectation): Given current abundance estimates θ , compute the probability that each read r originated from each transcript t :

$$P(r \in t \mid \theta) = \frac{\theta_t \cdot P(r \mid t)}{\sum_{t'} \theta_{t'} \cdot P(r \mid t')}$$

A uniquely-mapping read gets $P = 1$ for its transcript, 0 for all others. A multi-mapping read is split: θ_t acts as a prior weight; more abundant transcripts receive a larger fraction of the ambiguous read.

M-step (Maximization): Update transcript abundances from the fractional assignments:

$$\theta_t = \frac{\sum_r P(r \in t \mid \theta)}{\sum_{t'} \sum_r P(r \in t' \mid \theta)}$$

The numerator is the effective read count for transcript t ; the denominator normalizes across all transcripts.

Concrete example. Two isoforms sharing one exon, 10 total reads: 6 unique to isoform 1, 2 unique to isoform 2, 2 ambiguous (map to both). Initialize $\theta_1 = \theta_2 = 0.5$.

Iteration 1; E-step: the 6 unique reads assign fully to isoform 1, the 2 unique to isoform 2. Each ambiguous read splits: $P(\text{iso1}) = 0.5$, $P(\text{iso2}) = 0.5$. M-step: $\theta_1 = (6 + 0.5 + 0.5)/10 = 0.70$, $\theta_2 = (2 + 0.5 + 0.5)/10 = 0.30$.

Iteration 2 (E-step: ambiguous reads now split 0.70 : 0.30. M-step: $\theta_1 = (6 + 0.7 + 0.7)/10 = 0.74$, $\theta_2 = (2 + 0.3 + 0.3)/10 = 0.26$. Convergence to $\sim 0.75 : 0.25$ within 5 iterations) the ambiguous reads are distributed in proportion to the uniquely-mapping evidence.

The EM algorithm maximizes the likelihood $P(\text{all reads} \mid \theta)$; the probability of observing the entire read set given the transcript abundances, and guarantees monotonic increase toward a local maximum.

The transcriptome reference. EM depends on the completeness of the annotation. If a novel isoform exists but is absent from the reference, its reads are treated as ambiguous among the closest annotated isoforms, and EM distributes them across incomplete representatives, biasing all abundance estimates. This is why novel transcript discovery (StringTie, IsoQuant) should precede quantification in discovery-oriented studies.

Salmon and Kallisto: pseudoalignment. Rather than aligning each read to the genome, Salmon and Kallisto hash each read's k-mers against a transcriptome index. Reads sharing the same set of compatible transcripts form an **equivalence class** (e.g., "all reads compatible with transcripts {1, 3, 5}"). EM runs on class-level aggregate counts rather than individual reads, reducing the optimization from millions of objects to thousands; a 100x speedup with equivalent gene-level accuracy.

Normalization: Why RPKM/FPKM Are Wrong

RPKM/FPKM normalize by gene length and sequencing depth but are **not comparable across samples**:

$$\text{RPKM} = \frac{\text{counts}}{\text{gene length (kb)} \times \text{total mapped reads (millions)}}$$

The sum of RPKM values across all genes differs systematically between samples because the denominator (total mapped reads) is dominated by highly expressed genes, when one gene doubles, all others mechanically appear to decrease. **TPM (transcripts per million)** fixes this:

$$\text{TPM}_i = \frac{\text{counts}_i / \text{length}_i}{\sum_j \text{counts}_j / \text{length}_j} \times 10^6$$

The denominator $\sum_j (\text{counts}_j / \text{length}_j)$ is the sum of per-gene rates across all genes. Normalizing by this value rather than total mapped reads removes the artifact where a single highly expressed gene deflates all others: TPM is a proportion, so its sum is always 1 million regardless of sequencing depth or expression distribution.

Differential Expression: The Standard Workflow

```
library(DESeq2)

dds <- DESeqDataSetFromMatrix(countData = counts,
                              colData = metadata,
                              design = ~ condition)
dds <- DESeq(dds) # NB dispersion + Wald test
results <- results(dds, contrast = c("condition", "treated", "control"))
```

DESeq2 fits a Negative Binomial GLM per gene, shrinks dispersion estimates via empirical Bayes, and computes Wald test p-values with Benjamini-Hochberg FDR correction. The input must be raw integer counts; normalized values (TPM, FPKM) break the NB variance model.

Biological Interpretation

TPM is the correct relative abundance metric for within- and between-sample comparison. The EM algorithm depends on the completeness of the transcriptome reference: if a novel isoform is present but missing from the annotation, its reads are assigned to the nearest annotated isoform, biasing all estimates. This is why novel transcript discovery (StringTie, IsoQuant) should precede quantification in discovery-oriented studies.

RNA-seq quantification artifacts can mimic biological signals: GC bias (systematic underrepresentation of AT-rich or GC-rich fragments), 3' bias in degraded RNA (apparent downregulation of 5' exons), and fragment length bias in FFPE samples. These must be diagnosed before biological interpretation.

The choice of quantification method matters: alignment-based (STAR + featureCounts) is more accurate at gene level; pseudoalignment (Salmon, Kallisto) is faster and more accurate at transcript level. For standard gene-level differential expression, either is acceptable.

Summary and Required Reading

1. **Exon-exon junctions require splice-aware alignment:** DNA aligners (BWA) lose up to 50% of RNA-seq reads.
2. **EM resolves multi-mapping ambiguity:** Salmon/Kallisto use k-mer equivalence classes for 100x speedup.
3. **TPM is the correct normalization metric:** RPKM/FPKM are not comparable across samples.
4. **DESeq2 uses Negative Binomial GLMs with empirical Bayes dispersion shrinkage:** input must be raw counts.

Required Reading

- Dobin et al.: "STAR: ultrafast universal RNA-seq aligner" (Bioinformatics, 2013).
- Patro et al.: "Salmon provides fast and bias-aware quantification of transcript expression" (Nature Methods, 2017).

Johnson's Rule: A read that maps to three transcripts is not a problem; it is data. Ignoring it is the problem.

Chapter 12: Epigenomics and Hidden Signals

DNA is Hardware, Epigenetics is Software

Every cell contains the same 3.2 billion base pairs. A neuron, a liver cell, and a skin cell differ not in their DNA sequence but in which genes are accessible and expressed. A mutation is a hardware defect; an epigenetic change is a software bug; reversible, context-dependent, and programmable.

This analogy is more literal than it appears. DNA sequence (hardware) constrains what is possible; a gene cannot be expressed if it does not exist, and a regulatory element cannot function if its sequence is deleted. But the epigenome (software) determines what is actually executed, which promoters are accessible, which enhancers are active, which genes are silenced. Two cells with identical genomes can execute entirely different programs.

The techniques in this chapter are complementary probes of the software stack. No single technique tells the full story: ChIP-seq identifies which proteins are bound to DNA, ATAC-seq reveals where chromatin is accessible, motif analysis reads the regulatory code embedded in the DNA sequence itself, and DNA methylation measures a persistent chemical mark that stabilizes gene expression programs across cell divisions. Together, they form a multi-layered picture of how the same genome produces different cellular outcomes.

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Transposase-accessible DNA fragments (ATAC-seq), DNA fragments enriched by antibody capture (ChIP-seq / CUT&Tag), or chemically converted DNA (bisulfite sequencing).
- **Molecular Level:** The epigenetic modification and chromatin architecture levels. This includes DNA methylation, histone modification states, chromatin accessibility, and transcription factor occupancy.

Layer 1: Where Proteins Bind (ChIP-seq and CUT&Tag)

Chromatin Immunoprecipitation sequencing (ChIP-seq) maps genome-wide binding sites of a specific protein. The biochemical protocol is straightforward: crosslink proteins to DNA with formaldehyde, sonicate to 200-600 bp fragments, immunoprecipitate with an antibody targeting the protein of interest, reverse crosslinks, and sequence the enriched DNA fragments.

The signal interpretation depends entirely on the antibody target:

Target	Biological Meaning	Peak Shape
H3K27ac	Active enhancers and promoters	Narrow peaks
H3K4me3	Active promoters	Narrow peaks at TSS
H3K27me3	Polycomb-repressed chromatin	Broad domains (kb-Mb)
H3K9me3	Heterochromatin	Broad domains
CTCF	Insulator binding / chromatin looping	Narrow peaks
POLR2A	Active transcription	Narrow peaks at TSS + gene body

Antibody specificity is the dominant factor in ChIP-seq quality. A poorly validated antibody pulls down off-target proteins, producing clean-looking peaks at biologically irrelevant sites that pass all computational filters. The essential quality metric is the FRiP score (Fraction of Reads in Peaks): for a well-characterized transcription factor, FRiP > 5% indicates enrichment above background, while FRiP < 1% suggests the antibody failed. There is no computational method that can rescue a poor antibody; the enrichment simply is not there.

Broad vs. narrow peak calling. Histone modifications fall into two structurally distinct classes. Narrow marks (H3K27ac, H3K4me3, CTCF) produce focal enrichment at specific regulatory elements, typically 200-1000 bp. Broad marks (H3K27me3, H3K9me3) produce diffuse enrichment across tens of kilobases to megabases. MACS2 handles both via its `--broad` flag, which changes the background model to detect contiguous regions of enrichment rather than isolated peaks. Using narrow-peak mode on a broad histone mark cuts each domain into hundreds of fragmented “peaks” that violate the assumption of independent binding events; each “peak” is not an independent biological observation.

CUT&Tag (Cleavage Under Targets and Tagmentation) has largely replaced ChIP-seq for histone modifications in many applications. In this protocol, a protein-A-Tn5 fusion protein is guided directly to the target protein by a primary antibody under native (non-crosslinked) conditions, and the Tn5 is activated by adding magnesium, which cuts and tags DNA locally. By tagmenting under physiological conditions rather than crosslinking and sonication, it avoids epitope masking and reduces background genomic noise by >90%. Advantages: lower cell input (100–1,000 cells vs. 10^6 for ChIP-seq) and higher signal-to-noise ratio. The tradeoff: CUT&Tag requires a high-quality primary antibody with high affinity, and the short tagmentation fragments (~150 bp) are less informative for broad domain calls.

Layer 2: Where Chromatin is Open (ATAC-seq)

ATAC-seq maps open chromatin without targeting a specific protein. The Tn5 transposase simultaneously cuts DNA and inserts sequencing adapters, preferentially accessing nucleosome-free regions. The protocol is remarkably simple: ~3 hours from cells to library, no crosslinking, no antibody. Native reaction: intact nuclei are incubated with Tn5, which integrates adapters preferentially into accessible chromatin. The resulting fragments are PCR-amplified and sequenced.

Property	ChIP-seq	ATAC-seq
Target	Specific protein	Open chromatin
Antibody	Required	None
Protocol	2-3 days	~3 hours
Cell requirement	10^6	500-50,000
Signal interpretation	Binding site locations	Accessibility (regulatory potential)
Known bias	Antibody specificity	Tn5 sequence preference (GC-rich)

The Tn5 “Smart Molecular Scissors” Metaphor: Imagine the Tn5 transposase as a smart molecular scissor: it integrates adapters preferentially in nucleosome-free regions (<100 bp), leaving nucleosome-wrapped regions untouched. Thus, a successful ATAC-seq library displays a physical hallmark of chromatin architecture: a distinct fragment length periodicity.

ATAC-seq cannot identify which transcription factor is bound at each accessible site. Open chromatin is a necessary condition for transcription factor binding but not sufficient; an accessible region may be occupied by any of dozens of factors with motifs in that region, or by none at all. Determining which factor is bound requires either:

1. **Motif analysis** (below): which transcription factor binding motifs are enriched in the accessible regions
2. **Orthogonal ChIP-seq** for the specific factor of interest
3. **Footprinting:** Tn5 cutting is blocked where a transcription factor is bound, creating a depletion of cut sites at the motif position. The nucleotide-resolution cutting

pattern reveals “footprints” of ~15-30 bp where the factor protects DNA from integration; detectable as local dips in the Tn5 cut frequency within open peaks.

The Nucleosome Ladder as the Physical QC: The fragment size distribution is the ultimate physical diagnostic of your wet-lab execution. A successful assay displays a clear nucleosomal ladder: a prominent peak at < 100 bp representing nucleosome-free open regions, followed by a series of periodic peaks at ~200 bp (mononucleosomes), ~400 bp (dinucleosomes), and ~600 bp (trinucleosomes). If the distribution shows a monotonic drop-off with no nucleosome peaks, your Tn5 was over-active or your cells were dead (over-digestion). If there are no sub-nucleosomal fragments, your Tn5 was under-active or your permeabilization failed. Check this plot before writing a single line of alignment code.

MACS2 Peak Calling: Poisson Background and Signal Detection

MACS2 models the background read distribution as a Poisson process. As established in Chapter 4, the Poisson models sampling noise when events occur independently at a constant rate, which describes the null hypothesis for ChIP-seq: in the absence of binding, sonicated DNA fragments are randomly distributed across the genome. For a window of length w , the expected number of reads under the null hypothesis is:

$$\lambda_{\text{local}} = \frac{\text{reads in local region}}{\text{length of local region}} \times w$$

A peak is called if the observed read count exceeds λ_{local} with a Poisson p-value below a threshold (typically $p < 10^{-5}$ for narrow peaks). MACS2 applies double-peak correction (shifting each tag toward the 3' direction by half the estimated fragment length) to reconstruct the true binding site position from the sonicated fragment ends.

Significance vs. effect size. The Poisson p-value quantifies statistical confidence, but a peak with a tiny p-value may still have negligible biological effect. A deeply sequenced region (100x coverage) can produce $p < 10^{-10}$ for a peak with only 1.5-fold enrichment over background; statistically significant but biologically questionable. MACS2 reports both the p-value and a **fold-enrichment** score (observed signal $\div$ local background). The recommended filtering threshold for transcription factor ChIP-seq is fold-enrichment > 2 in addition to $p < 10^{-5}$. A peak that passes the p-value threshold but not the fold-change threshold is likely a high-coverage artifact rather than a genuine binding event.

The choice of λ estimator matters. MACS2 offers three options: λ_{local} (a window around the candidate peak), λ_{global} (genome-wide average), and $\lambda_{\text{per-chromosome}}$. λ_{local} is the most conservative and is recommended for transcription factors with focal binding. λ_{global} overestimates the expected background in open regions and underestimates it in closed regions, distorting the p-value distribution. For broad histone marks, MACS2's λ_{local} is averaged over a larger region (10 kb default) to match the broader enrichment scale.

IDR: Irreproducible Discovery Rate. The IDR framework quantifies cross-replicate consistency. It models the joint distribution of peak rankings from two replicates as a mixture

of reproducible signals and irreproducible noise, fitting a copula mixture model to separate the two components. The IDR score for each peak is the posterior probability that it belongs to the irreproducible component.

From IDR to consensus peaks. The IDR analysis produces three sets of peaks for each replicate: peaks below the IDR threshold (e.g., $IDR < 0.05$), peaks above the threshold but with significance in one replicate ($IDR \geq 0.05$ but passing the individual peak caller), and peaks with no significance (failed peaks). The final **consensus peak set** is typically defined as the union of IDR-passing peaks across both replicates, not the intersection, requiring a peak to pass IDR in both replicates is too stringent because genuine binding sites at the edge of detectability may pass IDR in one replicate and just miss it in the other. The union of $IDR < 0.05$ peaks from both replicates captures the maximal reproducible signal while excluding the “peaks” that appear in only one replicate.

Critical limitation: IDR corrects for technical noise but not for systematic antibody bias. If the same poor antibody is used in both replicates, IDR will produce perfectly reproducible but biologically wrong results; replicate consistency is necessary but not sufficient for biological validity.

Layer 3: Reading the Regulatory Code (Motif Analysis)

ChIP-seq tells you *where* a transcription factor binds; motif analysis tells you *what sequence features* recruit binding. These are complementary: ChIP-seq identifies binding sites genome-wide, and motif analysis reveals the DNA sequence preferences that drive binding.

Position weight matrices (PWMs) represent transcription factor binding preferences as a $4 \times L$ matrix of nucleotide frequencies or weights at each position. For a transcription factor with binding motif of length L , the PWM entry $W(b, i)$ encodes the log-likelihood ratio:

$$S(\text{sequence}) = \sum_{i=1}^L \log \frac{P(\text{base}_i \mid \text{PWM}, i)}{P(\text{base}_i \mid \text{background})}$$

The score S is the log-likelihood ratio of the sequence being generated by the PWM vs. by a background model (typically mononucleotide frequencies). A score of zero means the sequence is equally likely under both models; positive scores indicate a match to the motif.

Two modes of motif analysis:

1. **De novo discovery** (MEME, HOMER, STREME): Given a set of peak sequences, find overrepresented motifs without prior knowledge. MEME uses EM to fit a mixture model where some sequences contain the motif and others are background. The output is a set of PWMs ranked by statistical significance (E -value). De novo discovery is essential for studying poorly characterized transcription factors or species where known motif databases are incomplete.
2. **Known motif enrichment** (FIMO, AME): Scan peak sequences against a database of known PWMs (JASPAR, CisBP, HOCOMOCO). For each PWM, compute whether sequences matching that motif are enriched in the peak set relative to a matched background. Known motif analysis can identify which of hundreds of transcription factors with similar binding preferences is most likely driving the ChIP signal.

The relationship between motif strength and binding is not linear. A high-scoring motif match does not guarantee binding: the motif may be in closed chromatin, or the transcription factor may not be expressed in the cell type under study. Conversely, many

genuine ChIP-seq peaks contain weak motif matches or no identifiable motif at all; transcription factors can bind through cooperative interactions with other factors that are not captured by the PWM model. The PWM assumes each position contributes independently to binding energy, which is a simplification: nucleotide dependencies, DNA shape, and cofactor interactions all modulate binding affinity. Roughly 30-50% of ChIP-seq peaks for a typical transcription factor lack a detectable canonical motif, reflecting indirect binding via protein-protein interactions rather than direct DNA contact.

Positional enrichment analysis adds signal. Rather than scanning peaks uniformly, compute the spatial distribution of motif occurrences relative to the summit (peak center). For direct-binding transcription factors, the canonical motif should be enriched within ± 50 bp of the summit. Loss of this positional enrichment is a signature of indirect binding; the factor is present but not directly contacting DNA at that site.

Layer 4: Chemical Memory (DNA Methylation)

DNA methylation (5-methylcytosine at CpG dinucleotides) is the most stable epigenetic mark, persisting through cell division via maintenance methylation by DNMT1, which recognizes hemi-methylated CpGs after replication and methylates the daughter strand. This self-templating mechanism makes methylation a true epigenetic memory; it propagates information across cell generations without changing the DNA sequence.

Bisulfite sequencing (WGBS) treats DNA with sodium bisulfite, which deaminates unmethylated cytosines to uracil while leaving 5-methylcytosine intact. After PCR, unmethylated cytosines appear as thymines, while methylated cytosines remain as cytosines. Methylation level at a single CpG is estimated as:

$$M = \frac{C_{\text{unconverted}}}{C_{\text{unconverted}} + T_{\text{converted}}}$$

where $C_{\text{unconverted}}$ is the count of C reads and $T_{\text{converted}}$ is the count of T reads at that CpG position. The binomial confidence interval for M depends on coverage: at 10x coverage, a CpG with 8 C reads and 2 T reads has $M = 80\%$ with a wide confidence interval (~45-95%); at 100x coverage, the same proportion has a tight interval (~72-86%). This coverage dependence is why methylation analysis requires either deep coverage or aggregation across CpGs (tiling windows or regional averages).

Reduced representation bisulfite sequencing (RRBS) uses MspI digestion (cuts at CCGG sites) to enrich for CpG-rich regions before bisulfite conversion, reducing sequencing cost by focusing on promoters and CpG islands. RRBS covers ~5% of CpGs genome-wide but concentrates on the CpG islands most relevant for gene regulation. The tradeoff: RRBS misses distal regulatory elements and repetitive regions where methylation plays structural roles, and the MspI bias prevents interrogation of non-CCGG CpGs.

Genomic context determines functional interpretation. CpG islands (CGI, typically defined as >200 bp with >50% GC content and >0.6 observed/expected CpG ratio) in promoter regions are normally unmethylated. Methylation of promoter CGIs is robustly associated with transcriptional silencing. But the direction of causality is not resolved: methylation may silence the gene, or the silent gene may become methylated as a secondary consequence of losing active transcription. This ambiguity cannot be resolved from

static methylation measurements; it requires perturbation experiments (dCas9-TET1 for demethylation or dCas9-DNMT3A for de novo methylation) and measurement of the transcriptional response.

Gene body methylation follows a different pattern. Within transcribed regions, methylation is positively correlated with expression level; the opposite of the promoter relationship. This paradox arises because methylation in gene bodies may suppress spurious transcription initiation from cryptic internal promoters. The functional significance of gene body methylation remains an area of active investigation.

Non-CpG methylation (CHG, CHH) is rare in differentiated somatic cells but abundant in plants and in embryonic stem cells. In the human brain, non-CG methylation accumulates during postnatal development and is enriched in synaptic genes, suggesting a regulatory role in neuronal function that is independent of the canonical CpG paradigm.

Biological Interpretation

The four layers form an integrated picture. ChIP-seq identifies the hardware binding events. ATAC-seq reveals which regions of the genome are accessible for binding. Motif analysis reads the software code that determines regulatory logic. DNA methylation records a persistent memory of gene expression state. No single layer is sufficient: accessibility without motif analysis cannot identify the regulator; motif analysis without accessibility finds matches that are never bound; ChIP-seq of one factor reveals its binding but not the broader regulatory landscape; methylation captures persistent states but misses dynamic regulatory changes.

Of these, ChIP-seq is the most fragile. Antibody quality is the dominant factor in data quality, and computational filtering cannot rescue a poor antibody. The IDR framework corrects for technical noise but not for systematic antibody bias. A peak that passes IDR with high statistical significance is reproducible and biologically wrong if the antibody targets the wrong protein. The FRiP score is the simplest quality diagnostic: for a well-characterized transcription factor, FRiP > 5% indicates enrichment; FRiP < 1% indicates failure. This should be checked before any downstream analysis.

ATAC-seq signal is more robust (fewer antibody-related artifacts) but less specific. A differential ATAC peak between conditions identifies a change in accessibility but not the regulator driving the change. ATAC-seq quality is primarily assay-dependent: fragment size distribution (nucleosomal ladder), TSS enrichment (ratio of signal at transcription start sites vs. flanking regions, typically >5 for good data), and the ratio of reads in peaks to total reads. Each metric captures a different failure mode: TSS enrichment indicates successful targeting of regulatory DNA, fragment periodicity indicates correct Tn5 activity, and FRiP-like metrics reflect signal-to-noise.

DNA methylation analysis requires careful attention to context. Promoter CGI methylation correlates with repression, but causality is uncertain. Gene body methylation shows the opposite pattern (positive correlation with expression). Non-CG methylation, when present, is cell-type specific and developmentally regulated. The biological meaning of

a methylation change cannot be interpreted without knowing the genomic context and the cell type.

Summary and Required Reading

1. **ChIP-seq / CUT&Tag measure protein-DNA binding:** signal quality depends on antibody specificity. CUT&Tag requires fewer cells but both require careful antibody validation.
2. **ATAC-seq measures chromatin accessibility:** detects open regions but cannot identify which transcription factor binds them. Requires motif analysis or orthogonal ChIP-seq for factor identification.
3. **MACS2 uses Poisson background modeling** for peak detection; IDR assesses cross-replicate reproducibility but cannot correct for systematic antibody bias.
4. **Motif analysis (PWMs) reads the regulatory code:** de novo discovery for unknown motifs, known motif scanning for factor identification. The PWM independence assumption limits accuracy for cooperative binding.
5. **DNA methylation (bisulfite sequencing)** measures a self-propagating epigenetic mark; promoter CGI methylation correlates with repression, but causality is bidirectional. Genomic context determines functional interpretation.

Required Reading

- Zhang et al.: "Model-based Analysis of ChIP-Seq (MACS)" (*Genome Biology*, 2008).
- Li et al.: "Measuring reproducibility of high-throughput experiments" (*Annals of Applied Statistics*, 2011).
- Roadmap Epigenomics Consortium: "Integrative analysis of 111 reference human epigenomes" (*Nature*, 2015).

Johnson's Rule: A mutation is a hardware defect. An epigenetic change is a software bug.

Chapter 13: Metagenomics and the Human Microbiome

Microbiology is Compositional and Unculturable

The majority of microbial life on Earth, and in the human body has never been cultured. A standard microbiological plate selects for the survivable minority. Metagenomics is the only lens that reveals the unculturable majority. Before you can interpret a microbiome, you must first understand that the data you are analyzing is a compositional shadow—a relative, not absolute, measure of community structure.

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Total DNA extracted from an environmental or host-associated microbial community (shotgun metagenomics), or PCR-amplified marker genes (16S rRNA amplicon sequencing).
- **Molecular Level:** The taxonomic and metagenomic pathway levels. Metagenomics measures the relative abundances of microbial species and their functional metabolic genes in a mixed population.

The Uncultured Majority and Shotgun Metagenomics

Microbial culture plates capture only species capable of growing under laboratory oxygen concentrations, nutrient formulations, and temperature conditions. Estimates suggest fewer than 1% of environmental microbes are culturable. Shotgun metagenomics bypasses culture entirely: total DNA is extracted from a sample, fragmented, sequenced, and reads are assigned by computational classification. The resulting data represents the aggregate genomic content of all organisms in the sample; bacteria, archaea, viruses, fungi, and human host.

Two primary experimental strategies exist:

- **16S rRNA Amplicon Sequencing:** PCR amplification of the hypervariable V3-V4 region of the 16S ribosomal RNA gene using universal bacterial primers. Fast and

cheap; limited to taxonomic classification only, cannot resolve function. Susceptible to primer amplification bias; certain taxa are systematically under-amplified.

- **Shotgun Metagenomics:** Whole-genome sequencing of extracted community DNA. Enables both taxonomic profiling and functional gene annotation. More expensive; requires deeper sequencing depth to detect low-abundance taxa.

Taxonomic Classification: Two Competing Paradigms

Read-based classifiers assign taxonomy to individual sequenced reads.

k-mer Lowest Common Ancestor (LCA) Classifiers (Kraken2, Bracken): Each k-mer in a read is matched against a pre-built database of all reference genomes. The taxonomic classification of a read is the LCA of all matching k-mer taxa. Extremely fast (> 1 Gb/s classification speed). Bracken applies Bayesian re-estimation to produce abundance estimates at any taxonomic level. The critical limitation: unclassified reads represent the "dark matter"; novel species without reference genomes.

Marker-gene Profiling (MetaPhlAn4): Rather than classifying all reads, MetaPhlAn4 maps reads exclusively to species-specific single-copy marker gene databases. This avoids reference bias from human or food contamination reads. The resulting relative abundance profiles are compositionally defined: they sum to 100% across all detected species.

Metagenomic Assembly and MAG Reconstruction

When reference-free analysis is required, reads are assembled *de novo* into contigs. Two strategies:

- **Single-sample assembly:** Assembles reads from one sample independently (MEGAHIT, MetaSPAdes). Produces clean species-level contigs but misses low-abundance taxa without sufficient read depth.
- **Co-assembly:** Pools reads from multiple samples before assembly. Recovers rare organisms across samples, but risks generating chimeric contigs if closely related strains from different samples are incorrectly merged.

Assembled contigs are binned into **Metagenome-Assembled Genomes (MAGs)** using differential coverage (the same organism has consistent depth across samples) and tetranucleotide frequency (closely related organisms share compositional signatures). MAG quality is validated using **CheckM2**, which predicts completeness and contamination based on lineage-specific single-copy marker gene presence and copy number.

Functional Annotation

HUMAnN3 maps quality-controlled reads against a nucleotide database (ChocoPhlAn) and a translated protein database (UniRef90). The outputs are gene family abundances (reads-per-kilobase RPK) and pathway coverages (MetaCyc). This enables comparison of microbial community functional capacity across samples.

Metatranscriptomics: Total RNA shotgun sequencing measures active transcription in a community. Ribosomal RNA depletion is required to enrich for mRNA. Metatranscriptomics integrated with metagenomics reveals which genes are expressed, not just which genes are present.

Compositional Statistics

Microbiome data is **compositional**: sequencing produces relative, not absolute, abundances. The sum of all taxon proportions is always 1 (or 100%). Applying standard statistical tests that assume independence (t-test, Wilcoxon) to compositional data violates the Aitchison simplex geometry and produces spurious results.

Log-ratio transformations project compositional data into Euclidean space where standard linear statistics are valid:

- **Centered Log-Ratio (CLR):** $\text{CLR}(x_i) = \ln\left(\frac{x_i}{g(\mathbf{x})}\right)$, where $g(\mathbf{x})$ is the geometric mean of all components.

Compositionally robust differential abundance tools:

- **ALDEx2:** Monte Carlo Dirichlet samples to propagate compositional uncertainty through CLR transformation before differential abundance testing.
- **ANCOM-BC:** Models absolute abundance as the difference between observed log relative abundance and sample-specific bias; tests W-statistic (fraction of log-ratio comparisons significant).

Biological Interpretation

The human microbiome is not a single entity; it varies by body site (gut, skin, oral, vaginal), host genetics, diet, medication history, and disease state. A microbiome study reporting “healthy vs. disease” differences without controlling for antibiotic use, dietary fiber intake, geographic origin, and sequencing batch is not interpretable. The effect of dietary fiber on gut Bacteroidetes:Firmicutes ratio exceeds the difference between healthy and disease states in most inflammatory bowel disease studies.

The Reviewer Mindset: Never apply standard statistical tests to raw relative abundance columns. If *Bacteroides* increases as a proportion, every other taxon’s relative abundance decreases automatically; even if their absolute cell counts are unchanged. Alpha diversity (Shannon entropy, species richness) and beta diversity (Bray-Curtis dissimilarity, UniFrac distance) capture distinct signals: alpha diversity describes within-sample richness; beta diversity describes between-sample compositional distance. These must not be conflated. A single MAG does not constitute evidence of an organism’s functional role in the community; its ecological role requires metatranscriptomic confirmation and ideally a germ-free mouse colonization experiment.

Summary and Required Reading

The uncultured majority of the microbial world is accessible only through shotgun metagenomics. The primary analytical challenges are compositional statistics, assembly chimeras, and functional dark matter (unknown protein families). Every microbiome study must demonstrate control for confounders and compositional-aware statistical methods before its results are interpretable.

Required Reading:

- Quince et al.: "Shotgun metagenomics, from sampling to analysis" (*Nature Biotechnology*, 2017).
- Gloor et al.: "Microbiome datasets are compositional: and this is not optional" (*Frontiers in Microbiology*, 2017).
- Parks et al.: "CheckM2: a rapid, scalable and accurate tool for assessing microbial genome quality" (*Nature Methods*, 2023).

Chapter 14: Genome-Wide Association Studies (GWAS) and Statistical Genetics

Correlation is Not Causation in Genetic Association

Correlation between a genetic variant and a disease does not prove causation. Linkage disequilibrium means that the associated variant is almost never the causal variant; it is merely in statistical proximity to it. The causal variant must be fine-mapped. Before you can interpret a GWAS result, you must understand that a “hit” is a chromosomal region with correlated alleles, not a confirmed biological mechanism.

Core Concepts

What is Being Sequenced and at What Molecular Level

- **What is Being Sequenced:** Genomic DNA variants (typically single nucleotide polymorphisms, or SNPs) across thousands of individuals.
- **Molecular Level:** The population genotype level. GWAS correlates population-wide allele frequencies with disease states or continuous biological traits to identify statistical associations.

Linkage Disequilibrium and Haplotype Blocks

The human genome is not a collection of independent variants. Nearby SNPs are inherited together far more often than expected by chance, because recombination events are rare and localized to hotspots. This non-random co-inheritance is called **linkage disequilibrium (LD)**.

Two LD statistics:

- r^2 : Squared Pearson correlation between allele frequencies at two loci. $r^2 = 1$ means perfect co-inheritance. Used in imputation accuracy assessment.
- D' : Normalized covariance relative to the theoretical maximum. $D' = 1$ means no observed recombinant haplotypes in the sample.

LD structure defines **haplotype blocks**: chromosomal intervals within which recombination is suppressed. A single lead SNP in a GWAS may tag an entire block of hundreds of correlated variants. Fine-mapping is required to resolve which variant within the block is functional.

GWAS Study Design and Association Testing

A GWAS tests the statistical association of each of 500,000 to 10,000,000 SNPs with a phenotype of interest.

- **Case-control design:** Binary trait (disease/healthy). Logistic regression: $\text{logit}(P(\text{disease})) = \beta_0 + \beta_1 \cdot \text{genotype}_i + \gamma \cdot \mathbf{C}$, where $\mathbf{C}$ is a matrix of covariates (age, sex, ancestry PCs).
- **Quantitative trait:** Continuous phenotype (e.g., BMI, LDL, gene expression). Linear regression: $Y = \beta_0 + \beta_1 \cdot \text{genotype}_i + \gamma \cdot \mathbf{C} + \epsilon$.

Genotype is coded as dosage (0, 1, 2) under an additive model, assuming the effect of two copies is twice the effect of one.

Population Stratification

Genetic ancestry is a confounder in GWAS: it correlates simultaneously with disease prevalence (due to environmental and lifestyle factors differing across populations) and with allele frequencies. Without correction, ancestry will drive spurious associations; the statistical signal of “this allele is associated with disease” is actually “this allele is more common in the ancestry group with higher disease prevalence.”

Correction strategy:

1. Compute genotype PCA from LD-pruned SNPs. The top 10 principal components capture the major ancestry axes.
2. Include PC1–PC10 as covariates in all regression models.
3. Assess residual inflation using the **genomic control inflation factor** $\lambda_{GC} = \text{median}(\chi^2_{\text{observed}})/0.456$. $\lambda_{GC} > 1.1$ for a study of standard size signals residual confounding.

Multiple Testing and Genome-Wide Significance

Testing $\sim 10^6$ independent SNPs requires a Bonferroni-corrected significance threshold of:

$$P < 5 \times 10^{-8} \approx \frac{0.05}{10^6}$$

The standard $P = 5 \times 10^{-8}$ threshold is conservative for studies using imputed genomes with higher SNP counts, and liberal for studies restricted to targeted arrays. A

Manhattan plot visualizes the $-\log_{10}(P)$ landscape across all chromosomes; a Q-Q plot compares observed vs. expected $-\log_{10}(P)$ values to diagnose inflation.

Fine-Mapping and Causal Variant Identification

A GWAS lead SNP identifies a genomic region, not a causal variant. Fine-mapping uses Bayesian methods (SuSiE, FINEMAP) to compute a **credible set**: the smallest set of variants that contains the causal variant with 95% posterior probability. LocusZoom visualizes the LD structure and association statistics around a locus.

The causal variant is often not the protein-coding variant. Most GWAS signals map to non-coding regulatory elements (promoters, enhancers) and affect gene expression rather than protein function.

Expression Quantitative Trait Loci (eQTL)

eQTL mapping identifies genetic variants that affect gene expression levels in specific tissues. GTEx (Genotype-Tissue Expression) provides tissue-specific eQTL maps across 54 human tissues.

Colocalization tests whether a GWAS signal and a tissue-specific eQTL share the same causal variant, providing mechanistic insight into how the risk allele alters biology (coloc, SMR). A GWAS hit that colocalizes with a brain cortex eQTL for a specific gene provides strong evidence that the gene mediates the disease association through regulatory effects in that tissue.

Polygenic Risk Scores

The vast majority of complex disease heritability is distributed across thousands of common variants, each with tiny individual effects. Polygenic Risk Scores (PRS) aggregate this distributed signal:

$$\text{PRS} = \sum_{i=1}^M \hat{\beta}_i \cdot g_i$$

where $\hat{\beta}_i$ is the GWAS effect size estimate and g_i is the allele dosage at SNP i . PRS are now clinically validated for coronary artery disease, breast cancer, and type 2 diabetes risk stratification.

Biological Interpretation

A GWAS hit is a chromosomal region, not a gene. The closest protein-coding gene to the lead SNP is rarely the causal gene; enhancer-promoter chromatin loops can bridge a variant to a target gene 500 kb or more away. Interpreting a GWAS hit requires integration

with tissue-specific eQTL databases, chromatin accessibility (ATAC-seq, open chromatin enrichment), and 3D genome contact maps (Hi-C TAD boundaries).

The Reviewer Mindset: Population stratification is the primary confounding threat in GWAS. The Q-Q plot and genomic inflation factor (λ_{GC}) are the first quality checkpoints. $\lambda_{GC} > 1.1$ for a study of standard size indicates residual stratification or cryptic relatedness, not genuine polygenic signal. Polygenic Risk Scores demonstrate predictive value within the ancestry group in which they were trained. Applying a European-ancestry PRS to African-ancestry populations loses 30–80% of predictive power because LD structure and allele frequencies differ systematically across ancestry groups; a critical equity limitation of current PRS approaches. This is not a minor technical footnote: it represents a structural health disparity, because African-ancestry individuals are simultaneously most underrepresented in GWAS discovery cohorts and least well-served by the resulting risk scores.

Summary and Required Reading

GWAS identifies genomic regions associated with complex traits by testing millions of variants simultaneously. The primary analytical challenges are population stratification, multiple testing correction, and interpreting statistical associations as regulatory mechanisms rather than protein-coding mutations. Fine-mapping and eQTL colocalization are required to move from statistical signal to biological mechanism.

Required Reading:

- Visscher et al.: “10 years of GWAS discovery: biology, function, and translation” (*American Journal of Human Genetics*, 2017).
- Price et al.: “Principal components analysis corrects for stratification in genome-wide association studies” (*Nature Genetics*, 2006).
- Martin et al.: “Clinical use of current polygenic risk scores may exacerbate health disparities” (*Nature Genetics*, 2019).

TIER 3: High-Resolution & Systems

Chapter 15: Single-Cell RNA-seq and Dimensionality Escape

Bulk RNA-seq is a Fruit Smoothie

Bulk RNA-seq crushes 100,000 cells into a tube and sequences the average. A T-cell and a neuron co-ingested in a bulk sample produce an average expression profile that corresponds to no biological reality. Cell types are discrete physical identities, and averaging destroys the signal of rare but biologically critical populations.

Single-cell RNA-seq escapes this averaging problem but creates three new ones in its place: sparsity (90% of entries are zero), dimensionality (30,000 genes measured per cell), and ambiguity of measurement units (UMI counts vs. read counts vs. molecules). The statistical challenge of single-cell analysis is not the sequencing; it is recovering biological structure from a data matrix where most entries are zero, most genes carry no signal, and the unit of replication is a sample, not a cell.

Core Concepts

Microfluidics and Barcoding: The 10x Chromium

Three fluid streams converge in a microfluidic channel: cells in suspension, barcoded hydrogel beads, and oil. At the junction, droplets form spontaneously; each **GEM (Gel Bead-In-Emulsion)** contains approximately 0 or 1 cell plus 1 bead carrying a unique barcode.

Poisson loading. The number of cells per droplet follows a Poisson distribution. To minimize doublets (two cells in one droplet), the cell concentration is kept low so that most GEMs contain 0 cells. The fraction of GEMs containing at least one cell is $P(\geq 1) = 1 - e^{-\lambda}$. At $\lambda = 0.1$, only ~9.5% of GEMs have a cell, but among those, the doublet rate $P(\geq 2 | \geq 1) \approx \lambda/2 \approx 5\%$, meaning ~95% of occupied GEMs are singlets. This is a deliberate tradeoff: lowering the cell concentration reduces doublets but also reduces the number of cells captured per run. Standard 10x runs target ~10,000 cells with an expected doublet rate of ~4-8% at $\lambda \approx 0.05$ -0.1.

UMI collapse: from reads to molecules. Each mRNA molecule receives a unique molecular identifier (UMI) (a random 10-12 bp sequence) during reverse transcription. After

sequencing, reads sharing the same cell barcode, gene, and UMI are collapsed into a single count. This is conceptually transformative: 10 sequencing reads with the same UMI represent 1 original mRNA molecule, not 10. Without UMIs, PCR duplicates inflate apparent expression, and the count matrix reflects amplification history as much as biological abundance.

The consequence is that scRNA-seq count distributions differ from bulk RNA-seq in a subtle but important way. Chapter 4 established that bulk RNA-seq counts follow a Negative Binomial distribution: $\text{Var}(Y) = \mu + \alpha\mu^2$, where the quadratic term $\alpha\mu^2$ captures both technical (amplification) and biological (cell-to-cell) variation. In scRNA-seq with UMIs, amplification noise is largely removed by UMI collapse (a single molecule either is captured and receives a UMI or is not. This makes the capture process approximately Poisson: $\text{Capture} \sim \text{Poisson}(\lambda_{\text{capture}} \cdot \text{mRNA}_{\text{cell}})$. But the biological parameter λ itself varies across cells) some cells express a gene at 10 copies, others at 0. If λ follows a Gamma distribution across the cell population (the natural model for cell-to-cell expression variability), then the mixture of $\text{Poisson}(\lambda)$ with Gamma-distributed λ yields exactly the Negative Binomial from Chapter 4: $\text{Var}(Y) = \mu + \alpha\mu^2$. The critical difference from bulk RNA-seq: the dispersion α in scRNA-seq reflects biological variability alone (amplification noise is removed by UMI collapse), while in bulk RNA-seq α captures both technical amplification noise and biological variability. The UMI removes the amplification variance term but not the biological variance.

The Sparsity Crisis

Single-cell data is ~90% zero entries. A gene expressed at 10 mRNA copies per cell in bulk RNA-seq may appear in only 1-10% of individual cells. Understanding why requires distinguishing two sources of zeros:

- **Technical dropout:** The mRNA molecule existed in the cell but was not captured (failed to enter the droplet, was not reverse-transcribed, or was not sequenced). At typical capture efficiencies of 10-30%, a cell expressing a gene at 10 copies will on average capture 1-3 molecules and miss 7-9.
- **Biological zeros:** The gene is genuinely not expressed in this cell; the transcription rate is zero in that cell type or state.

The distinction matters because they require different handling. Technical dropout is a sampling problem (increase capture efficiency or sequencing depth), while biological zeros are the signal of cell-type specificity. Most scRNA-seq analysis does not attempt to separate them explicitly; instead, clustering methods rely on the observation that highly expressed cell-type marker genes have sufficient capture to define clusters, while dropout-affected genes simply contribute noise that the dimensionality reduction step filters out.

Capture efficiency and its consequences. Standard 10x Chromium captures ~10-30% of mRNA molecules per cell. This means for a gene with 10 mRNA copies per cell, the expected observed count is 1-3 UMI counts in expressing cells, and zero in 70-90% of cells that do express it. The expected fraction of cells with at least one UMI is $1 - e^{-\lambda}$, where λ is the true mean expression multiplied by capture efficiency. At $\lambda = 1$, ~63%

of cells show at least one count; at $\lambda = 0.1$, only ~9.5% do. The observed sparsity is thus a direct consequence of the physical capture efficiency; it is not a modeling choice.

The zero-inflated Negative Binomial (ZINB) models both sources of zeros as a mixture: a point mass at zero (dropout) plus an NB distribution for the expressed component. However, debate continues over whether zero-inflation is needed at all, with UMIs and moderate expression levels, the observed zeros may be adequately explained by Poisson sampling alone, without an explicit zero-inflation component. The practical implication: ZINB-based normalization methods (SCTransform, ZINB-WaVE) are standard but should be validated against simpler NB models for the specific dataset.

From Counts to Structure: The Dimensionality Escape

A typical scRNA-seq experiment produces a count matrix of ~30,000 genes $\times$ ~10,000 cells (300 million entries, 90% of which are zero. Working directly in 30,000-dimensional space is computationally intractable and statistically meaningless (the curse of dimensionality: distances between points converge in high dimensions). The solution is the dimensionality escape) successive compression steps that reduce the data to a biologically interpretable low-dimensional representation.

Step 1: Feature selection. Most of the 30,000 genes are not informative. The first step selects genes with high variance relative to their mean (dispersion), typically 2,000–5,000 highly variable genes (HVGs). The rationale: a gene expressed uniformly at 1 count per cell across all cells carries no information about cell identity; a gene with mean 1 count and variance 10 is biologically informative. Feature selection reduces the matrix from $30,000 \times 10,000$ to $3,000 \times 10,000$ while retaining the biologically relevant dimensions.

Step 2: Normalization. Raw UMI counts must be normalized to account for differences in sequencing depth across cells. A cell with 50,000 total UMIs and a cell with 5,000 total UMIs will have systematically different counts for every gene, ignoring this confounds biological variation with technical depth.

- **Library-size normalization (scran):** Compute size factors per cell by pooling cells with similar expression profiles, then normalize counts by dividing by the size factor. This is the scRNA-seq analogue of DESeq2's median-of-ratios normalization from Chapter 4.
- **SCTransform:** Fits a regularized Negative Binomial (ZINB) model per gene, regressing out sequencing depth. The Pearson residuals from the model are the normalized expression values. SCTransform handles the mean-variance relationship explicitly, removing the need for separate HVG selection; it replaces Steps 1 and 2 with a single regularized regression.
- **Log-normalization:** $\log(1 + \text{CPM})$; simple and fast, but distorts the mean-variance relationship and is not recommended as the sole normalization step.

Recommendation: SCTransform is the best default for most single-cell analyses; its ZINB-based residuals handle dropout and overdispersion jointly, remove the need for separate HVG selection, and are well-behaved for PCA. Use scran's pooling-based size factors when you need to preserve absolute expression levels for cross-condition differential expression, where residual-based values can be harder to interpret.

The normalization choice determines what signal the downstream steps see. Under-normalization leaves depth confounders in the data, and the first principal components will capture sequencing depth rather than biology. Over-normalization (aggressive regression of all technical factors) can remove biological signal correlated with library complexity, such as cell-cycle effects or activation states.

Step 3: PCA compression. The normalized matrix (3,000 genes $\times$ 10,000 cells) is reduced to ~50 principal components. Why does PCA work on a sparse, zero-inflated matrix? Because genes are correlated (the covariance structure exists in the sparse matrix because co-expressed genes vary together across cells. A gene expressed in only 5% of cells contributes to the covariance when it is expressed; the zeros simply add noise that the top PCs average out. The first PC typically captures a dominant source of variation (cell type, cell cycle, or technical artifact) inspect it before trusting it). The number of PCs to retain is chosen by the **elbow heuristic**: plot the variance explained per PC and keep the PCs before the curve flattens. For 10x data, this is typically 15-50 PCs. Too few PCs discard rare cell types; too many introduce noise that degrades clustering. PCA compresses 3,000 dimensions to ~50 while preserving the structure needed for clustering.

Step 4: Graph construction and clustering. The 50-dimensional PCA embedding is used to construct a K-nearest neighbor (KNN) graph: each cell is a node connected to its k nearest neighbors (typically $k = 10-30$) by Euclidean distance in PCA space. The graph is the key data structure because it captures neighborhood relationships without requiring cells to be close in the original 30,000-dimensional space; two cells of the same type that are far apart in gene space can be connected through intermediate cells.

The **Leiden algorithm** optimizes modularity on this graph by moving cells between communities until the partition is stable. The resolution parameter controls cluster granularity: higher resolution produces more clusters, but each cluster must be validated biologically. The algorithm will find clusters in random noise; cluster existence is a property of the algorithm, not necessarily of the data.

UMAP: Visualization, Not Inference

UMAP (Uniform Manifold Approximation and Projection) builds a fuzzy simplicial complex from the KNN graph in PCA space and optimizes a low-dimensional (2D) embedding. Compared to t-SNE, UMAP preserves more global structure, is computationally faster, and produces more deterministic embeddings. The key parameters: `n_neighbors` (higher = more global structure preserved) and `min_dist` (lower = tighter cluster packing).

Critical caveats:

- UMAP is a visualization, not a clustering. Different random seeds produce different visual arrangements even with the same data.
- Cluster sizes and distances in UMAP are not interpretable; the algorithm can produce visually separated clusters from continuous gradients.
- The same UMAP parameters applied to different random seeds or subsamples produce visually different plots, yet the biological interpretation should be the same.

t-SNE (t-distributed Stochastic Neighbor Embedding) is an older alternative that converts high-dimensional distances to probabilities and minimizes KL divergence. t-SNE preserves local structure well but has additional limitations: cluster positions and sizes are not interpretable, perplexity choice is non-obvious, and different random seeds produce substantially different outputs.

Doublet Detection

Doublets (two cells in one droplet) produce expression profiles that are combinations of two distinct cell types. They appear as intermediate cells in UMAP and form spurious clusters. Computational detection (Scrublet, DoubletFinder):

- Simulates doublets from random pairs of real cells
- Trains a classifier to distinguish simulated vs. observed cells based on the adjacency of their expression profiles to other cells
- The expected doublet rate is typically 3-8% per 10x channel

Doublets cannot be completely eliminated by loading concentration. Even at optimal loading, physical doublets occur. Joint analysis of combined channels amplifies the problem: a doublet between a T-cell and a macrophage in one channel appears as a valid intermediate state unless detected and removed.

Multi-Sample Integration and Seurat V5 Layer Design

When analyzing single-cell datasets across multiple biological conditions, batches, or patients, you face the challenge of **batch effects**. These are technical variations (e.g., cell capture efficiency, sequencing depth, sample viability) that systematically shift cell positions in the expression space, masquerading as novel cell types or states.

- **The Integration Dilemma:** The goal of integration is to align equivalent biological states across datasets while preserving true biological differences. Standard methods operate on the lower-dimensional PCA space:
 1. **Canonical Correlation Analysis (CCA):** Projects both datasets onto shared dimensions that maximize correlation, assuming that the major biological structures are conserved. It is highly effective for identifying equivalent states but can over-correct and erase dataset-specific responses.
 2. **Mutual Nearest Neighbors (MNN) / FastMNN:** Identifies cell pairs across datasets that are mutual nearest neighbors. It projects one dataset onto the other based on the batch correction vector, which is more conservative and less prone to over-alignment.
 3. **Harmony (PCA-based iterative clustering):** Projects cells into PCA space, performs soft clustering, and iteratively shifts cells to maximize cluster-level batch mixture. Harmony is extremely fast and scales to millions of cells.
- **Seurat V5 Assay Design:** Historically, batch integration required copying and splitting the full expression matrix, which doubled the memory footprint. Seurat V5 addresses this with a new **Assays v5** design: a single object houses raw counts in distinct **layers** (e.g., `counts.batch1`, `counts.batch2`). Normalization, variable feature selection, and dimensional reduction run across these layers independently

or jointly without splitting the object. Integration is executed by specifying an integration method directly on the reduction layer (e.g., `IntegrateLayers(method = CCAIntegration)`), which generates a corrected low-dimensional space (e.g., `integrated.cca`) while preserving the raw, un-integrated counts in the background. This prevents data leakage and makes it easy to compare integrated vs. un-integrated states.

Biological Interpretation

Never trust a cluster you cannot name. If you cannot find known marker genes for a cluster, it may be a doublet, a batch effect, or a technical artifact. UMAP is a visualization, not a clustering; different random seeds produce different visual arrangements, and the resolution parameter is a tuning decision that must be justified biologically, not computationally.

Pseudo-bulk aggregation (summing counts across all cells of a type per sample) restores the unit of replication; the biological sample, not the cell. Treating each cell as an independent replicate inflates statistical power by orders of magnitude and produces irreproducible results. Differential expression from single-cell data must use pseudo-bulk approaches or mixed models that account for the sample-level correlation structure. A dataset with 3 cases and 3 controls, each with 5,000 cells, has 6 replicates, not 30,000.

Cluster annotation is the most subjective step in scRNA-seq analysis. Expression of a single canonical marker (e.g., CD3D for T-cells) is suggestive but not definitive; CD3D fragments may align to a different gene in a different species, or a macrophage may have picked up T-cell mRNA from phagocytosis. Multi-marker validation, cross-reference with published atlases, and consistency across samples are required for rigorous annotation.

Summary and Required Reading

1. **Microfluidic barcoding + UMIs** enable molecular counting of 10,000+ cells per experiment, eliminating PCR bias. The capture process is approximately Poisson; remaining biological overdispersion motivates NB models.
2. **The dimensionality escape** proceeds in four steps: feature selection (HVGs) → normalization (scran/SCTransform) → PCA (50 dimensions) → KNN graph + Leiden clustering. Each step compresses the representation toward biological structure.
3. **Sparsity (90% zeros)** combines technical dropout (low capture efficiency) and biological zeros (cell-type specificity). ZINB models attempt to separate them, but debate continues on whether explicit zero-inflation is necessary with UMIs.
4. **PCA on sparse data works** because genes are correlated; the covariance structure exists, and the top PCs extract the dominant biological signals.

5. **Pseudo-bulk analysis** is required for valid differential expression; cells within a sample are not independent replicates.
6. **Cluster annotation requires multi-marker validation:** a single marker gene is not sufficient for cell-type identification.

Required Reading

- Zheng et al.: "Massively parallel digital transcriptional profiling of single cells" (*Nature Communications*, 2017).
- McInnes et al.: "UMAP: Uniform Manifold Approximation and Projection" (*Journal of Open Source Software*, 2018).
- Luecken & Theis: "Current best practices in single-cell RNA-seq analysis" (*Molecular Systems Biology*, 2019).

Johnson's Rule: Never trust a cluster you cannot name.

Chapter 16: Trajectory Inference and RNA Velocity

Biology is a Continuum

A hematopoietic stem cell does not instantaneously transform into a red blood cell. It passes through thousands of intermediate transcriptomic states. If you force developing cells into discrete clusters, you destroy the timeline of their evolution. Chapter 13 showed how to identify cell types; this chapter addresses a deeper question: within a cell type, how do cells transition from one state to another?

The challenge is observational: you have a single snapshot; tissue from a single time point, but you need temporal information. Pseudotime and RNA velocity recover this temporal dimension from static data by exploiting two different signals: transcriptional similarity between cells (pseudotime) and the kinetics of RNA splicing (velocity).

Core Concepts

Pseudotime: Ordering Cells Without a Clock

Pseudotime algorithms order cells along a developmental trajectory using transcriptional similarity, without knowledge of real time. The fundamental assumption is that cells captured at different points along a continuous process can be ordered by their expression profiles: a cell halfway through differentiation expresses a characteristic set of intermediate genes that places it between the starting and ending states.

Minimum Spanning Trees (Monocle). Builds a KNN graph in reduced-dimension space and finds the shortest path connecting all cells (the minimum spanning tree). With user-specified root cells (the “start”), pseudotime is the distance along the MST from the root to each cell. Changing the root changes all downstream interpretations; the root assumption is the critical user decision. Monocle also learns the “ordering” genes (those that change along the trajectory) as part of the algorithm, which can create circularity: the selected genes determine the trajectory, and the trajectory determines which genes are selected. This is not a bug; it is the algorithm’s best inference, but it means that different initial gene selections can produce different trajectories from the same data.

Diffusion maps (destiny). Models cell-to-cell transitions as a random walk. Cells con-

nected by many short paths are close in diffusion pseudotime. This approach is more robust to noise than MST-based methods because it considers all paths, not just the single shortest. Diffusion maps are particularly effective for datasets with continuous gradients (e.g., cell cycle) rather than discrete branching (e.g., differentiation with two daughter fates).

Sliding principal curves (Slingshot). Simultaneously identifies multiple trajectories and branch points. The algorithm constructs a principal curve through the data (the path of maximal variance) and projects each cell onto the curve for its pseudotime value. For branching lineages, Slingshot first identifies clusters, then constructs a minimum spanning tree between clusters, and finally fits principal curves along each branch. This hybrid approach (clustering followed by continuous trajectory) bridges the discrete-to-continuous gap that Chapter 13's clustering framework leaves open.

The root assumption. Every pseudotime method requires a root; a cell or set of cells designated as the starting point. The root is typically identified from prior biological knowledge (e.g., the earliest developmental stage in the dataset) or from expression of known marker genes. The critical caveat: different root choices produce different pseudotime orderings, and there is no computational way to validate the correct root from the expression data alone. RNA velocity (below) removes this requirement by inferring directionality from splicing kinetics.

RNA Velocity: The Physics of Splicing

Pseudotime orders cells but cannot determine direction; it requires the user to specify a root. RNA velocity solves this by adding a directional signal from RNA splicing kinetics.

RNA velocity infers transcriptional direction from the balance of unspliced pre-mRNA and spliced mRNA. When a gene is activated, transcription produces unspliced pre-mRNA, which is then spliced into mature mRNA and eventually degraded. A high ratio of unspliced to spliced mRNA for a given gene means transcription has recently increased; unspliced molecules have accumulated and have not yet been fully spliced. A low ratio means transcription has slowed and the remaining spliced mRNA is decaying.

The kinetics follow a coupled pair of ordinary differential equations:

$$\frac{du}{dt} = \alpha - \beta u \quad (\text{unspliced})$$

$$\frac{ds}{dt} = \beta u - \gamma s \quad (\text{spliced})$$

where u is unspliced counts, s is spliced counts, α is the transcription rate, β the splicing rate, and γ the degradation rate.

Steady state. When α is constant and enough time has passed, both u and s reach equilibrium ($du/dt = 0$, $ds/dt = 0$):

$$u^* = \frac{\alpha}{\beta}, \quad s^* = \frac{\alpha}{\gamma} = \frac{\beta}{\gamma} u^*$$

At steady state, cells lie along the line $s = (\beta/\gamma)u$ in the phase portrait (unspliced vs. spliced counts per gene, each point is a cell). The ratio $s/u = \beta/\gamma$ is constant, splicing and degradation are balanced.

Interpreting the phase portrait. When transcription changes, cells deviate from the steady-state line:

- Cells **above** the line ($s/u > \beta/\gamma$) have excess spliced mRNA; transcription was recently **downregulated**; spliced mRNA persists while unspliced decays.
- Cells **below** the line ($s/u < \beta/\gamma$) have excess unspliced mRNA; transcription was recently **upregulated**; unspliced is accumulating while splicing and degradation lag.
- Over time, the system relaxes back toward steady state, tracing a loop in the phase portrait. The direction of this loop (upregulation → below line → recovery, or down-regulation → above line → recovery) encodes the velocity signal.

The original RNA velocity model (La Manno et al., 2018) estimated β/γ from cells assumed to be at steady state, then computed each cell's velocity as its residual from the steady-state line. **scVelo** (Bergen et al., 2020) replaces this steady-state assumption with a **dynamical model** that estimates α , β , and γ per gene by fitting the full ODE system across all cells. This is essential for transient states; in a developing tissue, few genes are at equilibrium, and the steady-state assumption systematically underestimates velocity for recently induced genes.

Preprocessing. Unspliced and spliced counts are not available from standard scRNA-seq quantification. The preprocessing step (velocity, or spliced/unspliced quantification in alevin-fry and kallisto-bustools) classifies each read as unspliced (overlapping intronic sequence) or spliced (spanning exon-exon junctions) against an annotated transcriptome.

CellRank: Integrating Velocity with Connectivity

CellRank (2021) combines RNA velocity with transcriptomic similarity to model cell fate probabilities. It constructs a transition matrix where each entry T_{ij} is the probability that cell i transitions to cell j , combining:

- **Transcriptomic similarity:** cells with similar expression profiles are more likely to be adjacent in developmental time.
- **Velocity direction:** cells should transition in the direction of the velocity field, not against it.

The combined Markov chain is analyzed to identify absorbing states (terminal cell types) and compute the probability of reaching each terminal state from every starting cell. This identifies:

- **Root cells:** high probability of transitioning away from
- **Terminal states:** high probability of transitioning toward
- **Driver genes:** genes whose velocity best predicts the fate decision

Unlike pseudotime alone, CellRank does not require an explicit root assumption; directionality is inferred from the velocity field. The Markov chain formulation also provides a natural measure of fate commitment: cells with high entropy in their transition probabilities are multipotent progenitors; cells with low entropy are committed to a specific fate.

Validation: When the Trajectory is Wrong

Trajectory inference produces a compelling visual output; cells arranged along a smooth path from start to end, but this ordering is an inference, not an observation. The trajectory can be wrong in several ways that are invisible from the output alone:

- **False trajectories:** Pseudotime algorithms will order any dataset, even one with no biological trajectory. Clustering followed by trajectory inference on random noise produces a smooth path through the noise.
- **Cell cycle as dominant signal:** A velocity field that points in circles instead of along a trajectory suggests the dominant signal is the cell cycle, not differentiation. Before interpreting velocity as differentiation direction, validate against cell cycle phase markers; if the top velocity-driven genes are histone genes and cyclins, the signal is likely cycling, not development.
- **Splicing rate assumptions:** RNA velocity assumes constant splicing rates across cells. These fail in cell cycle, stress response, and metabolic transitions where splicing rates change rapidly. The phase portrait (unspliced vs. spliced per gene) is the most informative diagnostic: genes with clear loops are the best candidates for driving the trajectory; genes with diffuse clouds are consistent with steady-state noise or cell cycle variation.
- **Root sensitivity:** Pseudotime results depend on the root choice. A trajectory that appears robust to different root choices is more trustworthy than one that rearranges completely.

Velocity and pseudotime should be validated against known time points or orthogonal markers before trusting the inferred ordering. A trajectory that looks biologically plausible but disagrees with well-established developmental markers should be regarded with skepticism.

Biological Interpretation

Pseudotime requires an explicit root assumption; changing the root changes all downstream interpretations. RNA velocity removes the root assumption but adds assumptions

about constant splicing rates across cells; these fail in cell cycle, stress response, and metabolic transitions where splicing rates change rapidly.

The trajectory is only as good as the clustering it builds on. If the underlying cell types are incorrectly identified (Chapter 13), the trajectory connecting them is meaningless. Conversely, a good trajectory can reveal that what appeared as two distinct clusters are actually endpoints of a continuous gradient; a common case in immune activation, where resting and activated states of the same cell type are separated by a continuum of intermediate states.

Trajectory inference from a single time point is fundamentally underdetermined. Multiple distinct biological processes can produce the same observed expression patterns. The trajectory is a hypothesis about the developmental process, not a measurement of it. Independent validation (time series experiments, genetic perturbations, or orthogonal assays) is required to confirm the inferred ordering.

The phase portrait visualization (unspliced vs. spliced per gene) is the most informative diagnostic: genes with clear loops (hysteresis) are the best candidates for driving the trajectory; genes with diffuse clouds are consistent with steady-state noise or cell cycle variation. A trajectory driven entirely by genes with weak phase portraits should be treated as a tentative hypothesis.

Summary and Required Reading

1. **Pseudotime orders cells along a developmental trajectory** using transcriptional similarity; requires explicit root specification; changing the root changes all results.
2. **RNA velocity uses unspliced/spliced ratios** to predict future expression states from a single snapshot, removing the root assumption.
3. **scVelo's dynamical model** relaxes the steady-state assumption for more accurate velocity in transient states, fitting splicing kinetics per gene.
4. **CellRank integrates velocity with connectivity** to identify root and terminal states and driver genes without requiring an explicit root.
5. **Trajectory inference is fundamentally underdetermined:** validate against known time points, orthogonal markers, or perturbation data before trusting the inferred ordering.

Required Reading

- Trapnell et al.: "The dynamics and regulators of cell fate decisions" (*Nature Biotechnology*, 2014).
- Bergen et al.: "Generalizing RNA velocity to transient cell states through dynamical modeling" (*Nature Biotechnology*, 2020).

Johnson's Rule: Biology is a fluid continuum. If you force developing cells into discrete clusters, you destroy the timeline of their evolution.

Chapter 17: Spatial Transcriptomics and Coordinate Math

A Cell Without Spatial Coordinates is Headless

Dissolving a tissue to isolate single cells destroys the biological map. A macrophage surrounded by healthy epithelium behaves entirely differently than one surrounded by necrotic tumor. Chapter 13 taught you to identify cell types; Chapter 14 taught you to order them in time. But development and function unfold not only in time; they are embedded in three-dimensional tissue architecture. Without spatial context, single-cell data reveals the molecules but obscures the anatomy.

Spatial transcriptomics solves this by measuring gene expression while preserving tissue coordinates. The core operation is coordinate math, mapping every measurement to a position (x, y) in tissue space, then asking which genes vary across those coordinates, which cell types occupy which coordinates, and which neighboring coordinates communicate with each other.

Core Concepts

The Spatial Resolution Tradeoff

Spatial transcriptomics is defined by a fundamental tradeoff: whole-transcriptome coverage vs. single-cell resolution. No current technology achieves both. Choosing a technology means choosing where on this tradeoff curve to operate, and that choice determines what biological questions can be addressed.

Spatial barcoding (10x Visium, HD, Slide-seq) uses a slide printed with spatially bar-coded oligonucleotides. A tissue cryosection is placed on the slide; cellular mRNA diffuses vertically into the capture probes. The barcode encodes the (x, y) coordinate of each capture spot, enabling reconstruction of measurement positions after sequencing. Visium spots are 55 μm diameter at 100 μm spacing, each capturing RNA from ~1-10 cells. Visium HD (2 μm pixels) achieves near-single-cell resolution. The key property: whole-transcriptome coverage (all poly-adenylated RNA is captured), enabling discovery of unanticipated gene expression patterns.

In situ hybridization (MERFISH, Xenium, CosMx) uses fluorescent probes that hybridize directly to mRNA in the intact tissue section. Sequential barcoding cycles

decode transcripts: each gene receives a binary code, and 10-20 imaging cycles decode hundreds to thousands of genes at subcellular resolution (~100-200 nm). The key property: resolution sufficient to resolve individual transcripts, enabling subcellular localization and single-cell assignment. The tradeoff: limited panel size (hundreds to thousands of genes, not the full transcriptome).

Property	Spatial Barcoding	In Situ Hybridization
Coverage	Whole transcriptome (20K genes)	Targeted panel (100-5,000 genes)
Resolution	Multi-cell (55 μm spots) to near-single-cell (HD)	Subcellular (~200 nm)
Discovery power	High (unbiased gene discovery)	Low (only panel genes detected)
Bioinformatic challenge	Deconvolution of mixed spots	Molecular decoding, registration

Cell Segmentation: The Image-Based Bottleneck

Image-based spatial platforms (such as Xenium and MERFISH) detect the spatial coordinates of individual transcript molecules with sub-micron precision. However, to analyze this data similarly to single-cell RNA-seq, we must group these molecules into cells. This is **cell segmentation**, and it represents the primary bottleneck of image-based spatial analysis.

We run deep learning segmentation models (such as Cellpose or Mesmer) on nuclear stains (DAPI) and membrane/cytoplasm markers to draw physical boundaries around cells. * **The Segmentation Errors:** These models frequently make two critical errors: 1. **Undersegmentation:** Grouping transcripts from overlapping or adjacent cells into a single boundary, generating a false hybrid “cell type” that does not exist in nature. 2. **Oversegmentation:** Splitting a single large cell into multiple smaller compartments, creating artificial cell fragments (e.g., a “cell” containing only mitochondrial reads and no nucleus). * **Panel Constraint Bias:** Unlike sequencing-based methods, image-based methods rely on pre-designed targeted probe panels. What you do not probe remains invisible. A cell cannot be annotated if its marker genes are not on the panel, and downstream deconvolution or cell type assignment is strictly bounded by these probe design choices.

Deconvolution: Unmixing Mixed Spots

Each Visium spot captures RNA from multiple cells. Deconvolution is the inverse problem: given the mixed expression profile of a spot, estimate the cell type composition. Formally, each spot’s expression vector $\mathbf{y}_s$ is a weighted sum of cell type reference profiles:

$$\mathbf{y}_s = \sum_{k=1}^K \beta_{sk} \cdot \mu_k + \epsilon_s$$

where μ_k is the reference expression profile of cell type k , β_{sk} is the proportion of type k in spot s , and ϵ_s is residual noise. This requires a reference scRNA-seq dataset with annotated cell types from the same tissue. Deconvolution solves for the β_{sk} coefficients per spot.

RCTD (Robust Cell Type Decomposition) models each spot's expression as a linear combination of cell type profiles, with statistical significance testing to determine which cell types are present in each spot. The "robust" in the name refers to its handling of genes that are poorly fit by the linear model; it downweights genes whose expression deviates from the reference, preventing a few discordant genes from driving the deconvolution result.

Cell2location uses a Bayesian Negative Binomial model that accounts for both technical variation (NB count distribution) and biological variation (cell type abundance heterogeneity). Unlike RCTD, which estimates relative proportions, Cell2location estimates the absolute number of cells of each type per spot. This is more interpretable but requires stronger modeling assumptions about cell type expression profiles.

CARD incorporates spatial correlation into deconvolution, assuming that cell type abundances vary smoothly across tissue sections. This makes biological sense; a B-cell zone in the lymph node grades into a T-cell zone gradually, not in isolated spots, and improves deconvolution accuracy for spatially structured tissues.

The reference dependency problem. All deconvolution methods depend critically on the reference scRNA-seq dataset. If the reference lacks a cell type present in the tissue, that cell type's contribution will be misassigned to the most transcriptionally similar available type. If the reference and spatial data come from different tissue sections, different donors, or different processing protocols, batch effects between the two datasets will distort the deconvolution. A deconvolution result is only as good as the reference it builds on.

Neighborhood Analysis: From Spots to Tissue Organization

Once cell types are assigned to coordinates, the next question is tissue organization, which cell types are near which others, and how that proximity shapes function.

Spatial autocorrelation (Moran's I) quantifies whether neighboring spots are more similar than expected by chance:

$$I = \frac{n}{W} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

where w_{ij} is a spatial weight matrix (1 if spots i and j are neighbors, 0 otherwise), W is the sum of all weights, and n is the number of spots. Moran's I ranges from $[-1, +1]$: $I > 0$ indicates spatial clustering (neighboring spots are more similar than expected), $I < 0$ indicates a "checkerboard" pattern (neighboring spots are systematically different), and $I \approx 0$ indicates random spatial organization.

The null expectation. Under the null hypothesis of no spatial autocorrelation, the expected value of Moran's I is:

$$E[I] = -\frac{1}{n-1}$$

For large n (thousands of spots), $E[I] \approx 0$, but for small tissue sections with $n < 100$, the null expectation is meaningfully negative; a value near zero can actually indicate weak positive autocorrelation. Significance is assessed by permutation testing: randomly shuffling expression values across spots (breaking spatial structure), recomputing I thousands of times, and comparing the observed I to the empirical null distribution. A p -value is the fraction of permuted I values more extreme than the observed I .

Neighborhood definition. The weight matrix w_{ij} encodes which spots are considered neighbors. The choice of neighborhood definition determines W and directly affects the computed I value. Two common approaches exist: **k -nearest neighbors** (each spot connects to its k closest neighbors by Euclidean distance, ensuring every spot has the same number of neighbors) and **radius-based** (all spots within a fixed distance threshold are neighbors, which can leave isolated spots on tissue edges with few or no neighbors). For Visium's hexagonal grid, $k = 6$ is the natural choice; each spot has exactly six immediate neighbors corresponding to the six faces of a hexagon. Using fewer than 6 neighbors underweights spatial structure; using more than 6 dilutes local signal by including spots beyond the immediate tissue neighborhood.

Moran's I is widely used to identify spatially variable genes, but the signal must be interpreted cautiously. A gene may appear spatially variable due to: biological compartmentalization (e.g., crypt vs. villus in intestine), tissue edge artifacts (higher signal at tissue periphery due to probe penetration), or technical variation (uneven permeabilization across the section). Comparing Moran's I across genes within the same tissue section controls for the last two sources.

Ligand-receptor inference (NicheNet, CellChat, SpaTalk) predicts cell-cell communication from co-expression of ligands and receptors in adjacent cell types. The approach: if cell type A expresses a ligand gene and adjacent cell type B expresses its cognate receptor, then signaling from A to B is plausible. Scores are based on prior knowledge databases of known ligand-receptor pairs and downstream signaling targets.

The communication inference gap. Detecting mRNA for a ligand in one cell type and its receptor in an adjacent cell type is consistent with signaling but does not prove it. mRNA abundance correlates weakly with protein secretion, and secretion does not guarantee binding or downstream pathway activation. Ligand-receptor inference identifies candidates for experimental validation, not confirmed interactions.

Registration: Aligning Data to Anatomy

Spatial transcriptomics data measures gene expression, but the biological interpretation depends on knowing which anatomical region each spot corresponds to. Registration aligns the sequencing data to the tissue histology image (H&E or immunofluorescence):

an affine transformation that rotates, scales, and translates the spot grid to match the tissue image coordinates.

Registration is not a biologically neutral preprocessing step. An affine transformation maps original coordinates $\mathbf{x}$ to transformed coordinates $\mathbf{x}'$ via a 2×2 rotation-scaling matrix and a translation vector:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

The six parameters (a, b, c, d, t_x, t_y) are optimized to minimize mismatch between the spot grid and the tissue boundary in the histology image. The quality of this transformation directly determines whether transcriptomic measurements map to the correct anatomical regions. A misregistration of 100 μm (one Visium spot) can shift a spot from tumor epithelium to stroma, entirely changing its biological interpretation. For in situ methods with subcellular resolution, registration errors of a few microns can misassign transcripts to the wrong cell.

The registration challenge is compounded by tissue deformation during sectioning. The tissue on the slide is not identical to the tissue in the fresh block; it has been stretched, folded, or torn. Computational registration must account for non-linear warping, not just affine transformations, to accurately map sequencing data to histology.

Biological Interpretation

Spatial data is a hypothesis generator, not proof. Detecting ligand-receptor pairs in adjacent cells generates candidates for cell-cell communication, but confirmation requires perturbation, knocking out the ligand or receptor and measuring the effect on the recipient cell. A correlation between a gene's spatial expression pattern and a tissue compartment is suggestive of compartment-specific function, but the same correlation can arise from technical artifacts (tissue edge effects, permeabilization gradients) or from correlated but non-causal biological processes.

The deconvolution of multi-cell spots is a statistical inference, not a measurement. Results depend heavily on the quality and relevance of the scRNA-seq reference used. A deconvolution that assigns 30% T-cells to every tumor spot likely reflects a reference artifact (e.g., shared stress-response genes between T-cells and tumor cells) rather than true T-cell infiltration. Validating deconvolution results against independent methods (immunohistochemistry, flow cytometry) is essential but rarely done.

Spatial autocorrelation (Moran's I) identifies spatially variable genes, but biological interpretation requires distinguishing genuine compartment-specific expression from technical artifacts. A gene with high Moran's I that encodes a known structural protein of the extracellular matrix is plausibly compartment-specific; a gene with high Moran's I that is uniformly expressed in bulk RNA-seq is more likely an artifact of uneven tissue permeabilization.

Summary and Required Reading

1. **Spatial transcriptomics faces a fundamental tradeoff:** whole-transcriptome coverage (Visium, Slide-seq) vs. single-cell resolution (MERFISH, Xenium). Technology choice determines what questions can be addressed.
2. **Deconvolution (RCTD, Cell2location, CARD)** estimates cell type composition per spot using scRNA-seq references; results depend critically on reference quality and relevance.
3. **Moran's I quantifies spatial autocorrelation** of gene expression; **ligand-receptor inference** identifies candidate cell-cell communication. Both require validation against artifacts.
4. **Registration aligns sequencing data to histology** via affine transformations; registration quality directly determines anatomical interpretability.
5. **Spatial data is a hypothesis generator:** correlations between expression patterns and tissue compartments suggest function but do not prove it. Perturbation experiments are required for confirmation.

Required Reading

- Ståhl et al.: "Visualization and analysis of gene expression in tissue sections by spatial transcriptomics" (*Science*, 2016).
- Cable et al.: "Robust decomposition of cell type mixtures in spatial transcriptomics" (*Nature Biotechnology*, 2022).

Johnson's Rule: Without spatial context, single-cell data is headless; you know the molecules but not the anatomy.

Chapter 18: Systems Biology and Network Graphing

No Gene Acts Alone

A protein that physically interacts with 100 other proteins is biologically more important than a protein that interacts with 3, regardless of fold-change. A transcription factor at the junction of DNA damage sensing and apoptosis is central to chemotherapy response, even if its expression never changes. Phenotype emerges from network interactions, not isolated gene lists.

The differential expression analysis from Chapters 4 and 6 produces lists of genes ordered by p-value. But biology does not operate through lists; it operates through networks. A gene with moderate fold-change but high network centrality may be more biologically relevant than the most differentially expressed gene, which could be a downstream bystander. This chapter shifts the unit of analysis from individual genes to the relationships between them.

Core Concepts

Network Theory: Nodes, Edges, and Centrality

A biological network $G = (V, E)$ consists of nodes V (genes, proteins) and edges E (interactions). Edge types vary by data modality: physical binding (protein-protein interaction), co-expression correlation (transcriptomics), genetic interaction (CRISPR screens), or phosphorylation (kinase-substrate).

Three centrality metrics quantify node importance in fundamentally different ways, and each has a distinct biological interpretation:

- **Degree:** The number of edges incident to a node. High-degree nodes (hubs) are connected to many partners. **Biological meaning:** Essential genes tend to have higher degree. Knockout of a hub is more likely to be lethal because it affects more pathways simultaneously. Degree is the simplest metric but highly sensitive to literature bias; well-studied proteins have more reported interactions regardless of biological centrality.

- **Betweenness:** The fraction of shortest paths passing through a node. High-betweenness nodes are bottlenecks; their removal fragments the network into disconnected components. **Biological meaning:** Bottlenecks are often signaling integrators (e.g., TP53, AKT1) that sit at the convergence of multiple upstream pathways and diverge to multiple downstream effectors. A bottleneck protein is a particularly effective drug target because perturbing it disrupts the information flow through the entire pathway.
- **Eigenvector centrality:** A recursive definition; a node is important if it is connected to other important nodes (the PageRank analogue). **Biological meaning:** Eigenvector centrality captures position within the network hierarchy. A transcription factor connected to highly central signaling proteins ranks higher than one connected to isolated peripheral proteins, even if their degrees are equal.

Weighted Gene Co-expression Network Analysis (WGCNA)

WGCNA (Langfelder & Horvath, 2008) constructs a gene co-expression network from expression data. Unlike PPI networks which use curated physical interactions, WGCNA infers networks from correlations in expression data across samples:

1. **Correlation matrix:** Compute pairwise Pearson correlations between all gene pairs across samples. For 20,000 genes, this produces a $20,000 \times 20,000$ correlation matrix; 200 million correlations.
2. **Soft thresholding:** Raise the absolute correlation to a power β (typically 6-12): $a_{ij} = |\text{cor}(x_i, x_j)|^\beta$. This suppresses weak correlations toward zero while preserving the rank order of moderate-to-strong correlations. Why not use a hard threshold (keep only correlations > 0.8 , set others to zero)? A hard threshold creates a discontinuity: a correlation of 0.79 is zero while 0.81 is full strength, which is biologically arbitrary and makes the network unstable to small perturbations. Soft thresholding preserves the continuous nature of co-expression.

The power β is not an arbitrary parameter. WGCNA evaluates a range of powers (typically 1-20) and selects the lowest β where the network approximately follows scale-free topology, assessed by the R^2 of the linear regression $\log_{10} P(k) \sim \log_{10} k$; the degree distribution should follow a power law $P(k) \sim k^{-\gamma}$ with $R^2 > 0.8$. Powers below this threshold produce networks that are too dense (all genes appear connected) to reveal modular structure. Higher β punishes weaker correlations more aggressively, producing smaller, tighter modules at the cost of potentially fragmenting genuine biological pathways.

The degree distribution diagnostic plot (log-log of degree vs. frequency) is the first quality check in any WGCNA analysis. A downward-curving rather than linear plot indicates the data lacks scale-free structure, and WGCNA modules may be artifactual; this is common in datasets with strong batch effects or low sample sizes (typically $n < 15$).

3. **Topological overlap matrix (TOM):** Transform the weighted adjacency matrix $A = [a_{ij}]$ to TOM, where the similarity between two genes accounts for their shared neighbors. Formally:

$$w_{ij} = \frac{l_{ij} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}, \quad l_{ij} = \sum_u a_{iu} a_{uj}, \quad k_i = \sum_u a_{iu}$$

where $a_{ij} = |\text{cor}(x_i, x_j)|^\beta$ is the soft-thresholded adjacency, l_{ij} sums the products of adjacency weights through all shared neighbors u (indirect connectivity), and k_i is the total connectivity of node i (the sum of all edge weights incident to it). The numerator combines direct adjacency (a_{ij}) with indirect connectivity through shared neighbors (l_{ij}). The denominator normalizes by the minimum connectivity of the two nodes, preventing hub genes from achieving artificially high TOM with all network members simply because they are highly connected.

Biologically, TOM captures the idea that genes participating in the same pathway are connected through a chain of intermediate genes; the direct correlation may be weak, but if they share most of their network neighbors, they are functionally related. A concrete example: two genes with direct correlation of only 0.3 but each co-expressed with the same 15 intermediate pathway genes will have a high TOM, correctly placing them in the same module despite the weak direct correlation.

4. **Hierarchical clustering:** Cluster TOM dissimilarity ($1 - \text{TOM}$) into modules. Each module is a group of genes with high topological overlap, interpreted as a pathway or functional unit.
5. **Module-trait correlation:** Reduce each module to its eigengene (first principal component of all module genes' expression), then correlate the eigengene with external traits (disease status, treatment response). This identifies which modules are disease-relevant.

The eigengene is the first principal component (PC1) of the module's expression matrix, directly applying the SVD framework from Chapter 5. By the Eckart-Young theorem, PC1 captures the maximum possible variance explained by a single vector; it is the optimal one-dimensional summary of the module's coordinated expression pattern. The fraction of total variance explained by the eigengene (the first singular value squared divided by the sum of all squared singular values) quantifies **module coherence**: a module whose eigengene explains >50% of its total variance is highly coordinated, while one explaining <30% is loosely associated and warrants examination for spurious clustering of unrelated genes.

The scale-free assumption. WGCNA assumes biological networks follow a scale-free topology: $P(k) \sim k^{-\gamma}$, where a few hubs dominate. This is approximately true for many but not all biological networks. A network with uniform degree distribution (e.g., a metabolic pathway where every enzyme interacts equally with its immediate neighbors) is not scale-free, and WGCNA's soft thresholding will produce spurious modules in such cases. The scale-free fit ($R^2 > 0.8$) should be verified before interpreting modules.

Protein-Protein Interaction Networks

PPI networks (STRING, BioGRID) catalog physical protein-protein interactions from experimental and computational sources. STRING scores interactions from 0-1, integrat-

ing evidence from co-expression, co-occurrence in the literature, text mining, and high-throughput screens. The score is not a confidence in a specific interaction but an aggregate of heterogeneous evidence; two proteins may have a STRING score of 0.9 because they are consistently co-expressed, not because they physically bind.

Network propagation (HotNet2). Given a set of seed genes (e.g., significantly mutated genes from a cancer study), propagate signal through the network to identify additional genes that are functionally connected:

$$F^{t+1} = \alpha A F^t + (1 - \alpha) F^0$$

where F is the heat vector (node relevance), A is the normalized adjacency matrix, and α is the restart probability (typically 0.5-0.8). At each step, a fraction α of the heat diffuses to neighboring nodes, and $(1 - \alpha)$ returns to the seed nodes. After convergence, nodes with high heat are prioritized as network-influential, even if they were not in the seed set.

The biological rationale: mutated cancer genes form subnetworks of interacting proteins that collectively drive tumorigenesis. A gene that is not itself mutated but is highly connected to mutated genes may be functionally important; its pathway is disrupted even if its sequence is not. Network propagation identifies such “passenger-by-proximity” genes.

Guilt-by-Association

Uncharacterized genes are assigned function based on their network neighbors. If a gene of unknown function is connected to three ribosomal proteins, it is likely involved in translation. This principle is the basis for most automated functional annotation pipelines (GO term prediction, gene function databases).

The limitation is circularity: well-studied genes have more known interactions and will appear as higher-priority nodes regardless of biological relevance. A gene connected to TP53 is not necessarily functionally related to TP53; it may be connected because TP53 is the most studied protein in biology and interacts with many partners. Network propagation amplifies this bias: seeds are typically well-studied genes, and heat propagates to their well-studied neighbors. Comparing results against degree-matched controls (randomly selecting hubs as seeds) quantifies the extent of literature bias in the propagation result.

Pathway Enrichment Analysis: ORA, GSEA, and the Background Gene Set Pitfall

Once you have identified differentially expressed genes or network modules, you must translate these lists of gene symbols into biological pathways.

- **Over-Representation Analysis (ORA):** Uses the hypergeometric distribution (Fisher’s exact test) to determine if a pathway is enriched in your gene list compared to what you would expect by chance.

- **The Background Gene Set Pitfall:** This is the most common statistical error in enrichment analysis. When running Fisher's exact test, you must specify the "background gene set" (the gene universe). If you default to using the entire human genome (~20,000 genes) as background when your experiment (e.g., cell-type-specific RNA-seq or a targeted proteomics panel) was only capable of detecting 10,000 genes, your analysis is invalid. Using an artificially large background deflates the probability of random overlap, creating highly inflated, false-positive p-values for common pathways. Always restrict your background set to only those genes that were actually detected or assayed in your experiment.
 - **Gene Set Enrichment Analysis (GSEA):** Avoids arbitrary differential expression thresholds by ranking the entire transcriptome based on a metric (e.g., signal-to-noise ratio or Wald statistic) and calculating a running enrichment score (ES) using a Kolmogorov-Smirnov-like statistic. GSEA captures subtle, coordinated shifts in entire pathways that single-gene tests miss.
-

Biological Interpretation

The hub gene of a disease module is a better drug target than the most statistically significant gene; this is the central insight of systems biology. A gene with moderate fold-change but high network centrality may be more biologically relevant than the most differentially expressed gene, which could be a downstream bystander. But this principle presupposes that the network reflects functional biology, not literature bias.

WGCNA modules should be validated for biological coherence (enrichment of known pathways, consistency across independent datasets) before interpretation. A module that correlates with disease but contains genes from multiple unrelated pathways may be driven by a confound (batch effect, sample quality) rather than biology. The module eigengene-trait correlation is a single number with no inherent biological meaning; it must be interpreted in the context of the module's functional annotation.

Network propagation amplifies signal but also amplifies bias: well-studied genes accumulate more known interactions and consistently rank higher in propagation analyses. This literature bias is systematic and must be accounted for by comparing against degree-matched controls. A propagation result that ranks only well-studied genes should be interpreted as a validation of known biology, not a discovery.

The contrast with Chapter 6 is instructive. Differential expression identifies individual genes by statistical significance; network analysis identifies functional modules by connectivity. These approaches are complementary: differential expression identifies the components that change, and network analysis reveals the system architecture within which those changes operate. A differentially expressed gene that is a network hub is more biologically interesting than one that is isolated.

Summary and Required Reading

1. **Network centrality metrics** (degree, betweenness, eigenvector) quantify node importance in fundamentally different ways; each has a distinct biological interpretation.
2. **WGCNA** constructs weighted co-expression networks via soft thresholding (preserving continuous edge weights) and TOM (capturing shared-neighbor relationships), identifying disease-relevant modules.
3. **Network propagation** prioritizes genes connected to known drivers via heat diffusion, but amplifies literature bias; well-studied genes consistently rank higher.
4. **Guilt-by-association** assigns function from network neighbors; powerful for novel genes but circular for well-studied ones.
5. **Differential expression and network analysis are complementary**: DE identifies the components that change; networks identify the system architecture they operate within.

Required Reading

- Langfelder & Horvath: "WGCNA: an R package for weighted correlation network analysis" (*BMC Bioinformatics*, 2008).
- Szklarczyk et al.: "STRING v11: protein-protein association networks" (*Nucleic Acids Research*, 2019).

Johnson's Rule: The hub gene of a disease module is a better drug target than the most statistically significant gene.

Chapter 19: Multi-Omics Integration

Integrating Bad Data from Two Assays Yields Twice as Much Bad Data

Intersecting lists of statistically significant genes from RNA-seq and proteomics does not integrate data; it compounds error rates. The overlap is dominated by the most highly expressed, easiest-to-measure genes, not by the biologically most relevant ones. True integration requires dimensionality reduction algorithms that mathematically identify shared variance between molecular layers, independent of arbitrary significance thresholds.

A gene with p-value 10^{-10} in RNA-seq and p-value 10^{-10} in proteomics may appear to be a biologically validated hit. But if the top 100 genes from each dataset overlap by only 20 genes, and those 20 are the most abundant housekeeping genes, the “validated” list is dominated by measurement convenience rather than biological importance. This chapter covers the methods that avoid this trap.

Core Concepts

The Three Axes of Integration

- **Horizontal integration:** Same data modality, different batches or cohorts (e.g., RNA-seq dataset A + RNA-seq dataset B). Challenge: batch effects (Chapter 6). Correction: ComBat, Harmony, scVI for single-cell data. This is the simplest integration axis because all data share the same distributional assumptions.
- **Vertical integration:** Different modalities, same biological samples (e.g., RNA-seq + ATAC-seq + proteomics on the same cells). Challenge: different data distributions (Poisson for RNA-seq, Gaussian for log-processed proteomics, binomial for ATAC-seq) and different missingness mechanisms (dropout in scRNA-seq vs. detection limits in proteomics). This is the axis where MOFA and CCA are most useful, because the shared sample structure constrains the solution.
- **Diagonal integration:** Different modalities, different samples (e.g., scRNA-seq from one study + spatial transcriptomics from another). Challenge: no paired measurements exist, so integration relies on shared features (genes) and assumed biological similarity between datasets. Methods must infer a joint latent space without any direct correspondence between samples. The most challenging and most common real-world integration scenario; no experiment measures everything in every cell.

Selecting the Integration Method

The choice of integration method depends primarily on whether you have paired samples and whether you seek shared or complementary variation:

- **If you have paired samples and want shared variation:** MOFA+ factorizes the data into shared and modality-specific factors, with interpretable variance decomposition per factor per modality. The output tells you “how much of RNA variance and how much of protein variance is explained by this shared factor.”
- **If you have paired samples and want correlational structure:** CCA finds linear combinations that are maximally correlated across modalities. The output tells you “which RNA and protein measurements co-vary across samples.”
- **If you have unpaired samples (diagonal integration):** Seurat’s anchor-based CCA or optimal transport methods (FGOT, MultiVIT) infer a matching between datasets based on shared features. These do not require any paired observations.
- **If you need causal inference:** eQTL or pQTL analysis (below) uses genetic variants as instrumental variables.

Canonical Correlation Analysis (CCA)

CCA finds linear combinations of variables from two datasets that are maximally correlated with each other. For two data matrices X and Y (e.g., RNA expression of 20,000 genes and ATAC peaks at 100,000 regions across the same 100 samples), CCA finds vectors w_x and w_y such that:

$$\rho = \max_{w_x, w_y} \frac{w_x^T X^T Y w_y}{\sqrt{w_x^T X^T X w_x} \sqrt{w_y^T Y^T Y w_y}}$$

The numerator $w_x^T X^T Y w_y$ measures the cross-covariance between the two linear combinations; the denominator normalizes by the variance of each combination. The result ρ is the canonical correlation; the Pearson correlation between the two projections.

The $p \gg n$ problem. The CCA objective requires inverting $X^T X$ and $Y^T Y$, which are $p \times p$ covariance matrices. With 20,000 genes and 100 samples, these matrices are rank-deficient (singular); the optimization has no unique solution because the sample covariance cannot reliably estimate $p(p+1)/2 \approx 200$ million unique entries from only 100 observations. In practice, this is resolved either by applying PCA to each modality first (reducing each to 30–50 PCs, as established in Chapter 5, then running CCA on the PC scores) or by using regularized CCA with a ridge penalty to stabilize the covariance estimates. The Seurat integration pipeline uses the PCA-then-CCA approach: it decomposes each dataset via SVD, projects into PC space, and finds canonical correlates in the compressed latent space where $n > p$.

The housekeeping trap. CCA components with near-perfect cross-modality correlation ($\rho > 0.9$) often reflect technical artifacts rather than biological coordination. If both modalities are dominated by abundant housekeeping genes (GAPDH, ACTB), the first

CCA component may simply capture “both measurements are working correctly” rather than a biological signal. The more informative components are typically the ones with moderate correlation ($\rho \sim 0.3 - 0.7$) that capture specific biological processes.

Multi-Omics Factor Analysis (MOFA)

MOFA (Argelaguet et al., 2020) extends factor analysis to multiple data modalities:

$$Y_m = ZW_m^T + \epsilon_m$$

where Y_m is the data matrix for modality m , Z is the shared factor matrix (across all modalities), W_m is the modality-specific weight matrix, and ϵ_m is modality-specific noise. The factors Z capture the sources of variation shared across data types. Each factor is a latent variable that explains a certain amount of variance in each modality.

The sparsity prior that makes MOFA interpretable. MOFA places an automatic relevance determination (ARD) prior on each weight column $W_{:,k}^{(m)}$, which shrinks irrelevant factors toward zero in a modality-specific manner. A factor that captures biological variation present in RNA but absent in ATAC will have large weights in the RNA modality and near-zero weights in the ATAC modality; the ARD prior automatically prunes it from ATAC. This is what produces the interpretable variance decomposition that distinguishes MOFA from standard factor analysis, where every factor loads on every modality with no built-in sparsity mechanism.

Key outputs:

1. **Variance decomposition per factor per modality:** “Factor 1 explains 30% of RNA variance and 50% of ATAC variance.” This tells you which modalities contribute to each biological process.
2. **Factor values per sample:** Each sample gets a score for each factor, enabling clustering, time-series analysis, or association with clinical outcomes.
3. **Feature weights per factor:** For each factor, the genes, peaks, or proteins with the highest absolute weights are the ones driving that factor.

The dominant-factor caveat. The first factor in any MOFA analysis is almost always a technical confound (batch, age, sequencing depth, sample quality). This is not a bug; it reflects the reality that technical variation often dominates biological variation across modalities. Before interpreting factors biologically, verify that they correlate with the variable of interest and not with known confounders. The number of truly informative biological factors is typically small (3-10), regardless of the number of input modalities.

eQTL and pQTL: Genetic Anchors for Causal Inference

Expression quantitative trait loci (eQTL) use genetic variants as causal anchors to infer that gene expression changes are causal rather than confounded. The logic: if a SNP is associated with both gene expression (as an eQTL) and disease (as a GWAS hit), and the eQTL effect is plausibly causal (the variant affects a regulatory element), then the gene

is a candidate causal mediator of the disease association. This is Mendelian randomization; the genetic variant is randomized at conception, independent of confounders like lifestyle or environment.

cis-eQTL vs. trans-eQTL. A cis-eQTL affects expression of a nearby gene (typically within 1 Mb of the variant). These are the most common and interpretable; the variant likely lies in a promoter, enhancer, or other regulatory element of that gene. A trans-eQTL affects expression of a distant gene (different chromosome or >1 Mb away). These are rarer, harder to validate, and typically have smaller effect sizes, but they reveal regulatory networks that cross chromosomal boundaries.

The GTEx consortium provides the most comprehensive eQTL atlas: ~50 tissues, >800 donors, millions of cis-eQTL associations. The key finding: most GWAS variants are enriched in non-coding regulatory regions and act through gene expression regulation, not protein-coding changes. For a typical complex disease, ~60-80% of GWAS loci have a nearby eQTL in the relevant tissue, implicating gene expression dysregulation as the causal mechanism.

pQTL extends this logic to proteins. Protein QTLs are more directly relevant to phenotype (proteins are the functional molecules) but are harder to identify due to smaller sample sizes in proteomics studies and higher measurement variability. The overlap between eQTL and pQTL signals is only ~20-40%; consistent with the mRNA-protein decoupling established in Chapter 17. A variant that affects protein abundance without affecting mRNA is evidence of post-transcriptional regulation.

Mendelian randomization assumptions. The causal interpretation of eQTL-mediated GWAS colocalization rests on three assumptions that are violated more often than satisfied: (1) **no horizontal pleiotropy**: the genetic variant affects the disease only through the gene, not through independent pathways; (2) **no linkage disequilibrium confounding**: the true causal variant is the eQTL, not a correlated variant in LD that affects disease through a different gene; and (3) **reliable exposure prediction**: the variant is a strong predictor of gene expression (the F -statistic for the variant-expression association exceeds 10). When these assumptions fail, colocalization produces a false causal inference: the variant and the gene expression appear causally linked to the disease when they are independent correlates of a hidden third factor.

Weighted Nearest Neighbor (WNN) and the Integration Benchmark

For single-cell multi-omic datasets where multiple modalities are measured in the *same* physical cells (e.g., 10x Multiome: RNA + ATAC; CITE-seq: RNA + Surface Protein), we do not need to align separate cells. Instead, we must construct a single joint representation of cell similarity that weights the relative contribution of each modality per cell.

- **The WNN Algorithm:** Seurat's Weighted Nearest Neighbor (WNN) calculates cell-specific modality weights. For each cell, it computes its nearest neighbors in RNA space and predicts its protein/ATAC expression, and vice versa. If a modality's neighbors accurately predict the other modality's expression, that modality gets a high weight for that cell. The cell's final distance to its neighbors is the weighted sum of its distances in the individual modality spaces, enabling the identification of cell states defined by a combination of both signals (e.g., distinguishing CD4+ and

CD8+ memory T cells using surface proteins when their mRNA profiles are nearly identical).

- **The Integration Benchmark:** When integrating different cells across datasets (diagonal or horizontal multi-omic alignment), researchers frequently rely on anchor-based alignment (like Seurat's `FindTransferAnchors` or standard CCA). However, you must approach these methods with extreme skepticism. A rigorous **2026 benchmark study (Genome Biology)** evaluated multi-modal single-cell integration algorithms on complex tissues. The authors found that standard anchor-based workflows mislabeled up to **45% of cells** in complex lineages, leading to a staggering **59% false-positive rate** in downstream cis-regulatory network inferences. The benchmark demonstrated that **optimal transport-based alignment** (which treats integration as an entropy-regularized probability matching problem) significantly outperformed heuristic anchoring, preserving true lineage boundaries without over-aligning distinct populations.
-

Biological Interpretation

Before integrating, ask: does this integration solve a biological question that single-omics cannot? The most successful multi-omics studies ask questions that are inherently multi-modal: "Which chromatin accessibility changes drive expression changes?" (requires ATAC + RNA), "Which post-translational modifications regulate protein activity independently of expression?" (requires proteomics + phosphoproteomics), or "Which genetic variants regulate gene expression in disease-relevant tissues?" (requires GWAS + eQTL).

CCA components with near-perfect cross-modality correlation should be examined critically; they may reflect technical artifacts (sample quality, batch, sequencing depth) rather than biological coordination. The number of truly informative factors is typically small (3-10), regardless of the number of modalities.

The dominant factor in any multi-omics analysis is almost always a technical confound (batch, age, sequencing depth). Before interpreting factors biologically, verify that they correlate with the variable of interest and not with known confounders. A MOFA factor that strongly separates cases from controls but also separates batch 1 from batch 2 is dominated by batch; the biological and technical signals are confounded, and the factor cannot be interpreted as biological.

Summary and Required Reading

1. **Three integration axes:** horizontal (same modality, different batches) uses batch correction (`ComBat`, `Harmony`); vertical (different modalities, same samples) uses MOFA/CCA; diagonal (different modalities, different samples) uses optimal transport or anchor-based methods.

2. **CCA** finds maximally correlated linear combinations; the foundation of Seurat integration; caution: first components often capture technical artifacts, not biology.
3. **MOFA** factorizes variance into shared and modality-specific components with interpretable variance decomposition per factor per modality.
4. **eQTL/pQTL** use genetic variants as causal anchors linking genotype to molecular phenotype; the ~20-40% overlap between eQTL and pQTL signals reflects post-transcriptional regulation.
5. **The dominant factor is almost always technical:** verify factor-biological associations are not driven by batch, depth, or sample quality before biological interpretation.

Required Reading

- Argelaguet et al.: "MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data" (*Genome Biology*, 2020).
- GTEx Consortium: "The GTEx Consortium atlas of genetic regulatory effects across human tissues" (*Science*, 2020).

Johnson's Rule: If this integration does not solve a question that single-omics cannot, you are adding complexity, not insight.

Chapter 20: 3D Genome Architecture

The Genome is Folded in Three Dimensions

Linear proximity on a chromosome is a false indicator of regulatory interaction. A promoter and its enhancer may be separated by 500 kilobases of linear sequence but brought into physical contact through chromatin looping. The linear genome is a coordinate system; the 3D genome is the functional reality.

Chapter 12 introduced the epigenome as software running on the DNA hardware. But software execution depends not just on which proteins are bound and which regions are accessible; it depends on which genomic elements are physically brought together in 3D space. An enhancer cannot activate a promoter if it is in a different physical compartment of the nucleus, regardless of how accessible the chromatin is at each element individually. Three-dimensional genome architecture is the spatial constraint that regulatory logic operates within.

Core Concepts

Proximity Ligation and Hi-C Physics

Hi-C measures physical proximity across the genome. The biochemical logic is simple: crosslink spatially adjacent chromatin with formaldehyde (fixing proteins and DNA that are within ~1 nm of each other), digest with a restriction enzyme (typically HindIII or MboI), ligate the crosslinked fragments back together (creating chimeric junctions between loci that were physically close in 3D space), reverse crosslinks, and sequence the junctions. Each sequencing read pair maps to two different genomic loci; the closer they are in 3D space, the more frequently they are ligated together.

The output is a contact matrix M_{ij} , where each entry records the number of ligation events observed between locus i and locus j across millions of cells. The matrix is symmetric ($M_{ij} = M_{ji}$) because proximity ligation is bidirectional; only the upper triangle contains unique information. For a genome binned at 1 Mb resolution, this produces a $3,000 \times 3,000$ contact matrix. At the highest sequencing depth (5-10 billion reads), kilobase-resolution contact maps are possible in targeted regions.

Resolution scales with sequencing depth. Resolution is defined by the bin size, and the required read depth scales quadratically as bins shrink: 1 Mb bins need ~20 million read pairs, 100 kb bins need ~200 million, and 10 kb bins need ~5 billion. Halving the bin size doubles the number of bins along each axis, quadrupling the total entries; most of which remain sparse at high resolution. In practice, compartment analysis uses 100–500 kb bins, TAD boundaries are visible at 10–50 kb, and individual loop detection requires 5–10 kb bins with correspondingly deep sequencing.

The decay law and normalization. Contact frequency decays with genomic distance following a polymer physics scaling law: $M_{ij} \sim |i - j|^{-1}$ for interphase chromatin (or $|i - j|^{-2}$ for mitotic chromosomes). This means that nearby loci (1–10 kb apart) always have high contact frequency regardless of biological loop formation. This distance-dependent background must be removed before biological signal can be interpreted. The standard approach is **observed-over-expected** (obs/exp) normalization: for each genomic distance $s = |i - j|$, compute the expected contact frequency as the average over all locus pairs at that distance, then divide each entry by its distance-matched expectation. After obs/exp normalization, a value > 1 means “more contacts than expected given the genomic distance between the two loci”; this is how TAD boundaries and loop anchors appear as enriched features in the normalized matrix.

Iterative Correction and Eigenvector decomposition (ICE) and Knight-Ruiz (KR) matrix balancing normalize the contact matrix by removing systematic biases: GC content bias (GC-rich regions are more mappable), restriction site density bias (fragments with more restriction sites produce more ligation products), and fragment length effects (longer fragments have higher ligation probability). Both methods model biases as multiplicative factors: the observed contact count M_{ij} equals the true contact frequency T_{ij} multiplied by bias factors b_i and b_j for each locus ($M_{ij} = T_{ij} \cdot b_i b_j$). Under the assumption that all loci have equal “visibility” in the absence of biological structure, the correction solves for b_i such that each row sum becomes uniform. After dividing by the bias factors, the normalized matrix $M_{ij}/(b_i b_j)$ reflects biological contact frequency independent of sequence-level biases. KR is faster; ICE handles sparser matrices better.

Topological Structures

The normalized contact matrix reveals hierarchical 3D structures at three scales:

A/B Compartments are the largest scale (multimegabase, typically 5–100 Mb). The first principal component of the normalized contact matrix (or the first eigenvector of the correlation matrix) partitions the genome into two states: A compartments are gene-rich, transcriptionally active, early replicating, and enriched for activating histone marks (H3K27ac, H3K4me3). B compartments are gene-poor, transcriptionally silent, late replicating, and enriched for repressive marks (H3K9me3, H3K27me3). The compartment signal is visible as a “plaid” pattern in the contact matrix; checkerboard enrichment indicating that A loci preferentially contact other A loci, and B loci preferentially contact other B loci.

Topologically Associating Domains (TADs) are megabase-scale (typically 200 kb – 2 Mb) regions within which contact frequency is enriched relative to the background. TADs are visible as triangular blocks along the diagonal of the contact matrix. TAD boundaries

are sharply demarcated by convergent CTCF motifs bound by CTCF and cohesin. Disruption of TAD boundaries (by deletion of the CTCF site or mutation of CTCF itself) causes ectopic enhancer-promoter contacts across the boundary, leading to misregulation of developmental genes. This mechanism is directly implicated in developmental disorders (such as limb malformations from boundary deletions in the *EPHA4* locus) and cancer (TAD boundary disruptions causing oncogene activation by distal enhancers).

Enhancer-Promoter loops are sub-Mb contacts identified by significant contact enrichment relative to the local TAD background. These are the functional units of gene regulation; a physical connection between a distal enhancer and its target promoter through the cohesin-CTCF extrusion machinery. Methods for loop identification (Fit-Hi-C, HiCCUPS, Mustache) differ primarily in how they model the background distribution: Fit-Hi-C fits a spline to the distance-decay curve and tests each contact against it; HiCCUPS searches for local enrichment relative to the surrounding region in the contact matrix; Mustache uses a binomial model with FDR control.

Loop Extrusion Model

The dominant mechanistic model for loop and TAD formation: the cohesin complex loads onto chromatin and actively extrudes a loop, reeling the chromatin fiber through its ring structure until it encounters CTCF proteins bound in a specific orientation (convergent motifs). The CTCF-cohesin interaction stalls extrusion, creating a stable loop anchored at the two CTCF sites. This model explains three key observations:

1. **Directional bias:** CTCF motifs at loop anchors are almost exclusively in convergent orientation ($\rightarrow \leftarrow$), never divergent ($\leftarrow \rightarrow$). The extrusion model predicts this; cohesin can only be stopped by CTCF when it encounters the motif from the correct direction.
2. **Nested hierarchy:** TADs contain sub-TADs which contain finer loops. The extrusion model explains this as partial extrusion; a loop can be stalled at early CTCF sites or continue to more distant ones.
3. **Size distribution:** Loops show a characteristic distribution peaking at 100-500 kb, consistent with the processivity of cohesin extrusion before stalling.

TAD Boundary Detection: The Insulation Index

To identify the coordinates of TAD boundaries from a normalized Hi-C contact matrix, we use the **Insulation Index** algorithm (Crane et al., 2015).

- **The Insulation Principle:** Since TADs are self-interacting domains with high internal contact frequency but low contact frequency crossing their boundaries, the boundary represents an “insulator” of chromatin interaction.
- **The Diamond Window Operator:** The algorithm slides a square “diamond” window (typically 100-200 kb on a side) along the diagonal of the contact matrix. At each genomic bin i , the insulation score is the sum of all contact frequencies within the diamond centered at that bin.
- **Local Minima Detection:** Within a TAD, the score is high because the diamond captures dense intra-domain contacts. When the diamond sits directly on a TAD

boundary, the score drops to a local minimum because it captures the depleted inter-domain contacts crossing the boundary. The algorithm identifies TAD boundaries as the local minima of this insulation score curve, using a delta threshold (difference between the minimum and adjacent peaks) to control for noise.

Biological Interpretation

A contact matrix is a population average across millions of cells. A strong contact in bulk Hi-C may represent a constant loop in all cells, or a statistically significant contact that occurs in only 5% of cells but is enriched far above the background expectation. Single-cell Hi-C and imaging-based methods (DNA-MERFISH, Oligopaint) have shown that TADs and compartments are stochastic at the single-cell level: any given cell has only a fraction of the loops detected in bulk, but the population average converges to the structures we call TADs.

The biological interpretation question is not “does this loop exist?” but “in which cells, at what frequency, and under what conditions does this loop form?” Loops that are constitutive (present in most cells across conditions) are likely structural; they organize the genome into a stable architecture. Loops that form only in specific cell types or after stimulation are likely regulatory; they mediate cell-type-specific or signal-responsive gene expression.

A/B compartment status correlates with transcriptional activity, but the direction of causality is bidirectional. Active transcription can drive a locus into the A compartment (transcription opens the region), and compartment assignment can influence transcription (spatial proximity to active regions promotes expression). Disentangling directionality requires perturbation experiments, for example, tethering a locus to the nuclear lamina (forced B compartmentalization) and measuring the transcriptional effect.

The linear-to-3D mapping challenge is substantial. A GWAS variant 500 kb from the nearest gene may regulate that gene through a loop, or it may regulate a completely different gene brought into contact by the 3D architecture. Variant-to-function mapping that only considers linear distance to the nearest TSS will systematically misattribute non-coding variants to the wrong target gene.

Summary and Required Reading

1. **Hi-C measures 3D proximity via proximity ligation:** contact frequency decays as $\sim 1/s$ with distance; ICE/KR normalization removes sequence biases to reveal biological structure.
2. **Three hierarchical structures:** A/B compartments (multimegabase, active vs. silent), TADs (megabase, regulatory neighborhood), enhancer-promoter loops (sub-Mb, functional contacts).

3. **The loop extrusion model** explains convergent CTCF motif orientation, nested TAD hierarchy, and loop size distribution through cohesin-mediated loop extrusion stalled by CTCF.
4. **Bulk Hi-C is a population average:** single-cell methods reveal that TADs and loops are stochastic, not deterministic, structures.
5. **3D genome interpretation requires perturbation:** compartment status and transcription have bidirectional causality; static measurements cannot resolve direction.

Required Reading

- Lieberman-Aiden et al.: "Comprehensive mapping of long-range interactions reveals folding principles of the human genome" (*Science*, 2009).
- Rao et al.: "A 3D map of the human genome at kilobase resolution reveals principles of chromatin looping" (*Cell*, 2014).
- Fudenberg et al.: "Formation of chromosomal domains by loop extrusion" (*Cell Reports*, 2016).

Johnson's Rule: Linear distance does not equal regulatory distance. If you are only looking at genes and their nearest neighbors, you are missing the regulatory architecture.

Chapter 21: Perturb-seq and Functional Genomics

Correlation is Not Causation

Observational multi-omics identifies correlation. A gene co-expressed with a phenotype across 500 patients may be a driver, a bystander, or a compensatory response. Only direct physical perturbation (knocking out the gene and measuring the consequence) can definitively map causal biological dependency networks.

The preceding chapters of this tier have built a sophisticated toolkit for observing molecular states. Chapter 16 taught you to build networks from correlations; Chapter 17 to measure proteins; Chapter 18 to integrate across modalities. But every one of these methods is fundamentally observational. You can correlate expression with phenotype, build co-expression networks, and integrate across omics layers, and still not know which genes are causal. Perturbation resolves this ambiguity by changing the system and measuring the response.

Core Concepts

CRISPR-Cas9 Biology

The CRISPR-Cas9 system: a guide RNA (gRNA) directs the Cas9 nuclease to a specific genomic locus via sequence complementarity. Cas9 creates a double-strand break ~3 bp upstream of the protospacer adjacent motif (PAM, NGG for SpCas9). The break is repaired by either:

- **Non-homologous end joining (NHEJ):** Error-prone repair that introduces small insertions or deletions (indels). When the indel shifts the reading frame (frameshift), the result is a gene knockout via premature stop codon and nonsense-mediated decay. ~65-80% of indels are frameshifts.
- **Homology-directed repair (HDR):** High-fidelity repair using a template. Requires co-delivery of a repair template (ssODN or plasmid). HDR efficiency is low (1-10% in most cell types) and cell-cycle dependent (G2/S phase only).

gRNA design principles. The gRNA must be uniquely complementary to the target gene and not match anywhere else in the genome. Design rules: target early constitutive ex-

ons (frameshifts in early exons are more likely to produce complete loss of function), avoid exons with alternative splicing or partial functional redundancy, and design multiple gRNAs per gene (3-5) to distinguish on-target effects from off-target edits. Tools: CRISPick (Broad Institute), CasOFFinder for off-target prediction.

Pooled CRISPR Screens

A pooled screen scales perturbation to the entire genome. A library of ~100,000 gRNAs targeting all ~20,000 genes (5 gRNAs per gene, plus negative controls) is synthesized as an oligonucleotide pool, cloned into a lentiviral vector, packaged into lentivirus, and transduced into cells at low multiplicity of infection ($\text{MOI} < 0.3$) to ensure most transduced cells receive exactly one gRNA. The number of gRNAs per cell follows a Poisson distribution (Chapter 4) with parameter $\lambda = \text{MOI}$. At $\text{MOI} = 0.3$, the probability of receiving two or more gRNAs is $P(k \geq 2 \mid \lambda = 0.3) \approx 0.04$; acceptably low for screens where assigning phenotype to a single perturbation is essential.

Screen types:

- **Positive selection (enrichment):** gRNAs that confer a growth advantage become enriched over time. Used to identify drug resistance genes (transduce cells, apply drug, sequence gRNAs from survivors).
- **Negative selection (dropout):** gRNAs targeting essential genes become depleted over time. Used to identify genes required for cell survival under specific conditions. Dropout screens require careful normalization; gRNA abundance at day 0 vs. day 14 across control and treatment conditions.
- **Fluorescence-activated screens:** Cells are sorted by marker expression (e.g., a surface protein for a phenotype of interest), and gRNA abundance in the high vs. low bins identifies genes that regulate that marker.

gRNA quantification from sequencing. After the selection period, genomic DNA is extracted from the cell population, and the integrated gRNA cassette is amplified by PCR. The resulting amplicon is sequenced, and reads are mapped back to the original gRNA library sequences. Each gRNA's read count is proportional to its representation in the surviving cell population; depletion relative to the initial time point (day 0) indicates that cells carrying that gRNA were lost (negative selection), and enrichment indicates a growth advantage (positive selection). The output is a count matrix (gRNAs $\times$ time points or conditions) that feeds into the statistical models below.

Statistical analysis. The Negative Binomial distribution (Chapter 4) models gRNA count data, where the mean-variance relationship $\text{Var}(Y) = \mu + \alpha\mu^2$ captures both sequencing noise and biological variation across cells. Tools (MAGeCK, BAGEL) test each gRNA's abundance change against the negative control gRNAs (non-targeting sequences) to estimate a per-gene effect size and false discovery rate. MAGeCK uses a modified NB model with mean-variance estimation pooling across all gRNAs; BAGEL uses a Bayesian framework with a prior derived from known essential and non-essential gene sets.

Copy number bias correction. In cancer cell lines with extensive chromosomal amplifications, a gRNA targeting an amplified region has additional target copies per cell, producing systematically higher gRNA abundance that mimics an essential gene signature;

the gRNA does not appear to drop out because there are more copies to cut, not because the gene is required for survival. The CERES algorithm (Meyers et al., 2017) models each gRNA's count as a function of both gene essentiality and local copy number, using copy number data (from WGS or SNP arrays) as a measured covariate to decouple the two effects. Without this correction, genes in amplified regions show false positive essentiality rates of 30-50%; a systematic bias that invalidates uncorrected essentiality screens in aneuploid cell lines. This is the most common computational artifact in cancer CRISPR screens and should be checked before interpreting any essentiality hits.

Perturb-seq: CRISPR + Single-Cell Readout

Pooled screens measure only a single phenotype (survival, marker expression). Perturb-seq marries CRISPR perturbation with scRNA-seq readout: each cell receives a gRNA, and the full transcriptome is measured simultaneously. This enables measurement of the complete transcriptional consequence of each knockout; not a single phenotype but thousands of expression features per perturbation.

The experimental pipeline. Cells are transduced with a gRNA library, then processed through the 10x Chromium (Chapter 13). The gRNA is detected as an additional feature (the gRNA transcript is captured by a specific primer during reverse transcription and sequenced alongside the cell barcode). Each cell's transcriptome is linked to its gRNA, identifying which gene was knocked out in that cell.

The computational challenge is detecting the perturbation signature against background variation. A knockout of a transcription factor may affect hundreds of genes, but each gene's individual expression change is small and noisy. The signal is in the coordinated response across multiple genes, not in any single gene's fold change.

- **Mixscape** (Seurat) classifies each cell as “perturbed” or “non-perturbed” for each gRNA using a regularized NB model. A knockout with 50% cutting efficiency means half the cells carrying a gRNA have no functional knockout, classifying them correctly is essential for detecting the perturbation signature. Mixscape uses the expression of a set of target genes (known or learned) to compute a “perturbation score” per cell.
- **scMAGeCK** applies NB testing per gene per perturbation, computing enrichment scores for gene sets rather than individual genes. It aggregates evidence across all cells carrying the same gRNA using a weighted sum approach that accounts for varying knockout efficiency.

Guide-Cell Assignment and sgRNA Leakage

In Perturb-seq, mapping a cell to its transduced sgRNA is not a simple binary lookup. Because multiple guides can contaminate a droplet (due to physical co-encapsulation or ambient sgRNA leakage from lysing cells), we must model the guide assignment statistically.

- **The Assignment Heuristic:** We count the number of UMI reads for each sgRNA in each cell. Standard pipelines apply a thresholding heuristic: a cell is assigned

to guide g if the count of g exceeds a minimum UMI threshold (typically ≥ 3) and comprises a significant fraction (e.g., $\geq 80\%$) of the cell's total sgRNA counts.

- **Ambient sgRNA Contamination:** In large-scale screens, cell lysis releases free lentiviral sgRNA transcripts into the cell suspension. These ambient transcripts are captured inside droplets containing unrelated cells. Without probabilistic models (such as those in **Mixscape** or **SCEPTRE**) that calculate the probability of guide capture versus ambient background, this “sgRNA leakage” results in mislabeling up to **10-15% of cells**, creating false-negative results and dampening the detected knockout signature.

Interpreting Screen Results

Essential genes: gRNAs that deplete over time identify genes required for cell survival. These are potential cancer drug targets, but a gene essential in one cell type may be dispensable in another. The “essentialome” is context-dependent, not universal.

Contextual essentiality and synthetic lethality. A gene may be essential only in specific genetic backgrounds. The classic example: BRCA1/2 mutations sensitize cells to PARP inhibition because the two DNA repair pathways (homologous recombination and base excision repair) compensate for each other. Loss of both is lethal, loss of either alone is viable. This is the **synthetic lethality** concept: two genes are synthetically lethal if mutation in either alone is viable but mutation in both is lethal. Cancer cells accumulate hundreds of mutations; each creates a vulnerability to synthetic lethal targeting.

The computational challenge: distinguishing true synthetic lethality from additive effects of two independent perturbations (each causes 20% growth reduction; together they cause 40%; this is additive, not synergistic). The difference between additive and synergistic effects requires modeling the expected combined effect under independence (typically Bliss independence: $E_{AB} = E_A + E_B - E_A E_B$) and testing for significant deviation.

Biological Interpretation

A Perturb-seq result must be interpreted through the lens of specificity and penetrance. A knockout that causes 50% of cells to change their expression state and 50% to remain unchanged suggests either: incomplete editing (Cas9 did not cut in all cells), compensatory mechanisms (paralogous genes buffer the loss), or a probabilistic threshold effect (the expression change requires additional signals present in only a fraction of cells). The response distribution (not just the mean effect) contains the biological information. A gene with a large mean effect driven by a small fraction of strongly responding cells and a majority of non-responders is fundamentally different from one with a moderate but uniform response across all cells.

Cutting efficiency is not knockout efficiency. A gRNA that cuts the genome with 90% efficiency may produce only 60% functional knockout; the remaining 30% of cuts were in-frame indels that preserve protein function. This discrepancy between molecular editing (detected by sequencing) and functional perturbation (detected by phenotype) is the

most common source of false negatives in CRISPR screens. Measuring protein abundance by FACS or western blot is the gold standard for confirming functional knockout.

Synthetic lethality is the most clinically actionable perturbation concept, but the gap between in vitro synthetic lethal hits and in vivo therapeutic efficacy is large. A gene pair that is synthetically lethal in a 2D culture may not be synthetically lethal in a 3D tumor microenvironment with hypoxia, immune pressure, and stromal interactions. In vivo CRISPR screens using barcoded tumor libraries in mice are bridging this gap, but at substantially higher cost and complexity.

Summary and Required Reading

1. **CRISPR-Cas9 creates double-strand breaks** repaired by NHEJ (frameshift knockout) or HDR (precise editing). gRNA design requires early constitutive exons and multiple guides per gene.
2. **Pooled screens** measure enrichment or depletion of gRNAs under selection; the Negative Binomial model (Chapter 4) provides the statistical framework for identifying significant hits.
3. **Perturb-seq** combines CRISPR with scRNA-seq readout, measuring the complete transcriptional consequence of each knockout; the signal is in coordinated gene responses, not individual fold changes.
4. **Synthetic lethality** identifies drugs targeting cancer-specific vulnerabilities, distinguishing synergy from additive effects requires modeling expected combined effects under independence.
5. **Cutting efficiency $\neq$ knockout efficiency**: in-frame indels preserve function; measure protein to confirm knockout.

Required Reading

- Shalem et al.: "Genome-scale CRISPR-Cas9 knockout screening in human cells" (*Science*, 2014).
- Dixit et al.: "Perturb-Seq: dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens" (*Cell*, 2016).
- Meyers et al.: "Computational correction of copy number effect improves specificity of CRISPR-Cas9 essentiality screens in cancer cells" (*Nature Genetics*, 2017).

Johnson's Rule: If you want to know whether a gene matters, don't look at its expression; knock it out.

TIER 4: Mass-Based Omics

Chapter 22: Mass Spectrometry and Proteomics

mRNA is a Blueprint, Not a Machine

The correlation between mRNA abundance and protein abundance is approximately 40–60%. Translation rate, protein half-life, post-translational modifications, and active secretion decouple protein levels from transcript levels. If you want to understand what the cell is actually doing, measure proteins. RNA is just noisy intent.

This gap between transcript and protein is not a measurement error; it is a biological reality. A strongly regulated mRNA may have no detectable protein change due to long protein half-life (days for histones, hours for metabolic enzymes). Conversely, a protein may be activated by phosphorylation or degraded by ubiquitination without any change in its mRNA. Chapters 4 and 13 taught you to measure RNA; this chapter teaches you to measure the actual molecular machines.

Core Concepts

The Physics of Mass Spectrometry

Mass spectrometry measures the mass-to-charge ratio (m/z) of ionized molecules. The entire pipeline (from peptide separation to ion detection) is designed around this single measurement, and each component introduces its own constraints on what can be detected.

Ionization (ESI). Electrospray ionization creates charged droplets from the liquid chromatography (LC) effluent. As solvent evaporates, droplets undergo Coulombic fission, releasing gas-phase ions with multiple charges. The key consequence: a peptide of mass 2,000 Da with 2 charges appears at $m/z = 1000$; the same peptide with 3 charges appears at $m/z \approx 667$. Multiple charge states mean the same peptide appears at multiple m/z values, complicating quantification but also enabling detection of high-mass peptides on instruments with limited m/z range.

Mass analyzers. The choice of analyzer determines resolution, speed, and accuracy:

- **Quadrupole:** Four parallel rods with alternating RF/DC voltages. Acts as a mass filter; only ions within a specific m/z window pass through. Low resolution

(\$1,000 FWHM) but fast scanning. Used for precursor selection in MS/MS (isolate the ion of interest for fragmentation) and for DIA isolation windows.

- **Orbitrap:** Ions oscillate around a central electrode. The frequency of axial oscillation is measured and converted to m/z via Fourier transform. High resolution (up to 1,000,000 FWHM) capable of resolving isotopic fine structure. The Orbitrap is the dominant high-resolution analyzer in modern proteomics because it combines high resolution with reasonable scan speed.
- **TOF (Time-of-Flight):** Ions are accelerated by a known voltage and their flight time to the detector is measured. Lighter ions arrive first. TOF has effectively unlimited m/z range and fast acquisition speeds, making it ideal for DIA and for detecting intact proteins (top-down proteomics).

MS/MS and acquisition modes. In **data-dependent acquisition (DDA)**, the instrument selects the top N most abundant precursor ions from each MS1 scan, fragments them by collision-induced dissociation (CID) or higher-energy collisional dissociation (HCD), and records the fragment ion spectrum (MS2). DDA is stochastic: the same peptide may be selected in one run but missed in the next due to minor abundance fluctuations. This “missing data problem” is the primary limitation of DDA; across multiple runs, only ~70% of detectable peptides are consistently identified.

Data-independent acquisition (DIA) solves the stochasticity problem by fragmenting all precursor ions within sequential isolation windows (typically 25–50 Da) across the full m/z range. Every peptide is fragmented regardless of abundance. The cost is spectral complexity; each DIA spectrum contains fragments from thousands of co-eluting peptides, requiring computational decoding against a **spectral library**: a collection of known peptides with their fragmentation patterns and retention times, built from prior DDA runs or synthetic peptides. The library provides a template for each peptide’s MS2 spectrum; matching algorithms (Spectronaut, DIA-NN, OpenSWATH) decode the multiplexed DIA spectra by finding the combination of library entries that best explains the observed fragment ion intensities. This library-based decoding is the key conceptual innovation that makes DIA work; it replaces stochastic precursor selection with deterministic, comprehensive fragmentation. DIA achieves >10,000 protein identifications per run with <1% missing values across replicates, making it the current standard for quantitative clinical proteomics.

Bottom-Up Shotgun Proteomics

Proteins are digested with trypsin, which cleaves after lysine (K) and arginine (R) (unless followed by proline. The choice is deliberate: K and R are abundant and evenly distributed across most proteomes, producing peptides of 5–30 amino acids) the ideal length for MS/MS sequencing. Trypsin also leaves a basic residue at the C-terminus of each peptide, which promotes positive-charge retention during ESI and produces predictable fragment ion series (y-ions) in MS/MS.

The digested peptides are separated by reverse-phase liquid chromatography on a C18 column with a gradient from aqueous (0.1% formic acid in water) to organic (0.1% formic acid in acetonitrile). Peptides elute in order of increasing hydrophobicity (short, hydrophilic peptides elute early; long, hydrophobic peptides elute late. This elution order

provides an orthogonal dimension to m/z : two peptides with similar mass can be resolved if their hydrophobicity differs. A typical 2-hour gradient resolves ~50,000 peptide species, though peak capacity scales with gradient length) longer gradients (4+ hours) are used for deep discovery proteomics. After separation, peptides are ionized by ESI and analyzed by MS/MS. Each peptide yields a fragmentation spectrum that is searched against a protein sequence database.

Database searching (SEQUEST, Andromeda) compares the observed MS2 spectrum to theoretical spectra derived from in silico digestion of the protein database. The SEQUEST XCorr score measures the cross-correlation between observed and theoretical spectra, accounting for the fact that fragment ion intensities are not perfectly predicted; the correlation is computed across multiple alignment offsets to find the best match. Andromeda (the MaxQuant search engine) uses a probabilistic scoring model based on the probability of matching fragment ions by chance, given the precursor mass tolerance and fragment mass tolerance.

The fundamental limitation: database search can only identify peptides present in the reference database. Novel peptides from unannotated genes, sample-specific variants, or unexpected post-translational modifications will not match and remain unidentified. Open modification searches and spectral library searching partially address this.

The PSM-to-protein hierarchy. The path from a raw spectrum to a reported protein involves three levels of inference, each with its own error structure. At the bottom, a **peptide-spectrum match (PSM)** links one observed spectrum to one peptide sequence; this is the atom of proteomics identification. Multiple PSMs for the same peptide (from replicate scans, different charge states, or modified variants) are aggregated to a **peptide identification**. Finally, one or more peptides support a **protein identification**, typically requiring at least two unique peptides per protein (the two-peptide rule). Errors propagate upward: a single false PSM can falsely support a peptide that in turn falsely supports a protein. The target-decoy framework below addresses this by estimating FDR at each level; PSM-level, peptide-level, and protein-level FDR are reported separately, and the protein-level FDR is always the most stringent because it counts any protein with even one false peptide as a false discovery.

Target-Decoy FDR

Decoy sequences (reversed or shuffled protein sequences) are appended to the database. A PSM matching a decoy sequence is, by definition, a false positive. The false discovery rate at the PSM level is:

$$\text{FDR} = \frac{\# \text{decoy hits}}{\# \text{target hits}}$$

At 1% FDR threshold, 1% of the reported PSMs are expected to be false. This is the universally accepted standard for proteomics identifications. Without a decoy database, proteomics results are untrustworthy; the search engine will always find the best match for each spectrum, and the reported score is meaningless without knowing how many matches would be expected by chance.

The protein inference problem. A single peptide may match multiple protein isoforms (shared peptides). When a peptide is shared between isoform A and isoform B, the software cannot determine which isoform contributed it. Most search engines report “protein groups”; sets of proteins that cannot be distinguished by the detected peptides. The reported abundance of a protein group is a weighted sum of all isoforms sharing the detected peptides. This ambiguity is structural: it cannot be resolved without longer peptides, different proteases, or orthogonal data (RNA-seq isoform quantification).

Quantification

Label-free quantification (LFQ): Peptide intensities are integrated across LC-MS runs (area under the chromatographic peak) and matched between runs by retention time and accurate mass. The “match-between-runs” feature in MaxQuant transfers identifications from one run to another for peptides that were detected but not selected for MS/MS fragmentation, increasing quantification depth by ~30-50%. LFQ assumes consistent LC performance across runs; retention time shifts of >1 minute cause matching failures.

TMT (Tandem Mass Tag): Isobaric tags label peptides from different samples (up to 18-plex with TMTpro). Each tag has the same total mass but different distribution of heavy isotopes. All labeled peptides co-elute and have the same precursor m/z , but upon MS/MS fragmentation, reporter ions at distinct m/z values (126-134 Da) are released. The reporter ion intensities encode relative abundance across samples.

Ratio compression. TMT suffers from a systematic bias: co-isolated peptide ions (precursors within the isolation window that are fragmented alongside the target peptide) contribute their reporter ions to the MS2 spectrum, diluting the true ratio. A true 10-fold change between conditions may appear as only 2-3 fold because the co-isolated peptides dilute the signal. Multi-notch MS3 (SPS-MS3) on the Orbitrap Fusion family reduces ratio compression by fragmenting the MS2 fragment ions a second time, isolating only the most abundant reporter ions, but at the cost of reduced sensitivity.

SILAC (Stable Isotope Labeling by Amino Acids in Cell Culture). Instead of labeling peptides after digestion (TMT), SILAC incorporates heavy isotopes metabolically: cells are grown in media containing light (Arg-0, Lys-0) or heavy (Arg-10, Lys-6) amino acids. After several doublings, all proteins in the “heavy” culture are fully labeled. Samples are mixed at the cell-culture stage, digested together, and analyzed in a single LC-MS run. Since labeling occurs before any processing, light and heavy peptides experience identical handling; there is no ratio compression because the labeled and unlabeled forms are chemically identical and co-elute. The mass shift (~6 Da for Lys-6, ~10 Da for Arg-10) is detected in the MS1 scan as a paired peak; the ratio of peak intensities is the relative abundance. SILAC is the gold standard for quantification accuracy but is limited to cell culture (not tissue) and classical 3-plex (though super-SILAC and SILAC with neutron encoding extend multiplexing). It remains the reference method against which LFQ and TMT accuracy are evaluated.

Phosphoproteomics and PTM Analysis

Phosphorylation is the most studied post-translational modification, but it illustrates a general principle that applies to ubiquitination, acetylation, and methylation: detecting

the modification is not enough, localizing it to the correct residue is essential.

Phosphorylation site enrichment (TiO₂, IMAC) isolates phosphorylated peptides from the non-phosphorylated bulk by affinity binding. After enrichment, site localization scoring (Ascore, PhosphoRS) determines which serine/threonine/tyrosine residue carries the phosphate. The Ascore computes the probability that the observed fragment ions are best explained by phosphorylation at site A vs. site B. An Ascore > 19 ($p < 0.01$) is considered confident localization.

Why localization matters: a phosphopeptide with ambiguous site assignment cannot be interpreted biologically; phosphorylation of Ser473 vs. Thr308 on AKT have completely different functional consequences (Ser473 is required for full AKT activation; Thr308 phosphorylation alone produces partial activity). Reporting "AKT phosphorylation detected" without site localization is biologically meaningless.

Biological Interpretation

Without a decoy database at 1% FDR, proteomics identifications are untrustworthy. The peptide-to-protein inference problem (shared peptides across protein isoforms) means many reported proteins are actually "protein groups"; the software cannot distinguish between isoforms, and the reported abundance is a weighted sum across all isoforms sharing the detected peptides.

Phosphoproteomics requires careful site localization scoring. A peptide with multiple serines where the phosphorylation site is ambiguous cannot be assigned to a specific signaling pathway. Many published phosphoproteomics datasets report sites at resolution insufficient to distinguish which kinase was active.

The correlation between mRNA and protein abundance (~40-60%) means that transcript-level differential expression cannot substitute for proteomics. A gene with unchanged mRNA but strongly altered protein abundance may be regulated at the translational or post-translational level. Conversely, a strongly regulated mRNA may have no detectable protein change due to long protein half-life. This is not a limitation of either technology; it reflects the orthogonal biological information each provides.

The dynamic range of the proteome spans ~10 orders of magnitude from the most abundant protein (albumin in plasma) to signaling molecules and transcription factors at the low end. A mass spectrometer can measure only 4-5 orders simultaneously; the instrument detects the most abundant ions while the low-abundance ones are suppressed by the high-abundance background. This is the fundamental detection bias of shotgun proteomics: the instrument always prioritizes abundant peptides, and low-abundance proteins (transcription factors, kinases, cell-surface receptors) are systematically under-sampled. Prefractionation (SCX, high-pH RP) extends coverage by reducing sample complexity before LC-MS analysis.

DDA and DIA produce systematically different missing data patterns. DDA has high missingness (~30% of detectable peptides are not identified in any given run), driven by the stochastic precursor selection and the dynamic range limit; the top-speed precursor se-

lection always picks the most abundant ions, and low-abundance peptides are rarely fragmented. DIA has low missingness but higher quantification noise from the complex multiplexed spectra. The choice between acquisition strategies depends on the biological question: DIA for discovery across many samples (clinical cohorts, biomarker screens) where consistent coverage matters more than absolute depth; DDA with fractionation for deep proteome coverage where quantification consistency across runs is less important; PRM (parallel reaction monitoring) or SRM (selected reaction monitoring) for targeted quantification of specific proteins where sensitivity and reproducibility are paramount.

Summary and Required Reading

1. **Mass spectrometry measures m/z :** ESI ionization produces multiply charged peptides; Orbitrap/TOF/quadrupole analyzers offer different resolution and speed tradeoffs.
2. **DDA vs. DIA** are complementary acquisition modes: DDA is stochastic (high missingness), DIA fragments everything (low missingness, complex deconvolution via spectral libraries). DIA is now standard for clinical proteomics; DDA with fractionation for deep coverage; PRM/SRM for targeted quantification.
3. **Target-decoy FDR** is the minimum quality standard; without a decoy database, identifications are untrustworthy. The PSM-to-protein hierarchy requires FDR estimation at the PSM, peptide, and protein levels, with protein-level FDR being the most stringent constraint.
4. **The proteome spans ~10 orders of magnitude in abundance**, but an instrument covers only 4-5; prefractionation extends coverage, and low-abundance proteins (kinases, transcription factors) are systematically under-sampled.
5. **Protein inference is ambiguous:** shared peptides mean reported “proteins” are often protein groups. Isoform-level inference requires orthogonal data.
6. **TMT ratio compression** systematically underestimates fold changes due to co-isolated peptide contamination; SPS-MS3 reduces but does not eliminate this.
7. **PTM site localization is essential for biological interpretation:** detecting phosphorylation (or ubiquitination, acetylation) is not enough; the specific residue determines functional meaning.

Required Reading

- Cox & Mann: “MaxQuant enables high peptide identification rates” (*Nature Biotechnology*, 2008).
- Elias & Gygi: “Target-decoy search strategy for increased confidence in large-scale protein identifications” (*Nature Methods*, 2007).

Johnson's Rule: mRNA is a blueprint; the protein is the machine. RNA is just noisy intent.

Chapter 23: Metabolomics and Mass-Based Profiling

Genomics Predicts What Might Happen, but Metabolomics Tells You What Is Actually Happening

I suggest you think of the cell not as a collection of static genetic blueprints, but as a dynamic, running chemical engine. DNA (genomics) represents potential; RNA (transcriptomics) indicates transcriptional intent; proteins (proteomics) represent the physical machinery; but the small molecule metabolites (metabolomics) are the actual downstream endpoints of all upstream regulation and environmental interactions. To understand the immediate physiological state of a cell, tissue, or organism, you must measure its metabolites.

Core Concepts

What Is Being Measured and at What Molecular Level

- **What Is Being Measured:** Small molecules (metabolites, amino acids, sugars, organic acids, lipids) typically under 1.5 kDa.
- **Molecular Level:** The cellular or biofluid biochemical concentration level. Metabolomics captures the active physiological processes and phenotypic endpoints of cellular metabolism in real time.

Analytical Platforms: Mass Spectrometry vs. NMR

I recommend you understand the physical and chemical tradeoffs between the two dominant platforms in metabolomics:

1. **Mass Spectrometry (MS):** Typically coupled with Liquid Chromatography (LC-MS) or Gas Chromatography (GC-MS).
 - *Sensitivity:* Extremely high. It can detect picomolar to nanomolar concentrations, capturing thousands of small molecule features in a single run.

- *Ionization constraints:* Molecules must be ionized (typically via ESI or chemical ionization). This makes the system susceptible to **matrix effects** (co-eluting compounds suppressing or enhancing the ionization of the target analyte).
2. **Nuclear Magnetic Resonance (NMR):** Measures the magnetic properties of atomic nuclei (typically ^1H or ^{13}C).
- *Reproducibility:* Superior to MS. It is non-destructive, highly quantitative, and requires minimal sample preparation.
 - *Sensitivity:* Low. It requires micromolar concentrations, meaning it only detects the top 50–100 highly abundant metabolites.

Targeted vs. Untargeted Workflows

I will teach you to distinguish between these two experimental strategies, as they require entirely different analytical frameworks:

- **Targeted Metabolomics:** Focuses on a pre-defined panel of chemically characterized molecules. You optimize the chromatography and mass spectrometer parameters (such as Multiple Reaction Monitoring, or MRM) to detect specific parent-to-product ion transitions. This yields absolute quantification (e.g., micromoles per liter) using internal stable-isotope standards, but is blind to any molecule outside the target list.
- **Untargeted Metabolomics:** Profiles all detectable chemical features (peaks with unique m/z and retention times) in a sample. This is a discovery-driven approach. The output is a matrix of relative peak intensities. The primary bottleneck is metabolite identification: converting a list of thousands of mass peaks into verified chemical structures.

Metabolite Identification and Reference Databases

In untargeted workflows, identifying a metabolite requires matching its accurate mass (MS1), isotopic pattern, retention time, and fragmentation pattern (MS2) against reference databases:

- **HMDB (Human Metabolome Database):** The primary source of truth for human metabolite annotations, containing detailed physiological concentrations and disease associations.
- **METLIN and GNPS:** Databases of tandem mass spectra (MS/MS) used to match fragmentation profiles.
- *Confidence levels:* Annotations are classified from Level 1 (identified via comparison with an authentic chemical standard analyzed on the same instrument) down to Level 4 (unknown feature with molecular formula only).

Multivariate Statistics: PCA vs. PLS-DA

Because untargeted metabolomics yields thousands of chemical features across a relatively small number of samples ($p \gg n$), you must use dimensionality reduction to interpret the data:

- **Principal Component Analysis (PCA):** An unsupervised method that rotates the data matrix to maximize variance along orthogonal axes (principal components). I suggest using PCA first to identify batch effects, outliers, and natural biological groupings without bias.
- **Partial Least Squares-Discriminant Analysis (PLS-DA):** A supervised method that rotates the matrix to maximize the covariance between the spectral features and the class labels (e.g., healthy vs. diseased).

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

Here, the algorithm actively guides the projection to separate the groups.

[!WARNING] **The PLS-DA Overfitting Trap.** Because PLS-DA is supervised, it will find a hyperplane that separates your groups even if the data consists of pure random noise. You must never report PLS-DA separation without validating it. I recommend using two metrics: R^2 (fraction of variance explained by the model) and Q^2 (predictive variance estimated by cross-validation). A model with high R^2 but low or negative Q^2 is overfitted and biologically useless. Furthermore, you must run permutation tests—shuffling the class labels 1,000 times to construct an empirical null distribution of Q^2 —to prove your model's classification is statistically significant.

Biological Interpretation

The Reviewer Mindset: The Quenching Mandate

Metabolic activity operates on a millisecond timescale. Enzymes do not stop working when you place a tissue sample on ice; indeed, cold shock can activate stress pathways and completely alter the metabolite profile within minutes.

Therefore, I suggest you adopt a critical mindset when reviewing metabolomics papers:

- **Quenching:** Was metabolic activity stopped instantaneously? I suggest looking for immediate liquid nitrogen snap-freezing or addition of cold organic solvents (like methanol/acetonitrile at -20°C).
- **The Ice Pitfall:** If the authors harvested cells by centrifugation at room temperature or left tissues on ice for 30 minutes before extraction, their metabolic profiles represent harvesting stress, not the physiological state of the organism. Do not trust their biological conclusions.

Confounders and Compositional Shadows

Metabolic profiles are highly sensitive to external variables. I suggest you control for these major confounders:

- **Dietary and Circadian Cycles:** Metabolites in biofluids fluctuate wildly based on fasting status, diurnal cycles, and physical activity. Lacking standardized collection times makes the data uninterpretable.
 - **The Microbiome:** A significant fraction of metabolites in human plasma (including short-chain fatty acids and bile acids) are synthesized or modified by the gut microbiota. A change in blood metabolites may reflect a shift in microbial ecology rather than host biology.
 - **Compositional Normalization:** Like sequencing data, total metabolite signal is bounded by the instrument's loading capacity. I recommend normalizing peak intensities by total signal (probabilistic quotient normalization, or PQN) or using internal standards to avoid compositional biases.
-

Current Landscape

- **Spatial Metabolomics (MALDI-MSI):** Matrix-Assisted Laser Desorption/Ionization imaging mass spectrometry allows mapping of small molecule distributions directly across thin tissue sections at near-single-cell resolution.
 - **High-Resolution MS (Orbitrap Astral):** Recent instruments achieve sub-ppm mass accuracy and extremely high scan speeds, allowing untargeted workflows to identify thousands of previously undetectable low-abundance metabolites.
 - **MetaboAnalyst 6.0:** The primary open-source suite for metabolomics statistical analysis, pathway enrichment, and multi-omics integration.
-

Summary and Required Reading

1. **Metabolites are dynamic endpoints:** They reflect the real-time physiological execution of the system, distinct from the potential state captured by genomics.
2. **MS has sensitivity; NMR has reproducibility:** Choose your platform based on whether you need deep discovery (LC-MS) or absolute quantification of major pathways (NMR).
3. **Untargeted workflows require identification validation:** Convert peaks to metabolites using databases like HMDB, paying attention to annotation confidence levels (Level 1–4).
4. **PLS-DA overfits easily:** Always validate supervised projections using Q^2 cross-validation and label-shuffling permutation tests.

5. **Quenching must be instantaneous:** Avoid harvesting artifacts by snap-freezing samples immediately.

Required Reading

- Wishart et al.: "HMDB 5.0: the Human Metabolome Database for 2022" (*Nucleic Acids Research*, 2022).
- Gloor et al.: "Microbiome datasets are compositional: and this is not optional" (*Frontiers in Microbiology*, 2017).

Johnson's Rule: Genomics is the blueprint; proteomics is the machinery; metabolomics is the exhaust. If you want to know how the engine is running, you must analyze the exhaust. Never assume transcription correlates with biochemical output.

Chapter 24: Lipidomics and Other Mass Spectrometry Assays

Lipids Are Active Cellular Gatekeepers, Not Passive Fat Storage

I suggest you discard the textbook view of lipids as mere energy-storing fat droplets. Lipids are structural barriers, signaling mediators, and anchors for membrane-spanning proteins. They dictate membrane fluidity, organize microdomains (lipid rafts), and initiate signaling cascades that control cell survival and death. Because lipids are highly hydrophobic and structurally complex, they require specialized extraction and analytical protocols. If you want to understand membrane biophysics or lipid-mediated signaling, you must measure lipids using dedicated lipidomic workflows.

Core Concepts

What Is Being Measured and at What Molecular Level

- **What Is Being Measured:** Hydrophobic small molecules (fatty acids, glycerolipids, glycerophospholipids, sphingolipids, saccharolipids, polyketides, sterols, prenols).
- **Molecular Level:** The cellular membrane structure and lipid signaling mediator levels. Lipidomics characterizes the diversity of lipid species, tracking their specific fatty acid tail lengths and saturation states.

The Chemical Complexity of Lipids

I recommend you appreciate the structural complexity that makes lipid identification particularly challenging:

- **Isobaric Species:** Many lipids share the exact same nominal mass but differ in their elemental composition (isobaric) or their structural configuration (isomeric). For example, PC(34:1) represents a phosphatidylcholine with 34 carbons and 1 double bond in its fatty acid tails, but this could be composed of PC(16:0/18:1) or PC(18:0/16:1).
- **Double-Bond Positions:** The position ($sn - 1$ vs $sn - 2$) and stereochemistry (*cis* vs *trans*) of double bonds alters the lipid's physical properties but is difficult to

resolve using conventional mass spectrometry.

Specialized Extraction Protocols

Unlike hydrophilic metabolites, lipids require organic solvents for separation from proteins and polar molecules. I will teach you the three major extraction methods:

1. **Folch Extraction:** Uses a mixture of chloroform and methanol (2:1, v/v) with water. The lipids partition into the lower chloroform phase. It is the historical gold standard for total lipid recovery but chloroform is toxic and the lower phase is difficult to pipet without contamination.
2. **Bligh and Dyer Extraction:** Similar to Folch but optimized for biological samples with high water content, adjusting the solvent ratio to achieve a monophasic mixture before adding water to split phases.
3. **MTBE (Methyl-tert-butyl ether) Extraction:** Uses MTBE and methanol. The major advantage: lipids partition into the upper organic phase. This makes automated pipetting and high-throughput workflows much cleaner, as you can draw off the lipid layer without disturbing the protein pellet or the aqueous phase at the bottom.

Analytical Strategies: Shotgun vs. LC-MS

I recommend you understand the two competing paradigms for analyzing extracted lipids:

- **Shotgun Lipidomics (Direct Infusion):** Lipids are infused directly into the mass spectrometer (often using nanoelectrospray) without chromatographic separation.
 - *Advantages:* Fast, highly reproducible, and enables constant spray conditions for quantitative comparisons.
 - *Limitations:* Susceptible to severe ion suppression, and cannot separate structural isomers.
- **LC-MS Lipidomics:** Lipids are separated using reverse-phase (C18 or C8) or normal-phase (HILIC) liquid chromatography before entering the mass spectrometer.
 - *Advantages:* Reduces ion suppression, resolves many isomeric lipid species, and provides retention time as an orthogonal identification metric.
 - *Limitations:* Longer run times, and retention time drift requires alignment algorithms.

Resolving Isomers: Ion Mobility Spectrometry (IMS)

An emerging dimension in lipidomics is Ion Mobility Spectrometry. IMS separates ions in the gas phase based on their size, shape, and charge as they drift through a buffer gas under the influence of an electric field.

- **Collision Cross Section (CCS):** The physical cross-sectional area of a lipid ion. CCS is a highly reproducible physical property that depends on the lipid's three-dimensional shape. By combining m/z , retention time, and CCS, you can confidently distinguish lipid isomers (such as determining the double-bond position) that are identical by mass spectrometry alone.
-

Biological Interpretation

The Reviewer Mindset: The Oxidation Warning

Lipids containing unsaturated fatty acids are highly susceptible to auto-oxidation when exposed to oxygen, light, or heat. This process (lipid peroxidation) creates artifactual oxidized lipid species and destroys the native lipids.

If you are reviewing a lipidomics study, I recommend you look for these quality control steps:

- **Antioxidants:** Did the authors add antioxidants like BHT (butylated hydroxytoluene) during extraction?
- **Storage Conditions:** Were the samples stored at -80°C under an atmosphere of nitrogen or argon? Storing lipid extracts at -20°C in standard air for months will lead to extensive oxidation, rendering the dataset invalid.

Dynamic Membrane Remodeling

The specific composition of lipids determines membrane biophysics. I suggest you focus on:

- **The Saturation Index:** Cancer cells often remodel their membranes by increasing the proportion of saturated and monounsaturated fatty acids relative to polyunsaturated fatty acids. This makes the membrane more resistant to lipid peroxidation and ferroptosis (an iron-dependent form of non-apoptotic cell death), protecting the tumor from chemotherapy.
 - **Signaling Cascades:** Cleavage of membrane lipids by phospholipases releases signaling mediators (such as arachidonic acid, the precursor for pro-inflammatory eicosanoids). Understanding these dynamics requires tracking the free fatty acids alongside the parent membrane phospholipids.
-

Current Landscape

- **LIPID MAPS Consortium:** The international classification system and reference database that standardizes lipid nomenclature, structural definitions, and mass spectra.

- **LipidSearch and MS-DIAL:** High-throughput software tools designed specifically for peak detection, alignment, and annotation of lipid species from LC-MS/MS data.
 - **Single-Cell Lipidomics:** Emerging microfluidic platforms coupled with high-resolution MS are starting to profile lipid compositions at single-cell resolution, though sensitivity limits remain a major bottleneck.
-

Summary and Required Reading

1. **Lipids are dynamic gatekeepers:** They dictate membrane biophysics and cell signaling, requiring distinct extraction and ionization protocols.
2. **Choose your extraction carefully:** MTBE extraction is preferred for high-throughput pipelines because the lipids partition into the easily accessible upper phase.
3. **Shotgun vs. LC-MS:** Shotgun is fast and reproducible; LC-MS resolves isomers and reduces ion suppression.
4. **Ion Mobility adds the CCS dimension:** CCS measurements resolve structural isomers based on physical shape rather than mass alone.
5. **Unsaturated lipids oxidize rapidly:** Protect samples with antioxidants (BHT) and inert gas storage to prevent technical artifacts.

Required Reading

- Fahy et al.: "Update of the LIPID MAPS comprehensive classification system for lipids" (*Journal of Lipid Research*, 2009).

Johnson's Rule: Lipids are the physical walls of the cell's compartments. If you do not measure the wall composition, you cannot understand how the molecular machines inside are isolated or regulated.